

The Economics of Public Procurement*

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Abstract. The economics of public procurement is an emerging subfield. Public procurement is large enough to matter for macro stabilization and long-run growth, yet it is not a frictionless component of “*government purchases*”: it is a rule-governed process that turns budgets into contracts, shaping prices, quality, delivery, and the distribution of public money. Using a five-stage procurement-cycle framework, this survey synthesizes evidence from four core areas: (1) how procurement demand transmits through composition, place, and production networks; (2) the size and sources of the wedge between authorized spending and delivered value; (3) how tender and award rules affect entry, bids, and value for money; and (4) how market structure, thin participation, and collusion risks condition performance. Three messages stand out: both public procurement design and contract awards have highly heterogeneous effects on firms and markets, the public procurement wedge is often large, and award prices alone are a noisy performance metric when quality and execution risks matter. We conclude with research priorities and a call for a cumulative “What Works for Public Procurement” evidence base.

Keywords: public procurement; public economics; fiscal policy; political economy; industrial organization

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1. Introduction

John Maynard Keynes's *The General Theory of Employment, Interest and Money* (1936) is the cornerstone of the macro case for countercyclical public works. It formalises how an increase in (net) investment raises income and employment via the multiplier, while warning that realised employment effects can be weakened by offsets and leakages. Yet Keynes largely treats implementation as a black box: “public works” enter as a net increment of government expenditures (G), with little attention to the public procurement (PP) and contracting institutions that translate spending plans into executed projects. Likewise, Acemoglu, Johnson, and Robinson's (2005) influential framework sets the rules of the game at the centre of growth, but it does not treat at all PP institutions. Yet this omission is non-trivial: PP is quantitatively large, 12.7% of GDP and 29.9% of total G across the OECD in 2023, and has several distinctive features (award rules, contract design, oversight, integrity risks) that warrant dedicated research rather than being treated as a neutral implementation detail (OECD, 2025).¹

Modern PP, understood as codified public advertising and competitive tendering, emerged around the 1830s, with major international harmonization beginning in the late 1970s and deepening in the mid-1990s.² For much of its scholarly development, PP research was dominated by legal scholarship, reflecting the centrality of statutory rules, interpretation, and implementation in governing buyer behaviour (Schooner, 2002; Trepte, 2004; Arrowsmith, 2014); under detailed procedural constraints and scrutiny by auditors, courts, and losing bidders, compliance and procedural defensibility were therefore first-order concerns for public buyers. From the late 1970s, and especially with New Public Management reforms in the 1980s, the evaluative focus in public administration shifted toward accountability, performance measurement, and value for money (VfM; Hood, 1991; Osborne and Gaebler, 1992; Robinson, 2015). This managerial turn did not supplant legal concerns, but it brought PP outcomes to the fore and sharpened interest in how tendering and contractual rules shape supplier participation, bidding behaviour, and competition. Economics could therefore draw on an already-mature body of auction and mechanism design to analyse tender formats and award rules (Che, 1993; Laffont and Tirole, 1993; Myerson, 1981; Vickrey, 1961), complemented by industrial organization to study entry, collusion, market structure, and performance (Bajari and Tadelis, 2001; Bajari, McMillan and Tadelis, 2009; Decarolis, 2014; Jofre-Bonet and Pesendorfer, 2003), moving PP beyond a narrowly legal or administrative topic and beyond a small applied subcategory within microeconomic theory and industrial organization.

Within economics, PP is formally recognised under the Journal of Economic Literature (JEL) code H57 (Procurement), which belongs to the broader Public Economics category (H). At the same time, much of the earlier economics literature approached PP primarily as a narrow application of auction theory (JEL category: D) and industrial organization (JEL category: L), rather than as a distinct domain with its own organizing questions. Recent research (e.g. Decarolis et al., 2020; Bosio et al., 2022; Chiappinelli et al., 2025a; Mazzucato et al., 2025), however, treats PP as an applied area with its own core questions, which we refer to as *the economics of public procurement*.³ We define *the economics of public procurement* as the study of how procurement rules and the choices of public buyers and suppliers shape ex ante outcomes (contract award and

¹ Over the past century, the scale of G has risen substantially in many economies, and in recent years it has reached high levels: for example, G accounted for approximately 57% of GDP in France, 45% in Brazil, 41% in Japan, 37% in Australia and Russia, and 36% in the United States (see Figure A1).

² For a brief historical overview of the evolution of public procurement frameworks, see Appendix A.

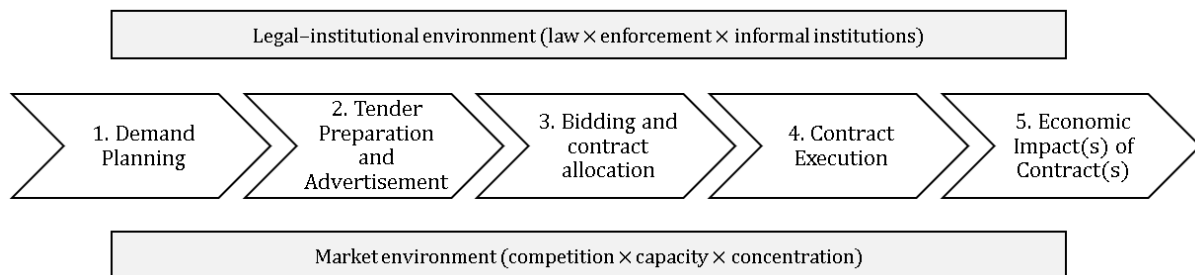
³ For supporting justification that the economics of public procurement can be treated as an applied subfield (Abbott (1988, 2001, 2005)), see Appendix B.

terms) and ex post outcomes (contract execution and performance), and how these in turn affect suppliers and, ultimately, industry-level and wider economic outcomes.

Following Bosio et al. (2022), we view many first-order features of PP as shaped upstream by the legal and institutional environment, including PP law, the broader regulatory framework, and how rules are implemented and enforced in practice. A second cross-cutting determinant is the market environment, including the degree of competition, capacity constraints, and supplier concentration. Against this backdrop, PP consists of repeated interactions around individual purchasing exercises among three sets of actors. The central government designs and enforces the legal and digital PP framework subject to political and budget constraints. Decentralised public buyers choose what to buy and how to buy it, plan purchases, run tenders, and manage contracts throughout their lifecycle, including monitoring, evaluation, and enforcement, to maximise VfM. Firms decide whether and how to compete for and execute public contracts, including choices during execution such as quality provision and subcontracting that shape realised costs, performance, and impacts.

These interactions unfold over five stages that organise the survey and locate where wedges between plans and outcomes arise: (1) demand planning; (2) tender preparation and advertisement; (3) bidding and contract allocation; (4) contract execution, including renegotiation, delays, fulfilment, realised quality, and enforcement; and (5) the economic consequences of contract awards and performance. Figure 1 summarises this structure and highlights that both the legal and institutional environment and the market environment operate throughout the cycle.

Figure 1. A Five-Stage Conceptual Framework



Existing reviews on PP, written by leading scholars in the field, provide careful and informative syntheses of key strands of PP research, typically organised around a single theme such as e-PP, PP reforms and savings, innovation PP, green PP, or government spending more broadly. However, because these contributions are theme-centred by design, they do not aim to offer a holistic account of the economics of PP (e.g. Fazekas and Blum, 2021; Mazzucato et al., 2025; Chiappinelli et al., 2025a; Ortega Carrasco et al., 2025).⁴ These reviews also suggest an uneven evidence base: rigorous causal evidence remains limited even for widely adopted “core” reforms aimed at cost efficiency, and is thinner still for non-price objectives such as innovation and environmental performance. This survey fills that gap by integrating theoretical and empirical evidence across themes and by identifying priorities for the next phase of work. We built the underlying literature using a structured database search and systematic citation tracking, and we iteratively validated our classifications through cross-checks and feedback.⁵

⁴ For a more detailed discussion of existing surveys (their scope and key findings) and how our contribution differs, see Appendix C, which also maps prior reviews to procurement-cycle stages and outcome families.

⁵ For more details on our Web of Science search code, see Appendix D.

The economics of PP has developed along two strands that are often studied separately. One treats PP as government spending that creates demand and affects firms, sectors, and places. The other studies the frictions and distortions that can prevent planned spending from translating into delivered goods and services of the intended cost and quality. We link these strands using the PP-cycle framework in Figure 1, which highlights where decisions and constraints matter from planning, to tender design and award, to execution and its consequences. Sections 2 to 6 follow this cycle, moving from what governments buy and how they plan purchases, to tendering and competition, to bidding and contract allocation, and then to execution outcomes and their broader effects. Section 7 concludes by summarising what is relatively settled, what is context-dependent, and where credible evidence remains sparse.

2. Public procurement transmission: plans to impacts

Demand planning is shaped by the interaction of three sets of actors: central government, individual public buyers, and suppliers. This section studies how these interactions determine (i) how much public spending is executed through PP contracts rather than other channels, and (ii) the composition of PP projects (what is bought, from whom, when, and where). Together, these choices govern the expected PP multiplier and, ultimately, the realized economic impact of PP spending. We connect two literatures. In the fiscal policy tradition, the “*public procurement multiplier*” is typically identified from aggregate shocks, but the estimated effect bundles together the pure spending impulse with a neglected PP wedge: the gap between planned or budgeted PP and the effective demand that is actually delivered through contracts. Newer work instead treats PP as a granular micro demand shock at the contract (or bundle-of-contracts) level (Srhoj and Dragojević, 2024; Ferraz et al., 2015). Once contracts are awarded and performance occurs, effects show up not only for direct winners but also upstream through production networks.⁶

A foundational determinant of the share of PP in G is the make-or-buy boundary. Governments do not procure everything they deliver: when tasks are difficult to specify, monitor, or verify, internal provision becomes relatively more attractive, whereas standardized or contractible tasks are more often outsourced. Empirically, this boundary is salient: in U.S. cities, a one-standard-deviation increase in perceived contracting difficulty is associated with roughly a 40% lower likelihood of outsourcing (Levin and Tadelis, 2010). Because make-or-buy choices determine which activities enter the PP pipeline at all, they shape the observed scale and composition of PP in the data. They also shape the stages that follow: only activities placed outside the government’s production boundary generate tenders, bids, contracts, and execution outcomes. Throughout the survey, therefore, PP should be interpreted as one implementation modality among several, with its own institutional logic and frictions.⁷

Procurement’s macro relevance is often framed through the fiscal multiplier literature, which estimates how changes in government purchases G translate into output Y . Across identification strategies, estimated purchases multipliers tend to fall in a central range around 0.7–1.5, while meta-regression evidence highlights a reference purchases multiplier around 0.7–1.0 and substantial heterogeneity (Ramey, 2011b, 2016, 2019; Gechert, 2015). This heterogeneity is consistent with variation in macro conditions (including mixed evidence on state dependence

⁶ Our analysis focuses on how procurement arrangements transmit economic effects across firms, sectors, and regions, rather than on the direct user benefits from the asset itself. Those user benefits belong to their specific fields, for example transport economics, health economics, education economics, and urban economics. Large infrastructure projects illustrate this boundary: they can reshape input markets, spatial economies, and future productivity, yet user-side benefits are outside our scope.

⁷ The institutional reality of most countries is more complex than a pure make-or-buy decision with mixed production models involving public-private partnerships or other models. We do not discuss such mixed models in this study.

and monetary constraints), openness and leakage, the composition of fiscal instruments, and the time horizon of adjustment (Ramey and Zubairy, 2018; Gechert, 2015; Boehm, 2020; Clemens, Michelsen, and Rieth, 2025; Antolín-Díaz and Surico, 2025). We use this benchmark while emphasizing a measurement distinction: multiplier estimates are typically based on national-accounts G , whereas PP research often focuses on contract-based procurement P observed in award and performance data. In national accounts, the link between G and contract-based procurement P is not one-to-one: “consumption” combines contracting with payroll, and “investment” captures capital formation while missing much routine contracting (operations, maintenance, and other current purchases). This motivates measuring P directly from contract-level awards and, where possible, the location of production and delivery.

As a guidepost for the discussion, we find it useful to distinguish four channels through which PP can shape aggregate outcomes. The formal decomposition appears in Appendix E, and here we use it only as a way to organize intuition.⁸ First, changes in non-PP spending such as public payroll, transfers, and administrative costs generate a baseline fiscal effect that can differ from the response induced by contract-based purchases. Second, holding total expenditure fixed, reallocating spending toward PP shifts the sectoral incidence of demand and thus the expected output response, generating a composition effect. Third, even when funds are authorised for PP, they may not translate into delivered activity or value. Losses can take the form of overpayment and cost overruns, quality shading, delays, or, in the extreme, partial or non-delivery, creating an efficiency wedge between planned spending and realised delivery. Fourth, PP effectiveness can vary with scale. Expanding or contracting the PP share of G may improve outcomes through learning and specialisation, or reduce them by straining administrative and market capacity, generating a recomposition wedge.

2.1. Public procurement: scale and composition

We now turn from aggregate fiscal multipliers to the structure of PP itself. This subsection focuses on Stage 1 and the composition of what governments buy. Procurement is sizable but varies across countries (Figure A1). For example, it is 14.5% of GDP in France and 9.4% in the United States. Measured relative to government purchases G , the share implemented through PP typically lies between 20% and 40% (median near 30%); for instance, it is 25.6% of G in the United States and 32.2% in France. A key source of compositional heterogeneity is what governments buy. Product level classifications such as the Common Procurement Vocabulary (CPV) show that PP demand is concentrated in a small set of categories and that the mix of works, goods, and services varies substantially across countries. Functional classifications such as COFOG point to equally large cross country differences in the public purposes implemented through contracts, ranging from defense intensive systems such as the United States (about one fifth of PP) to systems where defense is typically around 5% or less and PP is oriented toward health, environmental protection, or economic affairs.⁹ As a result, aggregate PP is not a uniform fiscal impulse, because its effects depend on the underlying product and functional mix.

2.2. Sectoral effects of public procurement

⁸ For the full decomposition, notation, and fiscal-transmission discussion underlying these four channels, see Appendix E (Sections E.1–E.3).

⁹ Figures A16–A17 show large cross-country heterogeneity in PP demand: CPV-based mixes of works, goods, and services are not proportional to domestic production, and COFOG profiles differ sharply, with U.S. defence around one fifth of procurement versus 5% or less elsewhere, and wide variation in health, environmental protection, and economic affairs.

These demand-side differences map directly into the supply side. Using the universe of U.S. federal contracts, Cox et al. (2024) document strong concentration across industries, with manufacturing (NAICS 33) and professional, scientific, and technical services (NAICS 54) accounting for a large share, and pronounced skewness of contract value across firms; this compositional specificity implies that PP shocks are sectoral, so their effects depend on which industries receive the demand injection.¹⁰ Using industry data, Nekarda and Ramey (2011) show that government spending expansions raise activity in directly exposed sectors, especially construction and public services, and are associated with lower real wages, higher markups, and modest declines in productivity, while untargeted sectors (e.g., parts of finance or high tech) respond much less. Consistent with this pattern, U.S. federal PP is disproportionately directed toward relatively sticky-price, high-markup industries (Cox et al., 2024), implying larger real effects when purchases hit those sectors because inflation responds less and monetary tightening crowds out less private demand.

Procurement shocks also propagate along supply chains, generating indirect effects on suppliers that can rival direct impacts in targeted sectors. Using U.S. data, Acemoglu et al. (2016) show that fiscal shocks transmit primarily to upstream suppliers of government-dependent industries, with more limited downstream effects. In a DSGE framework with nominal rigidities, Barattieri et al. (2023) similarly find that PP-driven demand raises upstream employment but can crowd out downstream activity through higher input costs when supply chains are tight. Three features govern the strength and sign of propagation: technological centrality, price flexibility, and contract design. Central upstream industries such as metals and machinery can act as hubs; sticky-price sectors tend to amplify real responses, while flexible-price sectors absorb shocks through prices; and long-term contracts can stabilise supplier demand but dampen cyclical responsiveness. Multi-sector evidence aligns with this distinction between direct effects and spillovers: Bouakez, Rachedi, and Santoro (2025) show that direct (origin) multipliers are larger when spending starts in downstream, labour-intensive, low-markup sectors and smaller when it starts in upstream, capital-intensive manufacturing. This origin ranking does not preclude large spillovers from upstream hubs: aggregate effects also depend on network propagation, so a small PP shock in a technologically central upstream sector can generate sizable economy-wide spillovers even if its local multiplier is modest.

2.3. Firm-level evidence

Firm-level research asks whether PP wins expand net demand or reallocate output across customer segments, and it is useful to frame the evidence around the constraints that public demand interacts with. One mechanism emphasizes relaxing demand or informational frictions (PP as a signal or relationship that supports later private growth), while the other emphasizes that a large public order can make capacity or liquidity constraints bind more tightly during execution. The latter reflects higher upfront working-capital needs that can rise faster than available finance, inducing short-run strain and potential crowding out of private activity (Hager and Huber, 2025). Across settings, winners typically gain revenues, but private-market spillovers and adjustment channels vary substantially.

Evidence from winner–runner-up and related designs shows both revenue gains and substitution. In Ecuador, winning a randomized construction contract raises contemporaneous revenues by about 22% and current assets by a similar amount, with effects fading after one year

¹⁰ Cox et al. (2024) document strong concentration in U.S. federal procurement: the largest two-digit NAICS sector is about 31% of contract value, the top two about 60%, and the top three about 70%; construction (NAICS 23) is 7–8% of obligations. At the firm level, value is concentrated among few suppliers and contract sizes are highly skewed.

(Fadic, 2020). In Croatia, winning sizeable works contracts increases total sales but shifts activity away from domestic private customers toward public demand, consistent with reallocation (Srhoj and Dragojević, 2024). Uganda points in the same direction: new government sellers increase total sales yet experience persistent declines in non-government sales and no improvement in value added per worker, suggesting crowding out rather than upgrading (Hoekman et al., 2024a). Spain adds a dynamic pattern, with short-run crowding out followed by recovery as firms accumulate net worth and capital (di Giovanni et al., 2022). Substitution is therefore more likely when PP is large relative to effective capacity and working capital, especially for financially constrained suppliers, and when public sales are relatively more remunerative. Two gaps remain central for cumulation: whether suppliers earn higher profit margins on public contracts than on comparable private sales, and whether such margins raise annual profitability. Addressing them likely requires better concordance tables linking PP categories to firm product codes such as CN or PRODCOM. Within this constraint-based interpretation, di Giovanni et al. (2022) emphasize limited-capacity reallocation, while Hoekman et al. (2024a) report survey evidence that many firms perceive government to pay higher prices and cite this as a reason to reduce private sales. Complementarity is more plausible when PP relaxes information frictions and provides reputation: in Korea, even short-term contracts are associated with longer-run growth and expansion of private operations (Lee, 2021), and related evidence from Brazil shows persistent growth and market participation consistent with learning and market expansion (Ferraz et al., 2015).

Labour-market responses differ sharply across contexts in persistence and adjustment margins. U.S. defence PP shocks (2001–2019) imply a three-year regional employment multiplier of about 0.1 and a cost of roughly \$284,000–\$305,000 per job-year, with only about 15% of the response at recipient establishments and about 85% arising through local spillovers via subcontracting, input–output linkages, and reallocation (Briganti et al., 2025). In Croatian construction, hiring is front-loaded and short-lived, peaking around ten weeks after award and largely dissipating within a year (Srhoj and Dragojević, 2024). In Brazil and Korea, wins increase employment by about 1–3 pp. with effects that can persist for several years, especially among younger firms (Ferraz et al., 2015; Lee, 2021). Composition differs as well: in Brazil, gains disproportionately reflect transitions from unemployment, informality, or non-employment into formal jobs (Ferraz et al., 2015), whereas in Croatia additional labour demand is met largely through foreign workers (Srhoj and Dragojević, 2024). Austrian construction shows state dependence, with post-award employment growth before the Great Recession but no differential headcount response during the crisis, consistent with labour hoarding (Gugler et al., 2020).

Capital and finance evidence supports a procurement-to-finance mechanism in which contract revenues raise investment directly and ease constraints as expected cash flows support borrowing, even though liquidity can be strained during execution when payments lag or guarantees are required. Ecuador shows short-lived increases in fixed assets (about 7%) and wage expenses (about 5%) in the award year (Fadic, 2020). In the United States, an extra dollar of federal PP induces 10–13 cents of additional capital investment among financially constrained firms (Hebous and Zimmermann, 2021). In Spain, selling to government raises leverage and expands credit lines, shifting borrowing toward earnings-based lending (di Giovanni et al., 2022). In Portugal, linked e-PP, tax, and credit-registry data show that wins increase credit and improve borrowing terms, with expansion concentrated in lending backed by guarantees (Gabriel, 2024). Crisis-era PP insulated contractors by sustaining capital expenditure and bank credit and reduced local employment losses in nearby retail and supplier industries (Goldman, 2020). A recurring

open question is whether these responses persist beyond contract completion and how they vary with contract size and duration and firms' balance-sheet conditions.

Productivity and upgrading effects are strongly context-dependent. In extreme defence mobilisations, binding constraints can induce organisational change: using newly digitised WWII aircraft data, Ilzetzki (2024) estimates that a 1% demand shock raises utilisation-adjusted, quantity-based TFP by about 0.4% within 12 months, with larger effects in capacity-constrained plants ("learning by necessity"). War PP can also have persistent downstream effects, as Vietnam War-era U.S. military PP in Korea increased later export participation (Barteska et al., 2025).¹¹ In contemporary non-war settings, winners more often gain revenues and survival with limited systematic TFP improvement, consistent with scale expansion and risk-sharing absent strong innovation pressure (Cappelletti et al., 2024). Germany shows that winning price-only contracts reduces subsequent product and process innovation and lowers innovation-related sales while increasing non-innovation sales (Krieger et al., 2024). Slovenia shows that PP participation attenuates productivity effects of public grants, consistent with relaxed constraints but weaker incentives to upgrade efficiency (Grajzl et al., 2024; Cepec et al., 2022; Abdurakhmonov et al., 2021). Reputation effects nevertheless matter, especially for young firms and information-scarce environments (Lee, 2021). Evidence on Matthew effects remains mixed: Brazil shows greater likelihood of future wins and persistent growth in size and survival (Ferraz et al., 2015), whereas in Croatian construction close winners tend to win fewer subsequent contracts, consistent with capacity constraints or bid rotation rather than cumulative advantage (Srhoj and Dragojević, 2024). Overall, the evidence suggests PP frequently increases scale and survival without pervasive productivity upgrading (Cappelletti et al., 2024; Hoekman et al., 2024a).

2.4. Strategic public procurement

The previous subsections examined downstream consequences of PP, that is, how contract awards affect firms, sectors, and production networks. This subsection turns upstream to demand planning, where public buyers decide what to purchase and whether to pursue objectives beyond core service provision. We distinguish instrumental PP, which delivers the core functions of government (e.g., operating hospitals, maintaining roads, providing schooling), from strategic PP, which embeds explicit additional objectives within those functions, such as innovation, environmental sustainability, industrial development, or socially responsible PP. Because these objectives are incorporated before tenders are written, they shape the composition of demand and therefore the channels through which PP transmits economic activity. Prominent examples include a) green PP, b) PP of innovation, and c) inclusion policies that target participation and awards across geography, firm size, and demography (Flynn, 2025; Hoekman and Sanfilippo, 2020; Laffont and Tirole, 1993; Lewis-Faupel et al., 2016).¹² Much recent attention has centred on PP of innovation, and Eurostat data suggest it is no longer marginal (Table A3). Countries therefore differ both in how they allocate PP demand across sectors and in how they steer demand within sectors across technologies and firm types, with implications for the planned PP-driven multiplier.

¹¹ Relatedly, PP can function as a geoeconomic instrument: supplier restrictions, tied aid, and state-directed finance steer foreign-financed contracts toward donor and allied firms (Mohr and Trebesch, 2025), reshaping the cross-border distribution of rents, capabilities, and strategic dependencies.

¹² The European Commission defines strategic PP as using PP to support policy goals such as innovation, green, and social objectives alongside VfM and service delivery: https://single-market-economy.ec.europa.eu/single-market/public-procurement/strategic-procurement_en [Accessed 29th August 2025].

2.4.1. Innovation-oriented procurement. Innovation-oriented PP commonly takes the form of PP of innovation (PPI) and pre-commercial procurement (PCP).¹³ These approaches use public demand to stimulate technological development and diffusion in contexts where market signals are weak or missing, making PP a demand-side instrument of innovation policy (Edler and Georghiou, 2007). Flagship programs such as DARPA, NASA, and Operation Warp Speed illustrate how large public buyers can accelerate technological development and shape technological trajectories (Mazzucato, 2011; Mazzucato et al., 2025). Specific public buyers repeatedly function as technologically sophisticated lead users. For example, PP by large research infrastructures can generate measurable learning-by-doing effects among suppliers. Using PP microdata from CERN's Large Hadron Collider, Castelnovo et al. (2018) document a statistically significant "CERN effect" on the long-run profitability of high-tech suppliers, consistent with technological learning and spillovers generated through demanding PP relationships. Similarly, public universities can induce demand-pull innovation among suppliers. Bianchini et al. (2019) show that suppliers of a large French public university exhibit a higher propensity to introduce new-to-the-market product innovations than comparable firms, suggesting that public scientific demand can foster novel product introduction and not only incremental upgrading.

Moving from illustrative cases to empirical estimates in more standard PP settings, evidence from Germany indicates that allowing innovation-specific selection criteria increased firms' turnover from new products and services, with gains driven mainly by firm novelties rather than market novelties (Czarnitzki et al., 2020). However, the broader literature does not uniformly imply incremental effects only, and outcomes appear sensitive to design choices, institutional context, and the surrounding policy mix (Guerzoni and Raiteri, 2015). Beyond firm-level outcomes, PP can shape the direction of technological change. Using U.S. data, Raiteri (2018) shows that innovative PP increases the technological generality of upstream technologies, consistent with broader diffusion rather than narrowly targeted innovation. Complementary evidence from Central and Eastern Europe suggests that impacts can be stronger when PPI is embedded in a broader policy mix. Stojčić et al. (2020) find that combining PPI with R&D support yields larger innovation additionality, particularly for SMEs. At the same time, interactions with other instruments can be nontrivial. Using panel evidence from Belgium, Czarnitzki and Prüfer (2024) find that while PPI and R&D grants each raise firms' R&D investment, their combined implementation exhibits meaningful crowding-out, implying that policy mixes can be substitutive when not well coordinated in design or timing.

Precisely because PPI and PCP target problems where market guidance is limited, they place demanding requirements on implementation. A recent synthesis concludes that PPI can affect firms' innovative activity and performance, but that impacts are highly sensitive to institutional design, implementation frictions, and buyer capabilities (Chiappinelli et al., 2025a). Effective PPI therefore depends on buyers' ability to articulate needs in functional terms, evaluate heterogeneous proposals beyond price, and manage risk and uncertainty through the PP cycle (Edler and Georghiou, 2007; Georghiou et al., 2014). Consistent with this, adoption and effectiveness remain uneven, reflecting well-documented barriers within public agencies,

¹³ PPI culminates in the purchase and deployment of an innovative good or service, typically requiring suppliers to undertake R&D to meet functional needs, whereas PCP procures R&D services only (often phased) without committing to a follow-on commercial purchase. Arora et al. (2025) use stock-market reactions to estimate that winners' expected private payoffs far exceed contract revenues (about 19× the maximum potential revenue), consistent with an implicit guaranteed-demand channel via downstream production and PP contracts, often noncompetitive and concentrated among large or vertically integrated firms. Belenzon and Cioaca (2025) similarly find R&D contracts crowd in firms' R&D spending and scientific publications (but not patents), with effects strongest when downstream procurement is noncompetitive and fading as R&D becomes more decoupled from downstream PP.

including risk aversion and organisational and procedural constraints that discourage experimentation and tilt procurement toward safer, incremental solutions (Uyarra et al., 2014).

2.4.2. Green procurement. Green public procurement (GPP) uses public purchasing power to advance environmental objectives, either by procuring greener goods and services or by weighting environmental criteria in awards based on the most economically advantageous tender. Evidence indicates that GPP shifts supplier behaviour along multiple margins. In Germany, winning firms are about 20 percentage points more likely to introduce environmentally friendly products, and these effects are confined to product innovation among SMEs, with no detectable impact on process innovation or on larger firms (Krieger and Zipperer, 2022). Complementary evidence shows improvements in environmental performance beyond innovation outcomes: energy efficiency increases through technology upgrading and cleaner inputs (Zhang and Jiang, 2022), and U.S. evidence indicates that green contract awards reduce firms' greenhouse gas emission intensity while increasing productivity, without crowding out private sales or merely selecting already-green firms (Chiappinelli et al., 2025b).

Effectiveness also hinges on implementation capacity. Larger and better-informed contracting authorities adopt green criteria more often (Testa et al., 2012), and EU co-financing increased GPP uptake in the Czech Republic by 2.7 pp (Nicolas et al., 2025). Spillovers include higher private green certification among suppliers (Simcoe and Toffel, 2014) and investment responses by competing firms that adjust to PP-driven demand while managing compliance costs (Chiappinelli and Seres, 2024). Where mandates, criteria, and monitoring are weak, the green premium is harder to justify and enforcement is uneven (Ortega Carrasco et al., 2025; Cheng et al., 2018). Beyond innovation and environmental objectives, governments also use PP to pursue distributional and social goals, and to steer demand toward domestic production (Flynn, 2025).

2.4.3. Socially oriented procurement. Governments use socially responsible PP policy (SRPP) to pursue distributional goals, such as inclusion and equal opportunity, by expanding access to contracts for targeted groups. While objectives are set upstream, effects largely materialize at award through eligibility and evaluation rules that shape who bids, who wins, and at what price. The two main tools are set-asides, which reserve contracts for eligible suppliers, and bid preferences, which keep contests open but give targeted firms an evaluation advantage. Because participation is endogenous, design is central: preferences aimed at marginal entrants can discourage entry, whereas modest handicaps applied to always-present incumbents can preserve rivalry and participation incentives (Jehiel and Lamy, 2015).

A prominent SRPP case is size-based inclusion targeting SMEs, often motivated by fixed bidding costs and financing constraints. Evidence on set-asides is mixed but sometimes favourable, including settings with no clear cost increase (Denes, 1997) and spillovers through strategic effects in adjacent markets (Nakabayashi, 2013). Bid preferences more often reallocate awards toward higher-cost bidders and can deter entry by low-cost rivals (Krasnokutskaya and Seim, 2011; Marion, 2007). Macro-consistent evidence illustrates the trade-off: using Spanish PP linked to firm outcomes, Di Giovanni et al. (2022) find that winning contracts expands credit access and sales, consistent with PP relaxing borrowing constraints, yet expenditure-neutral counterfactuals imply that tilting allocation toward smaller suppliers raises the relative price of public goods and reduces the real quantity delivered (about -0.85% of real GDP) even as participation rises. A related margin is lotting or reducing average contract size, which broadens participation but can lower aggregate efficiency in general equilibrium. This points to pairing inclusion goals with tools that relax constraints while preserving selection, such as newcomer-friendly lotting and execution-side instruments like milestone payments and faster payment cycles.

Demographic and status-based programs underscore implementation and enforcement margins. In U.S. road PP, disadvantaged business enterprise goals increased utilisation when enforcement was credible (Marion, 2011), while removal of race and gender preferences reduced average costs but lowered minority participation (Marion, 2009a, 2009b). Veteran set-asides provide a further benchmark: after an expansion of the practical set-aside use in the U.S. Department of Veterans Affairs, awards to service-disabled veteran-owned firms rose, entry increased, firms secured earlier first wins and higher survival, and there was no detectable deterioration in execution performance (Carril and Guo, 2023). SRPP tends to be less costly when performance is easier to verify and when entry responses can offset restricted eligibility. Finally, payment timing can matter as much as award rules: accelerating payments or using milestones can improve inclusion with fewer distortions to price competition (Marion, 2017; Marion and Rosa, 2025).

2.4.4. Industrial policy through procurement. Whereas SRPP targets inclusion and equal opportunity, industrial-policy PP targets geography and origin, for example through domestic-content rules and buy-national preferences. Industrial policy through PP uses purchasing power to steer demand toward domestic production, local employment, or strategic capabilities. The core instruments mirror those in SRPP but target geography or origin, for example through local and buy-national preferences and domestic-content requirements. As with SRPP, the key implementation levers are eligibility restrictions (who may bid) and scoring advantages (how bids are evaluated), and the central economic issue is the redistribution–efficiency trade-off: steering demand toward targeted suppliers can reallocate activity and jobs but often at the cost of weaker price discipline and higher PP spending (Craig and Sailors, 1988).

Quantitative evidence on buy-national provisions highlights the same mechanism in sharper form. Recent work on Buy American rules finds employment gains but a sizeable cost per job, and shows that tighter intermediate-input restrictions can raise costs further by limiting access to low-cost inputs (Bombardini et al., 2024). The broader lesson is that domestic-content rules can shift demand and employment toward targeted sectors and locations, but the channel frequently runs through higher PP prices and reduced input flexibility, which can lower the real quantity or quality delivered for a given budget. Across both SRPP and IP, evaluation benefits from separating whom the policy targets from how it is implemented (set-asides, bid preferences, domestic-content rules, and execution-side tools such as payment terms). Different designs shift participation, selection, and prices through distinct channels, determining whether distributional PP mainly reallocates awards ex post or relaxes constraints upstream while preserving rivalry.

2.5. Political control, incentives, and capabilities in demand planning

Public procurement is carried out by concrete public buyers, including ministries and agencies, municipalities, hospitals and schools, social security and health funds, state-owned enterprises, and other entities that commit public funds (Thai, 2001; Bosio et al., 2022; OECD, 2023). These organisations differ in legal form and governance but operate within political hierarchies: legislation and policy set mandates, elected officials appoint leadership, and budgets reflect negotiated priorities across levels of government (Acemoglu and Robinson, 2013; North, 1991). In demand planning, buyers decide whether needs can be met through regulation or internal reorganisation or require additional spending; conditional on spending, they choose timing, whether to allocate budgets now or later, and where to draw the make-or-buy boundary (Hart et al., 1997; Levin and Tadelis, 2010; Padgett, 1980; Wildavsky, 1964; Williamson, 1999). If they procure externally, they select contractual instruments and determine tender timing and

quantities, within this survey's focus on goods, services, and works rather than public-private partnerships and concessions (Bosio et al., 2022; Dimitri et al., 2006).

Budget constraints and funding rules structure these choices. Buyers typically combine core budgets financed by general revenues, local taxes, or user fees with earmarked intergovernmental transfers and, in many countries, supranational or external funds such as EU structural and investment funds, development-bank loans, and other co-financing mechanisms (Allen, Hemming, and Potter, 2013; Campos and Pradhan, 2007; OECD, 2023). Access to conditional and external resources requires projects that align with national or EU priorities, satisfy co-financing rules, and meet additional reporting and audit standards (Nicolas et al., 2025). As a result, demand planning reflects both operational needs and incentives embedded in funding regimes, shaping which projects enter the pipeline, how they are prioritised, and which objectives are attached to them.

Firms and industry associations can also shape upstream decisions. Early engagement may improve needs assessment, demand articulation, and cost benchmarks (Kelman, 1990), including through agenda setting and technical advice that effectively prespecifies key project features and helps buyers translate complex market knowledge into PP-relevant specifications (Selviaridis and Uyarra, 2025). But upstream influence is not neutral: political connections and lobbying can affect which projects are proposed, when they are initiated, and how they are structured, consistent with evidence on agenda setting and alignment-based resource allocation (Solé-Ollé and Sorribas-Navarro, 2008). Similar mechanisms, including early technical input and choices over contract sizing or bundling, can therefore either improve design or be used to favour particular suppliers.

Institutions shape both mandates and capacity. Planning capacity spans forecasting and aggregation tools, coordination across budget, line ministries, and PP units, and incentives that reward accuracy and timely delivery rather than blame avoidance or year-end budget exhaustion (Wildavsky, 1964; Padgett, 1980; Alesina and Perotti, 1996; Liebman and Mahoney, 2017; Coviello and Gagliarducci, 2017; Zamboni and Litschig, 2018; Kuroki and Sasaki, 2023). Capability demands also vary by task. For routine, standardised, high-volume categories, VfM hinges on spend analysis, forecasting, and aggregation institutions such as central purchasing bodies, framework agreements, and dynamic purchasing systems (Dimitri et al., 2006; Albano and Sparro, 2008; OECD, 2017; Bosio et al., 2022). For complex, project-based PP, planners must appraise and sequence large investments under uncertainty and coordinate design, lot structure, and multi-annual budgeting (Flyvbjerg and Gardner, 2023). When PP embeds objectives beyond service delivery, requirements rise further because goals must be translated into project selection, specifications, award criteria, and monitoring across policy domains (Edquist and Zabala-Iturriagagoitia, 2012; Lee and Malerba, 2017; Andreoni and Chang, 2019).

Political incentives and fiscal institutions affect how projects move through the pipeline. Politicians may favour visible, short-horizon projects and weaken oversight, contributing to cost overruns and higher prices (Brollo and Nannicini, 2012; Coviello and Gagliarducci, 2017). Spending can be steered toward electorally pivotal districts or politically connected firms (Golden and Picci, 2005; Mironov and Zhuravskaya, 2016; Lehne et al., 2018). Budget institutions add distortions: annual budgeting and weak carry-over provisions can induce end-year spending spikes, decentralisation can thin oversight, and excessive centralisation can concentrate rents (Allen and Tommasi, 2001; Mackie, Worsley and Eliasson, 2014). Bureaucratic incentives can reinforce these pressures. When mistakes are salient and punished but good planning is hard to observe and reward, officials become risk-averse and compliance-oriented, favouring defensible choices over innovative or complex projects (Alesina and Tabellini, 2007, 2008; Besley, 2006).

Conjoint experiments with PP officials in Finland and Germany support this mechanism: Tukiainen et al. (2024) show that officials prioritise avoiding unexpectedly high prices and suppliers with poor past performance over pursuing lower prices or higher quality, consistent with prevention-focused behaviour under weak extrinsic incentives and high accountability. Institutional correctives such as medium-term budgeting frameworks, transparent project pipelines, independent review bodies, and routine appraisal, cost-benefit analysis, and performance audits can improve selection and timing and support higher-quality planning (Allen and Tommasi, 2001; Mackie, Worsley and Eliasson, 2014). More broadly, systems that combine clear accountability with protected, well-defined discretion are better positioned to sustain effective planning and stable pipelines (Alesina and Tabellini, 2007, 2008; Besley, 2006).

Existing work provides a qualitative picture of how political control, multi-layered budgets, and bureaucratic incentives shape demand planning in PP, which unfolds within a nested delegation structure (Wildavsky, 1964; Besley, 2006; Bosio et al., 2022). By contrast, we still know relatively little causally about how improvements in buyer-side planning capacity, such as forecasting tools, more flexible budgeting, or stronger project appraisal, affect economic outcomes. Rigorous evidence remains comparatively thin on reforms targeting planning units rather than suppliers, even though supplier-side capability interventions can have large effects on market access and contract awards (Hjort et al., 2020).

3. The procurement wedge: sources, magnitude, and why it matters

This section asks how large the PP wedge is. We define the PP wedge as the gap between the social value PP could deliver for a given set of projects under an efficient process and the value actually realised when those projects are implemented through existing PP institutions. It materialises as cost overruns, delays, partial or non-delivery, and quality downgrades (Bandiera et al., 2009; Baltrunaite, 2020; Schoenherr, 2019), and as misallocation when contracts go to less productive firms despite the availability of more productive suppliers (Akcigit et al., 2023; Faccio, 2006). These losses shrink the effective fiscal envelope and reduce social welfare. Because buyers and firms anticipate them, they also feed back into expectations, make-or-buy decisions, and PP multipliers at both aggregate and contract or firm levels.

A central component of the PP wedge is politically mediated misallocation, whereby contracts are steered toward connected firms rather than the most cost-effective suppliers. Empirically, political connections are proxied using observable ties such as executives or owners who hold or have held political office, family links, overlapping networks such as school or party ties, campaign donations, or favoritism toward in-jurisdiction firms (Faccio, 2006; Goldman et al., 2013; Asai, Kawai, and Nakabayashi, 2018). These proxies likely understate true influence, yet they predict large advantages: in Brazil, firms donating to winning federal-deputy candidates from the ruling Workers' Party receive contracts worth roughly 14 times their contributions (Boas et al., 2014), while in Lithuania, banning corporate contributions reduces former donors' probability of winning by about 5 pp. relative to non-donors and is estimated to save taxpayers almost 1% of GDP in PP costs (Baltrunaite, 2020). Network channels also matter: in the United States, firms whose directors align with the party gaining control of Congress see large post-election increases in federal PP volume (Goldman et al., 2013), and in Japan, firms hiring retired senior officials from the relevant ministry raise contract revenue on the order of a few pp., consistent with revolving-door capture (Asai et al., 2018). Influence also operates through unelected political agents, meaning appointed advisers, senior party officials, and high-level political staff, and across 28 European democracies such links raise winning probabilities by about 5 to 8 pp., with effects larger and more robust than ties to elected politicians and

concentrated in upstream, low-visibility stages such as agenda setting, tender design, and internal approvals (He, 2025).

Beyond allocation, connections can degrade performance and may be electorally tolerated. In Korean state-owned enterprises, political appointments shift awards toward socially connected private firms whose construction contracts are 40.18 pp. more likely to be delayed and 24.50 pp. more likely to receive ex post cost increases, implying expected losses of about 32% of misallocated value and annual welfare costs around 0.4 to 1.1% of GDP, depending on network definitions (Schoenherr, 2019). In Croatian local governments, reelection probabilities peak when roughly 20% of PP is classified as suspicious, and incumbents tend to lose office only once about one half is suspicious (Vuković, 2020). Local favoritism can also arise without personal ties: Chinese PP auctions exhibit a strong bias toward local suppliers that is hard to explain by information frictions or classic corruption and instead appears linked to local leaders' career incentives (Wei et al., 2025). Consistent with entrenchment under repeated political control, one additional mayoral term in Italian municipalities reduces bidder participation by about 11.48%, lowers the winning rebate by about 5.7%, and increases both the probability the winner is local (about 5 pp.) and repeat awards (about 25.6%) (Coviello and Gagliarducci, 2017). These studies show that political and geographic connections distort both who wins and how contracts perform, creating a sizeable active component of the PP wedge as inefficiency deliberately chosen for private benefit.

Building on the preceding evidence on active waste through corruption, favoritism, and connected allocation, a key lesson is that these forces may account for only a minority of overall inefficiency. In a benchmark study of standardized goods in Italy, Bandiera, Prat, and Valletti (2009) decompose PP inefficiency into active waste and passive waste, where passive waste reflects mismanagement, weak administrative capacity, and low managerial effectiveness. They estimate that about 83% of total waste is passive rather than active. In PP-wedge terms, the distortions emphasized above are quantitatively important, but they sit on top of a much larger baseline of passive losses that is roughly five times larger. Complementary evidence comes from U.S. highway projects: Liscow, Slattery, and Nober (2025) document substantial heterogeneity in public engineers' quality with first-order cost consequences. Moving from the 25th to the 75th percentile of engineer quality reduces project costs by about 14%, holding project characteristics constant, and exogenous capacity shocks due to retirements raise costs by several multiples of engineers' wages, implying large social returns to public-sector human capital. These findings suggest that a substantial portion of the PP wedge reflects passive inefficiency driven by limited technical capacity, helping explain why similar PP rules can yield very different outcomes across agencies, sectors, and countries.

Cross-country evidence suggests that PP wedges are pervasive across income levels and legal traditions. Using World Bank survey data on a standardized open-tender road resurfacing contract of roughly USD 2.5 million implemented by a central government agency, Bosio et al. (2022) report expert assessments for a large set of countries. Using their data, we construct figures (Figure 3 and Appendix Figures A6 to A9) showing frequent reports of delays and cost overruns, opportunistic ex post renegotiation, delivery below tendered specifications, and restricted competition. For routine road works, experts often indicate that staying on budget, avoiding renegotiation, delivering promised quality, and maintaining open competition occur only occasionally or less often, implying that these problems are common features of PP practice rather than rare accidents. After documenting the wedge's aggregate magnitude, we turn to execution-stage proxies at the contract level and then to its determinants.

4. Public procurement outcomes

Once contracts are awarded, the allocative gains from competition and award design translate into VfM only if contractors actually deliver. Contract execution is therefore where PP outcomes are ultimately produced: costs are realized, schedules slip or accelerate, quality is delivered or degraded, and incomplete specifications are resolved through monitoring, renegotiation, and change orders rather than through the award rule alone (Decarolis, 2014).

4.1. Ex ante: price and non-price metrics

Because contract awards are typically observable and easily comparable, researchers and policymakers often use prices as a headline indicator of PP performance, sometimes treating lower prices as suggestive of stronger competition. For homogeneous goods with clear specifications, unit prices (for example, per kilogram or per license) are often the most informative efficiency measure because they track marginal cost. Where unit prices are unavailable or contracts are bespoke, many studies use the discount, defined as the final contract price relative to the buyer's estimate, as a proxy for savings (e.g., Srhoj and Dragojević, 2024). This proxy, however, inherits the properties of the estimate: estimation practices vary across buyers and over time (engineering models, historical prices, list prices, budget ceilings, vendor quotes), and estimates may be noisy or strategically set. As a result, an inflated estimate can mechanically generate a high discount without implying real savings, so comparisons of discounts across buyers or categories are informative only when conditioned on buyer, category, and time (for example, through fixed effects) or anchored to independent reference prices.

Prices capture only one dimension of PP performance. They are largely silent about quality, delivery, and renegotiation risk, and strong reliance on price-only award can induce lowball bids that are later recouped through change orders or quality shading (Laffont and Tirole, 1993; Decarolis, 2014). More broadly, realized payments can be distorted during implementation by renegotiation and corruption; for example, Chiappinelli (2020) documents substantial embezzlement through legitimate-looking expenses in Venice's MOSE flood-barrier project. For these reasons, PP practice typically reserves price-only auctions for settings in which technical, quality, and safety requirements can be specified with sufficient completeness; otherwise, buyers rely on multi-criteria procedures that explicitly score non-price attributes, with price as one component or, in some cases, not scored directly.

4.2. Ex post metrics

Ex post metrics assess VfM by comparing what was delivered, when, and under what contract modifications to what the buyer intended. We focus on renegotiation, fulfilment, timeliness, and realised quality, and then on the monitoring and enforcement institutions that shape them.

4.2.1 Renegotiations. Renegotiation refers to ex post contract modifications, typically via formal change orders. The scope for renegotiation increases sharply with complexity: for standardized purchases with clear specifications (e.g., a fixed quantity of a homogeneous good) there is little to revise beyond delivery, whereas in bespoke projects such as construction, uncertainty and incomplete contracting make changes to scope, design, and timelines routine. Renegotiation is a key channel through which the wedge between awarded terms and realized performance emerges during execution, and the governance problem is not whether contracts change but whether changes reflect efficient adaptation or opportunistic rent extraction. This tension is intrinsic to complex PP because competition can discipline the award price without disciplining total cost once contracts are adjusted. Since adaptation costs are large and fundamental, ex post

modifications are a core determinant of VfM rather than a residual detail (Bajari et al., 2014). This echoes classic critiques of competitive bidding for complex contracts (Goldberg, 1977) and contract-theoretic arguments that rigid fixed-price contracting performs poorly under substantial uncertainty when flexibility is efficiency-enhancing (Bajari and Tadelis, 2001). Public works evidence highlights both prevalence and interpretive difficulty. Using about 30,000 Italian projects, Decarolis and Palumbo (2015) show that non-trivial cost increases affect roughly half of contracts and time overruns are even more common, yet cost and time revisions are only weakly correlated, implying that price changes are not merely a mechanical by-product of delays. Time revisions matter because delays are costly and delay penalties are typically specified per day: in U.S. highway PP, delays can impose user costs exceeding \$1.75 million per day (Lewis and Bajari, 2011). Formats that explicitly reward timely delivery shorten completion times by 30–40% and generate welfare gains above \$6 million per contract, reflecting per-day savings over shorter delays (Lewis and Bajari, 2014). Delays can also carry political and human costs: in Venice’s MOSE project, corruption contributed to missing the planned timeline, leaving the city exposed during the 2019 flood that caused serious damage and fatalities (Chiappinelli, 2020). Across settings, observed renegotiation is consistent with both genuine adjustment to shocks and opportunism, including strategic underbidding and exploitation of informational asymmetries. In such environments *ex post* bargaining can reallocate rents even when the original award appears competitive (Decarolis, 2014; Laffont and Tirole, 1993), and renegotiation may be anticipated rather than exceptional, reshaping incentives *ex ante*. Many Latin American infrastructure concessions were renegotiated soon after award, with weak institutions facilitating large rent transfers (Guasch, Laffont and Straub, 2008; Guasch and Straub, 2009). A related mechanism arises in energy PP: widely anticipated renegotiation in Indian power purchase agreements encouraged underbidding and shallow indexation at award, shifting selection toward suppliers with greater bargaining power rather than lower underlying costs (Ryan, 2020). Which logic dominates depends crucially on the credibility of the buyer’s outside option. When costs rise or performance deteriorates, the buyer can accept revised terms, enforce strictly, or terminate and reprocur remaining works, but termination is often slow and costly due to litigation, administrative burdens, and delays from redesign, rebidding, and remobilization (Decarolis, 2014; Decarolis and Palumbo, 2015). Enforcement also operates through penalties (often liquidated damages), typically deducted at close-out from the final payment once total delay is verifiable; however, suppliers can litigate to overturn or recover penalties, making courts part of the outside option. In Italian public works, civil courts are the main forum, and litigation can interrupt completion and prolong delays, raising enforcement costs for buyers and weakening their bargaining position when suppliers threaten court challenges (Decarolis, Mattera and Menon, 2020). These frictions shape bargaining during execution and can make opportunistic amendments more attractive for suppliers (Decarolis, 2014; Laffont and Tirole, 1993). The enforcement environment also affects corruption risk: Arozamena et al. (2025) model renegotiation where suppliers bribe monitoring agents to obtain preferential treatment, which feeds back into bidding incentives and shifts awards toward favored suppliers; in equilibrium, low award prices can coexist with high realized costs once renegotiation and side payments are accounted for. The policy implication is therefore not to ban renegotiation, but to move it from bargaining toward rules and to make enforcement and the option to walk away credible so opportunistic change orders are unprofitable.

4.2.2. Fulfilment and realised quality. Even when delivery is timely, fulfilment can fail. A contract may be closed on time on paper while key components are missing, specifications are not met, or

the output is not usable, especially when verification is weak or acceptance is largely administrative. Contract fulfilment is therefore a minimum condition for value, and administrative records may overstate it. In India, politically connected suppliers were more likely to report completion even when inspections documented missing infrastructure (Lehne et al., 2018), underscoring the need for independent verification, particularly where local accountability is weak. Realised quality is often even harder to observe in administrative data because many important dimensions are difficult to specify, measure, or verify ex ante and may surface only during use or after failure. High-profile infrastructure failures with casualties underscore that weak oversight can surface as severe realised-quality problems, especially along non-contractible dimensions of performance.¹⁴ In most settings, quality must therefore be proxied using audits, inspections, or outcome-based indicators. In Indonesia, independent engineering verification revealed substantial missing road expenditures despite formal reporting (Olken, 2007), and in Brazil federal audits documented missing or incomplete public works despite reported compliance (Ferraz and Finan, 2011). In utilities and construction, researchers use measures such as safety violations, internal audits, and service interruptions to detect quality problems including pipe bursts, late deliveries, and work stoppages (Andreyanov et al., 2025; Decarolis et al., 2025). Because these dimensions are central to social value, post-execution monitoring, including safety metrics and citizen feedback, is essential for evaluation beyond what prices and timelines can reveal.

4.3. Oversight and enforcement in contract execution

Under incomplete contracts, oversight and enforcement are central to sustaining performance during execution. Targeted monitoring can reduce theft, cost overruns, and hidden quality failures (Olken, 2007; Decarolis and Giorgiantonio, 2022). Evidence also shows that monitoring instruments differ in what they deter: in Indonesia, higher government audit probability reduced overall leakage between reported spending and actual construction, whereas grassroots monitoring mainly reduced labor-related leakage, such as ghost workers, with little effect on material shortfalls, consistent with substitution across execution margins (Olken, 2007). In Peru, a randomized intervention combining civil society monitoring of identified worksites with credible signals of official follow-through reduced public-works costs by about 18% without delaying completion (Lagunes, 2017). In Brazil, a randomized 20 pp. increase in audit risk reduced both the share of audited resources with corruption and the share of PP processes exhibiting corrupt practices (Zamboni and Litschig, 2018).

Oversight can also be embedded in PP design through reputational incentives tied to audited performance. In construction contracts for the Italian electricity distributor Acea, a performance-based supplier rating based on randomized audits improved worksite safety compliance and service reliability, with external regulator data showing fewer and shorter blackouts, while bid discounts increased rather than prices rising (Andreyanov et al., 2025).

Monitoring and enforcement further shape renegotiation and termination during execution. When oversight is credible and sanctions can be applied and upheld, public buyers can more credibly reject opportunistic renegotiation requests, insist on performance, or terminate and

¹⁴ For example, the 1 November 2024 canopy collapse at Novi Sad's railway station in Serbia, killed 16 people and triggered public scrutiny of construction standards and oversight. See Reuters, "Serbia protests demand accountability one year after deadly roof collapse" (29 October 2025), <https://www.reuters.com/world/serbia-protests-demand-accountability-one-year-after-deadly-roof-collapse-2025-10-29/>; and the European Parliament (Strasbourg) event page describing the presentation of findings by the Independent Commission of Inquiry for Establishing the Responsibility for the Collapse of the Novi Sad Railway Station (21 October 2025), <https://www.socialistsanddemocrats.eu/events/serbia-turning-point-one-year-after-novi-sad>.

recover losses, reducing suppliers' incentives to treat renegotiation as a default continuation pathway (Decarolis and Giorgiantonio, 2022; Zamboni and Litschig, 2018). However, oversight design matters. In Chile, heavier auditing of auctions than direct awards induced officials to substitute toward direct contracting to reduce perceived sanction risk, lowering transparency and participation and increasing prices (Gerardino et al., 2024). Effective oversight therefore requires targeting genuinely high-risk behaviour with credible enforcement while avoiding protocols that penalise competitive procedures through disproportionate compliance costs. Safeguards do not end with bid data and auction monitoring. They also rely on the wider accountability system that makes enforcement possible, including audits, complaints mechanisms, and judicial review. Croatia's 2022 reform sharply increased appeal fees and was followed by a large decline in appeals, raising concerns about effective judicial protection (Turudić, 2024).

5. Bidding and contract allocation

5.1. Contract design in public procurement

Public procurement contract design shapes the PP wedge by allocating cost risk and hence balancing cost discipline against adaptability. Under fixed-price contracts, the contractor bears cost overruns (is the residual claimant), strengthening ex ante incentives to control costs; under cost-plus contracts, reimbursement shifts risk to the buyer, weakening cost discipline but facilitating adjustment to shocks and scope changes without immediate disputes over cost responsibility (Bajari and Tadelis, 2001). Because many projects require ex post modifications, competition can discipline the initial award without pinning down total costs once change orders become necessary. Fixed-price contracts therefore often turn adaptations into bargaining that can delay implementation, whereas cost-plus compensation can reduce the extent to which efficient adjustments are blocked by renegotiation frictions (Bajari and Tadelis, 2001). Contract form also shapes renegotiation leverage: fixed-price contractors facing losses may threaten delay, suspension, or exit, while cost-plus suppliers may disrupt performance via disputes over allowable costs, scope, or fees; when buyer outside options are weak, these channels can raise realised spending and delays. Consistent with these mechanisms, evidence from local public transport shows higher subsidies under cost-plus than under fixed-price contracts and persistent inefficiencies following cost-plus episodes (Gagnepain et al., 2013).

Contract design must also address quality and fulfillment when performance is multi-dimensional. High-powered cost incentives under fixed price can exacerbate the multitask problem, shifting effort toward measurable cost reductions at the expense of hard-to-contract dimensions such as quality, reliability, and safety. Hybrid arrangements therefore often combine price incentives with monitoring, performance provisions, and clear adjustment rules when quality is difficult to specify and verify (Holmström and Milgrom, 1991). These considerations underscore that contract design shapes the PP wedge through a common set of channels: the allocation of cost risk, the ease of adaptation, and the scope for quality shading during execution (Gagnepain et al., 2013).

5.2. Auction design in public procurement

Auction design matters in PP because the award mechanism shapes not only price bids, but also firms' ex ante decisions to enter (and whether to bid alone or in partnership) and their ex post choices over delivery, including quality provision and subcontracting. These margins affect execution outcomes and, ultimately, the PP wedge. Auction theory studies how mechanisms aggregate dispersed information into prices and allocations. In benchmark private-values

environments, Vickrey (1961) shows that truthful bidding is a dominant strategy in second-price auctions, and the Revenue Equivalence Theorem implies that standard formats yield the same expected revenue and allocation under independent private values, risk neutrality, symmetry, and common support (Vickrey, 1961; Myerson, 1981; Riley and Samuelson, 1981). In that benchmark, holding participation and reserve rules fixed, standard formats differ mainly in bidding strategies rather than expected revenue or allocative efficiency. Once values are common or signals are affiliated, equivalence becomes fragile: correlated information generates a winner's curse, affects participation, and makes information disclosure and the choice between open and sealed bidding first-order determinants of outcomes (Milgrom and Weber, 1982).¹⁵

Public procurement departs systematically from the conditions under which benchmark equivalence results are informative. First, PP is usually an auction of a contract, not a commodity: bids are often multidimensional, bundling price with quality, delivery time, service levels, warranties, sustainability criteria, or contract terms, so the allocation problem is not single-parameter. Second, bidders are often asymmetric in costs, capacity, and financing, and entry is endogenous because compliance and bidding costs are substantial, making "thin markets" common even when many firms are technically capable. Third, PP contracts are incomplete and adapted ex post through change orders, renegotiation, and discretionary enforcement, so the award auction disciplines the initial allocation but does not pin down total costs or performance when contingencies materialize. Fourth, PP is embedded in legal, audit, and protest constraints that make procedural defensibility a core objective: buyers must document and justify decisions, which favors simpler, verifiable rules (e.g., pay-as-bid pricing, auditable scoring dimensions) even when richer mechanisms could dominate in theory. Finally, information problems are often effectively two-sided (e.g., buyer's needs and supplier's costs/quality are imperfectly known) and, with limited enforceability, fully efficient trade can be infeasible without transfers in the background (Myerson and Satterthwaite, 1983). Together, these departures weaken any presumption of revenue equivalence and explain why format and information design can have first-order effects on participation, selection, and downstream execution.¹⁶

A large literature studies the choice between competitive bidding and negotiated contracting, jointly with contract form. Competitive bidding is often preferred for transparency and cost discipline, particularly for standardized tasks (Kelman, 1990; Tadelis and Bajari, 2006). When enforcement is credible and uncertainty is primarily on the supplier-cost side, auctions can limit rents through competition, and even one additional bidder can dominate bilateral negotiation (Laffont and Tirole, 1993; Bulow and Klemperer, 1996). For complex projects, rigid fixed-price tenders can invite costly renegotiation, and theory predicts fixed price when work is well specified and cost-plus when revisions are likely (Williamson, 1985; Bajari and Tadelis, 2001). Consistent with this, evidence suggests auctions deliver weaker ex post performance when designs are incomplete and entry is thin, while negotiated contracting can perform better by facilitating adaptation (see Subsection 4.2; Bajari and Tadelis, 2006; Bajari et al., 2009). The implication for award design is that format choice hinges on information structure, volatility, and enforceability: strong institutions and largely one-sided uncertainty favor auctions, whereas shared uncertainty and volatile environments raise the relative appeal of negotiation, albeit with greater risks of favoritism and rent shifting.

¹⁵ A brief glossary of the auction-theory terminology used in this section is provided in Appendix F.

¹⁶ Appendix F also discusses why second-price and VCG-style mechanisms are uncommon in PP and why multidimensional contracting weakens simple dominance comparisons (Ausubel and Milgrom, 2006; Kong et al., 2022).

5.3. Price-only formats and bidder asymmetries

In PP settings with thin entry, correlated costs, and asymmetric bidders, even price-only formats need not be equivalent. Different award rules induce different entry and bidding incentives, and the evidence shows that these differences translate into systematically different prices and execution outcomes. This subsection contrasts reverse first-price, closest-to-average-price, and reverse e-auctions. All three are price-only rules, but they translate bids into awards differently and therefore generate different incentives.

Lowest-price tenders are widely used because they are easy to verify and defend: payment equals the winning offer and budget control is straightforward. In standard models, lowest-bid PP yields well-behaved bidding, but with asymmetries and common sources of uncertainty that affect many bidders at once (e.g., input prices, site conditions, capacity constraints), strategies and expected payments become format-dependent and bids often bunch (Maskin and Riley, 2000a,b; Gugler et al., 2015). In reverse first-price PP, bidders frequently undercut to just below what they expect the lowest rival bid to be, producing clustering near the anticipated winning region (Maskin and Riley, 2000a; Gugler et al., 2015). Closest-to-average-price rules shift the strategic target from being the cheapest to forecasting others: bidders aim to be close to the expected average, which generates clustering around the anticipated cutoff and can create new scope for coordination and manipulation (Conley and Decarolis, 2016; Decarolis, 2018). Across formats, bids reflect not only costs but also beliefs about rivals and strategic responses to the rule, which matters for interpretation and for detecting collusion.

The comparison between lowest-price and average-bid rules therefore hinges on which margin the buyer prioritizes. Lowest-price rules typically offer stronger price discipline and sharper selection of low-cost suppliers when specifications are clear, entry is adequate, and enforcement is credible. Their weakness is that in complex projects with incomplete design and significant uncertainty, strong undercutting incentives can translate into worse execution. Reforms in Italian public works illustrate these trade-offs. In Turin, lowest-price PP produced sizeable discounts relative to reserve prices, but performance worsened: cost overruns increased and completion was substantially delayed (Decarolis, 2014). Screening “*abnormally low*” bids can mitigate these problems, but it raises administrative burden. Average-bid rules can broaden participation and reduce incentives to win by extreme undercutting, but they are cost-inefficient and invite strategic participation. In the Italian evidence, average-bid rules select the lowest-cost firm only about one third of the time and raise production costs by roughly 16% relative to lowest-price (Decarolis, 2018). Moreover, part of the extra participation can be strategic: collusion, shill bidding, and “average piloting” can inflate bidder counts without generating genuine competitive pressure (Conley and Decarolis, 2016). Consistent with these mechanisms, switching from average-bid to lowest-price reduced participation and subcontracting, increased joint ventures, and induced exit, including exit of firms linked to collusive groups, highlighting the trade-off between broader participation under average-bid rules and sharper selection under lowest-price rules (Branzoli and Decarolis, 2015; Conley and Decarolis, 2016).

Reverse electronic auctions (eRAs) keep a price-only award rule but change the bidding process by introducing dynamic online competition, typically after qualification, with suppliers able to revise bids downward in real time; anonymity and automatic time extensions can support price discovery and transparency (Soudry, 2004). The bid-revision stage also supports collusion screening: Baránek, Musolf and Titl (2026) show that suppressed undercutting between specific firm pairs, conditional on close initial bids, can reveal collusive networks in multistage auctions. The effects are not mechanical because bidders can start relatively high and cut only when

challenged, so additional price pressure depends on having enough active competition. End-game tactics often take the form of late, incremental undercutting (“sniping”) aimed at winning while revealing as little as possible about willingness to supply. Empirically, evidence points to modest but meaningful gains in some settings: in Slovakia, eRAs increased participation and reduced prices (Pavel and Sičáková-Beblavá, 2013). Broader e-PP reforms can also improve supplier quality and delivery performance even when price effects are limited (Lewis-Faupel et al., 2016), and ChileCompra reports administrative savings and average price reductions through standardized processes and wider access (Singer et al., 2009). These benefits are not automatic: low capacity, weak digital access, or limited enforcement can mute effects and encourage strategic behavior (Lewis-Faupel et al., 2016; Soudry, 2004). Overall, price-only formats can promote transparency and price discipline for standardized tasks, but their relative performance is highly sensitive to entry, common cost uncertainty, and the scope for manipulation.

5.4. Scoring auctions: price–quality trade-offs and reputation

Evidence on price-only formats highlights a key limitation of minimum-standard PP. Buyers can set technical thresholds, exclude non-compliant bids, and award to the lowest-priced compliant offer, but this can misallocate when performance above the minimum matters and suppliers differ in their cost–quality trade-offs. Scoring auctions address this adverse selection problem by awarding on “best value for money” through an explicit trade-off between price and non-price attributes (Che, 1993; Asker and Cantillon, 2008).

In a scoring auction, the buyer commits *ex ante* to an evaluation rule that maps each bid into a single score combining price with attributes such as quality or delivery time. Suppliers submit bundled offers and the contract is awarded to the highest score, so rivalry targets the buyer’s price–quality trade-off rather than price alone (Che, 1993; Asker and Cantillon, 2008). In practice, scoring is often implemented via point systems and non-linear or ratio rules, so design involves not only weights but also a scoring architecture that is defensible and auditable (Hanazono et al., 2016; Hortaçsu and Perrigne, 2021). Because suppliers privately know the cost of delivering higher quality, scoring creates information rents, and optimal scoring can place less weight on quality than its intrinsic value would suggest in order to preserve price competition (Che, 1993). These gains require credible commitment to the announced rule and sufficiently contractible or auditable attributes. When either condition is weak, design shifts toward simple and verifiable indices, constrained discretion, and complementary enforcement and information institutions (Che, 1993; Asker and Cantillon, 2008; Hanazono et al., 2016; Hortaçsu and Perrigne, 2021; Laffont and Tirole, 1993; Albano et al., 2025a). A useful distinction is between *ex ante* quality, which can be assessed and contracted at award, and *ex post* quality, which is realized in execution but is hard to verify to third parties. As PP moves toward *ex post* quality, buyers rely more on monitoring and reputational feedback (Hanazono et al., 2016; Hortaçsu and Perrigne, 2021; Albano et al., 2025a; 2025b). Certifications can provide auditable signals. In Italy, the Legality Rating improved access to PP and is associated with fewer cost overruns, modifications, and delays, with effects that depend on credibility and vary across settings (Pizzo, 2024).

When a non-price attribute is economically important and directly measurable, scoring can be especially transparent. In highway PP, A+B bidding combines price with completion time weighted by a user-cost rate and is often paired with incentive and disincentive clauses tied to promised time (Lewis and Bajari, 2011, 2014). Such designs can improve VfM while limiting discretion because the scored attribute is observable and enforceable (Hortaçsu and Perrigne, 2021; Lewis and Bajari, 2011, 2014).

Scoring also reshapes participation. Additional scored dimensions, heavier documentation, and evaluator judgment raise fixed bidding costs and can shrink entry, turning scorecards into participation filters (Bosio et al., 2022; Hanazono et al., 2016; Hortaçsu and Perrigne, 2021). Multi-attribute bidding raises cognitive and administrative demands and can induce mis-optimization, sometimes lowering efficiency despite stronger theoretical properties (Albano et al., 2024; Camboni et al., 2025; Laffont and Tirole, 1993). Non-price criteria also create predictable hazards: manipulability and litigation risk; incumbency bias via references and past performance; measurement error and metric gaming; misweighting of correlated attributes; and steering through over-specific technical screens (Bosio et al., 2022; Asker and Cantillon, 2008; Albano et al., 2024; Spagnolo, 2012; Albano et al., 2017). Buyer-side responses include more parsimonious scorecards, calibrated use of past performance, clearer definitions of “similar work,” and newcomer pathways such as probationary lots or reputation ramp-ups (Hanazono et al., 2016; Hortaçsu and Perrigne, 2021; Spagnolo, 2012; Albano et al., 2017).

Consistent with this logic, empirical evidence suggests that non-price attributes matter for selection and awards primarily under scoring rules. Krieger et al. (2026) find that innovation-related firm and founder characteristics predict participation and success only in scoring auctions, not in price-only formats. Similarly, Flammer (2018) shows that firms with higher corporate social responsibility are more likely to win government contracts, consistent with reputation operating as a valued non-price attribute.

When quality is weakly contractible, reputation systems become central because they let buyers condition future awards on realized performance. This shifts design toward information governance: how performance is measured, audited, and incorporated into future scoring in a credible and legally defensible way (Albano et al., 2025a; 2025b). Theory analyzes reputation-weighted scoring in repeated PP with unverifiable quality and compares it to alternatives such as low-price auctions with non-discriminatory reserve prices (Albano et al., 2025a; 2025b). Empirically, reputation-weighted scoring performs best when indicators are auditable and credibly mapped into future awards. Evidence from a large Italian utility shows that reputation based on audited performance indicators increased compliance and improved outcomes, with responses beginning already at announcement (Andreyanov et al., 2025; Decarolis et al., 2025). At the same time, scoring can expand the scope for discretion when non-price criteria are subjective or noisy, making evaluation harder to verify and easier to steer. In such settings, scorecards should be simpler and paired with safeguards so “quality” does not become an entry barrier or a vehicle for manipulation that widens the PP wedge by raising prices and weakening delivered performance (Che, 1993; Asker and Cantillon, 2008; Bosio et al., 2022; Hanazono et al., 2016; Hortaçsu and Perrigne, 2021; Albano et al., 2025a). Evidence from China underscores the risk: Chen (2025) documents patterns consistent with manipulation in tender preparation through customized scoring criteria and “zombie” bidders, with less productive firms more likely to win and winning prices higher than in comparable competitive tenders. Simulations suggest that trimming unnecessary scoring components and subjecting tender documents to independent review would lower prices and improve delivered quality, narrowing the PP wedge (Chen, 2025).

6. Pre-auction setting

6.1. Market structure and market conditions

A useful IO distinction is between market structure (buyer and seller concentration) and market conditions (entry barriers, information frictions, solicitation effort). Structure determines whether a contracting authority is effectively a price taker or has buyer power. With fragmented

demand and supply, buyers are close to price takers, while demand concentration enables buyer power through aggregation and volume reallocation. Supply concentration raises supplier market power, shifts outcomes toward markups and bargaining, and makes collusion easier. When both sides are concentrated, as in bilateral oligopoly, outcomes depend less on competitive bidding and more on strategic interaction and institutional details (Robinson, 1933; Manning, 2003).

6.1.1. Market structure. Within PP, spending is typically concentrated among a small set of large ministries, agencies, and municipalities, which account for a disproportionate share of total contract value. Network evidence reinforces this view. Studies using centrality measures identify “hub” buyers and CPV subnetworks with sharply different concentration profiles across product groups (Pliatsidis, 2024), and show that even when dispersion measures suggest a broad footprint, centrality can reveal dominance by a few institutions that account for most demand (Fountoukidis, Antoniou, and Varsakelis, 2023). Importantly, these measures describe concentration within the PP system: they show which contracting authorities account for most public spending and which buyers are “hubs” in PP networks. They do not by themselves establish that the public sector is a monopsonist or oligopsonist in the broader product market, which would require benchmarking public demand against total (public+private, domestic+foreign) demand for that market.

On the supply side, competitive pressure is shaped by the pool of firms that can realistically participate, which depends on concentration and on barriers to entry, expansion, and scaling in government markets. PP adds an institutional layer because investing in compliance capacity and procedural know-how lowers participation costs for specialized suppliers but can also entrench incumbents. As a result, realised participation is often thinner than the set of technically capable firms, and awards tend to concentrate among recurring suppliers (World Bank, 2017; Pliatsidis, 2024).

Aggregate indicators also point to thinning competition in the EU: average bidders per contract declined from about 5.7 in 2011 to around 3.2 in 2021, while the share of single-bid awards increased from roughly 24% to about 42%, with especially pronounced patterns in works and specialised goods (European Court of Auditors, 2023). This joint buyer–seller structure helps explain why identical PP rules perform differently across sectors and places. When markets are competitive, tendering is more likely to discipline prices, whereas concentration on either side shifts outcomes toward buyer power or supplier market power, and concentration on both sides makes bargaining and governance details central (Robinson, 1933; Manning, 2003). These distinctions map naturally into sectoral examples: defence, vaccines, and some medical products combine concentrated demand and supply with high fixed costs and capacity constraints, creating scope for bargaining and cost-based contracting (OECD, 2016; Baranov et al., 2017), whereas routine services typically feature many potential suppliers and weaker supply-side power.

Finally, evidence from defence PP illustrates how shifts in market structure translate into changes in governance and contract form. Mergers among U.S. defence contractors increased concentration and were followed by more non-competitive spending, greater reliance on cost-type contracts, and higher markups (Carril and Duggan, 2020). Classic bilateral PP models emphasise incentive design under asymmetric information and provide guidance on payment terms and risk-sharing when rich contracting is feasible (Laffont and Tirole, 1987, 1993). In practice, however, limited verifiability and thin participation imply that the central margin is often not the fine choice among auction formats, but the broader alignment of procedure, contract form, and entry policy with market conditions.

6.1.2. Market conditions. Having described market structure, we turn to market conditions that govern realised contestability in PP: entry and bidding costs, information frictions, uncertainty, capacity constraints, and buyer-side solicitation effort. Together they generate a participation wedge between technically feasible suppliers and active bidders: many firms could deliver ex post but do not bid ex ante because bidding is costly, risky, or predictably unprofitable. Buyers internalize this wedge when choosing procedures, qualification stringency, lot and contract size, transparency, and tender timing.

Participation is strongly related to performance, but nonlinearly: most gains come from moving away from single-bidder tenders. Early evidence suggests each additional bidder lowers the winning price by about 2% (Gaver and Zimmerman, 1977). In construction, the first few rivals matter most: discounts are about 9 pp. higher with two bidders than with one, while tenders with nine or more bidders have discounts about 26 pp. higher than single-bidder tenders (Srhoj and Dragojević, 2024). EU-wide diagnostics indicate PP often operates in this thin-competition region: roughly 40–50% of tenders attract a single bidder, while three or more bidders are associated with substantially lower prices and greater cross-border participation, especially for high-value and works contracts (World Bank, 2025). Beyond a small competitive set, marginal entrants may add less discipline because outcomes are already near competitive benchmarks and late entrants are often weaker or higher-cost (Gilley and Karels, 1981; Hendricks, Pinkse and Porter, 2003). Bidder counts should therefore be read as an equilibrium outcome of entry and selection, not a sufficient statistic for competition. Quasi-exogenous variation in publicity rules provides causal confirmation: adding one bidder lowers prices by roughly 6–10%, even for complex projects with incomplete contracting (Carril et al., 2022).

Market conditions are shaped not only by supplier decisions but also by buyer effort. Kang and Miller (2022) model PP as costly buyer search for additional sellers followed by negotiation, and document pervasive thin competition in U.S. federal PP: 44% of the PP budget was paid to contracts drawing a single bid in FY2015. They estimate expected bids under negotiation of about 1.6, versus 4.3 under a counterfactual first-price auction, and quantify marginal search costs of about \$1,700 per contract. The mechanism is straightforward: attracting and evaluating additional bids is costly, negotiation can partially substitute for auction competition, and the marginal entrant may provide limited discipline when suppliers are observationally similar.

Policy can still raise contestability when single-bidding reflects avoidable barriers. In the Czech Republic, restricting awards when only one bid is received lowered prices by about 6% without detectable quality losses (Titl, 2025). In Italy, stronger anti-mafia enforcement increased firm turnover and competition, consistent with organized crime sustaining entry barriers and collusive arrangements (Slutzky and Zeume, 2024). Together with Kang and Miller (2022), these findings imply that realised competition reflects not only technology and costs, but also buyer choices that shape the cost and risk of bidding and local institutions that deter collusion.

Even with modest entry costs, rivalry can be attenuated by relationship-based market organization. In U.S. highway PP, prime contractors frequently supply one another as subcontractors; each additional rival supplied as a subcontractor is associated with about a 1.4% higher prime bid (Marion, 2015). When a prior subcontracting partner is expected to participate as a prime, firms are 9–18 pp. less likely to bid, accounting for a sizable share of missing bidders (Gil and Marion, 2013). These dynamics imply that thin participation can be endogenous rather than purely regulatory and complicate inference from simple bidder counts (Hubbard and Paarsch, 2009).

Cross-border entry follows the same logic. Even under formally open regimes, administrative complexity, documentation requirements, and language costs deter non-local participation

(Bosio et al., 2022). When foreign entry occurs, it is typically led by larger firms and can generate learning and competitive spillovers (Hoekman and Sanfilippo, 2020). Legal openness alone is therefore not sufficient; buyer-side design that lowers participation costs can be decisive.

Overall, PP performance is often driven more by entry and participation design than by marginal differences in auction format. Auctions deliver their largest gains when participation is substantial and credible (Bajari and Hortacsu, 2008). When markets are concentrated or realised participation remains thin, negotiation or alternative mechanisms may dominate, and policies that reduce participation costs can improve contestability at limited efficiency cost (Athey et al., 2013). The practical prescription is therefore to match procedure choice and entry design to underlying market conditions, rather than to default to “always use auctions.”

When participation is thin, buyers rely on complementary instruments to discipline pricing and preserve rivalry. Entry assistance can reduce participation costs and increase bidding (Athey, Coey, and Levin, 2013). Benchmarking and reference pricing can limit overcharging when auctions do not reliably reveal costs in complex projects (Bajari and Tadelis, 2001). Lotting and multi-sourcing can sustain competition when a single large contract would otherwise eliminate the competitive fringe. More generally, targeted mechanism changes can reduce markups and adverse selection in specific environments (Cuesta and Tebaldi, 2025). These considerations help rationalize the prevalence of qualification and negotiated procedures. For standardized projects with a deep supplier pool, open or restricted auctions tend to perform well on price and transparency (Bajari et al., 2009; Chong et al., 2011). For complex or highly uncertain projects with few capable suppliers, negotiation can perform better by enabling screening, adaptation, and richer communication (Herweg and Schmidt, 2017). Finally, when bidding is costly, limiting entry can be welfare-improving, implying an optimal finite number of bidders (Samuelson, 1985).

6.2. Collusion and bid-rigging safeguards

Procurement collusion is most likely when the same firms meet repeatedly, demand is predictable, and institutions make deviations easy to detect and punish. In such settings, bidders can soften rivalry through bid rotation, market division, cover bids, phantom bidding, or designated-winner schemes that preserve the appearance of competition (Porter and Zona, 1993; Laffont and Tirole, 1993).

Bid rigging accounts for a large share of cartel cases pursued by U.S. and EU competition authorities, and related patterns arise in diverse settings, including Indian agricultural PP (Meenakshi and Banerji, 2005) and infrastructure PP in Lebanon (Mahmalat and Maktabi, 2024). Mahmalat and Maktabi (2024) document cartel organization around politically connected contractors and design consultants, with prices about 35% higher when both sides are linked to elites, highlighting consultants as brokers who facilitate coordination and corrupt deals, lower within-cartel transaction costs, and help sustain high-rent equilibria.

A practical response starts with detection. Bid screens apply statistical tests to bidding data to flag patterns unlikely under competition (Bajari and Ye, 2003). Common “fingerprints” include unusually low dispersion, suspicious bid similarity, and systematic rotation (Feinstein et al., 1985; Porter and Zona, 1993). Because these patterns can also arise in plausible competitive models, screens are best used for triage and to trigger deeper scrutiny rather than as stand-alone proof (Chassang and Kawai, 2025). Agencies can validate screens on known cartels (e.g., Sweden’s asphalt cartel) and deploy tools that do not require prior cartel identification, including designs targeting reserve manipulation, close-call patterns, and bid-threshold interventions (Kawai and Nakabayashi, 2022; Kawai et al., 2023; Ortner and Chassang, 2018).

Kawai et al. (2023) propose more robust tests that generate sharper predictions and are harder to reconcile with competition. With rebids triggered by secret reserve prices, collusion is indicated when the identity of the lowest bidder is unusually persistent from the initial auction to the rebid, even in near-ties where competitive bidding would often flip the winner. They also develop diagnostics based on “isolated winners” (missing mass of close losing bids near the winning bid) and “close-call” comparisons that test whether marginal winners and losers differ discontinuously around near-ties, as would be expected under rotation- or incumbency-based allocation rules but not under competition (Chassang and Kawai, 2025).

Multistage e-auctions also create a useful screening margin because bidders can revise bids downward after initial submission. Baránek et al. (2026) propose a network-based screen that flags firm pairs that systematically fail to undercut each other in the revision stage when initial bids are close, a pattern that is hard to reconcile with competition when initial bids are shaded. In their Ukraine application, flagged pairs are about nine times more likely to be sanctioned, and pairwise links can be combined into suspected collusive networks.

Design choices also shape cartel incentives. Changes in auction format can induce cartel exit and lower prices (Conley and Decarolis, 2016), and minimum thresholds can curb collusion in high-risk segments (Ortner and Chassang, 2018). Moving to scoring auctions can weaken cartel discipline because scoring creates imperfect monitoring: bidders observe bids and awarded scores but not underlying quality submissions, making deviations harder to detect and punish (Ortner et al., 2025). This points to a targeted transparency trade-off: broad disclosure need not be the problem, but releasing fine-grained information that improves cartel monitoring (e.g., highly mechanical scoring rules or detailed quality disclosures) can strengthen collusion. Deterrence can also backfire if cartels adapt (Kawai and Nakabayashi, 2024), so screening and format choices must be embedded in credible enforcement (Laffont and Tirole, 1993).

Many systems therefore rely on red-flag tools to prioritize scrutiny, not to prove wrongdoing. These indicators typically fall into a few families: competition risks (single bidding, non-open procedures), discretion risks (highly subjective criteria, repeated awards to the same supplier), process anomalies (departures from a buyer’s historical baseline), and VfM risks (change orders, cost overruns, renegotiations, and bid patterns such as bunching near reserve prices) (Ferwerda et al., 2017; Decarolis and Giorgiantonio, 2022; Lisciandra et al., 2022; Gnaldi and Del Sarto, 2024). Good practice combines signals into transparent risk scores, benchmarks across buyers and categories, validates flags against external measures of corruption and performance, and uses rankings for audit and civil-oversight triage rather than as stand-alone evidence (Fazekas and Blum, 2021).

Timing and portfolio design also matter: simultaneous tenders can be less conducive to tacit collusion than sequential PP, and first-price formats can be more robust in some collusive environments (Iossa et al., 2024). When formal monitoring is limited, reputation and relational contracting can partly discipline quality through repeated interaction, though they may also preserve incumbent advantages (Board, 2011; Albano et al., 2017). Overall, anti-collusion policy is most effective when detection and format choices operate within an enforcement system with auditing capacity, functioning appeals, risk-based triage, and credible sanctions (Laffont and Tirole, 1993). Complementarities matter: scoring rules linked to audited standards can improve quality without raising costs (Andreyanov et al., 2025), while heavy procedural scrutiny can induce substitution toward less transparent direct contracting and higher prices (Gerardino et al., 2024).

Public buyers pursue price, quality, and timeliness subject to budgets and supplier constraints, and many pre-award choices shape both entry and enforceability after award: solicitation

intensity, specification and scoring, contract size and lotting, disclosure and timing, and procedure and award rules (Section 4). These are the operational margins of discretion, affecting who participates and how verifiable performance is; where buyers can decide whether and how hard to solicit rivalry, thin participation is not only a market primitive but can also be a buyer outcome. Kang and Miller (2022) formalize this margin as costly buyer search for additional bidders in negotiated PP, helping explain why single-bid outcomes can persist even without strong administrative barriers.

6.3. Discretion, rules, and safeguards

Discretion is the authority to interpret rules, design tenders, and evaluate suppliers, and it operates through procedure choice, award criteria and scoring, qualification thresholds, timelines and documentation, invitee selection and rotation, and the scope of negotiation. In Decarolis et al. (2025), discretion is most salient in procedure choice and award criteria: below statutory thresholds officers may use simplified or negotiated procedures subject to minimum-invitee rules, whereas above thresholds competitive procedures are standard and negotiation is constrained. Discretion also governs how much competition buyers induce. Kang and Miller (2022) show that while more intensive seller search can increase expected offers, search is costly and can delay execution; when suppliers are relatively homogeneous conditional on observables and negotiation allows buyers to extract informational rents, the marginal benefit of additional bidders is limited. Consequently, low participation need not indicate wrongdoing, as inducing and evaluating extra offers may be costly and yield little additional discipline when suppliers are similar.

A recurring theme is that discretion tends to improve outcomes where administrative capacity is high, while more rigid rules perform better where capacity is weak (Laffont and Tirole, 1993; Bosio et al., 2022; Hoekman et al., 2024b). Organisational design mediates how discretion translates into performance. In a field experiment with 600 PP officers in Punjab (Pakistan), shifting authority from monitors to frontline officers lowered prices by about 9% without reducing quality, whereas performance pay had small average effects and worked only when monitors were not delaying approvals (Bandiera et al., 2021). This highlights interactions between autonomy, incentives, and monitoring, and the need to balance agency problems across organisational layers.

A useful distinction is between discretion over entry, the ability to shape who competes through restricted or invitational procedures and screening, and discretion over execution, the flexibility needed to manage incomplete contracts after award. Statutory value thresholds provide a common identification device because contracts just below and above a cutoff face different degrees of discretion. Threshold evidence illustrates the trade-off. In Italy, Coviello, Guglielmo, and Spagnolo (2018) exploit a threshold that makes competitive procedures more common above the cutoff and restricted procedures easier below it; greater discretion increases repeat contracting by about 25 pp. without worsening prices, delays, or cost overruns, with some evidence of improved execution. In Hungary, Szucs (2024) studies a reform that expanded access to high-discretion invitational procedures below a threshold and finds the opposite pattern: discretion exercised through bidder selection and reduced participation raises prices by 5–10%, lowers bidder counts, and shifts awards toward less productive and politically connected firms, alongside manipulation around the cutoff. U.S. evidence suggests that tighter procedural requirements can also be costly: using the simplified acquisition threshold, Carril (2021) finds that added documentation and oversight above the cutoff distort purchasing decisions and worsen ex post performance by 5–15%, while delivering only modest reductions in waste when

bureaucratic misalignment is limited. Read together, these studies imply that discretion is beneficial when it helps aligned, capable buyers manage execution and match contracts to needs, but harmful when it primarily restricts entry or steers awards, consistent with delegation frameworks (Aghion and Tirole, 1997).

Safeguards determine whether discretion functions as execution flexibility or entry restriction. In Italy, expanding negotiated-procedure thresholds increased negotiated awards without clear jumps in awards to investigated firms at cutoffs (Decarolis et al., 2025). The main efficiency gain was faster execution when minimum-invitee rules were respected; discounts were smaller, but fewer renegotiations reduced ex post cost growth. By contrast, bunching below thresholds is a common way to avoid competitive requirements (Palguta and Pertold, 2017). In Ecuador, connected winners are more common in discretionary awards and charge higher prices for homogeneous goods, implying meaningful overpricing (Brugués, Brugués, and Giambra, 2024). Related evidence links political connections to poorer quality and delays (Lehne et al., 2018) and documents politically timed inflation and diversion (Mironov and Zhuravskaya, 2016). Together with Kang and Miller (2022), the broader lesson is that low competition can reflect benign optimisation under costly solicitation or strategic restriction and capture. The policy object is therefore calibration and safeguards, not discretion per se: allow flexibility where capacity and transparency are strong, enforce rivalry within discretionary space (minimum invitees, non-discriminatory qualification, documentation and justification), and backstop with monitoring, auditing, and credible sanctions (Decarolis et al., 2025).

6.3.1. Administrative capacity as a moderator of rules and discretion. Administrative capacity, understood as the skills, experience, and institutional support available to PP offices, limits how effectively buyers can use discretion. With asymmetric information and incomplete contracts, capable buyers must translate needs into verifiable specifications, choose scoring rules, and match contract form to risk (Laffont and Tirole, 1993). Several studies indicate that capacity differences explain a large share of observed inefficiency. Bandiera et al. (2009) attribute most Italian waste to passive waste, consistent with weak capacity rather than active rent extraction. In the United States, measured office competence causally reduces delays, cost overruns, and renegotiations (Decarolis et al., 2020). Evidence from standardized medical devices in Italy similarly points to large buyer fixed effects. When mandatory reference prices reduced discretion, price dispersion fell, but the policy raised prices for more competent buyers and lowered them for less competent buyers, implying gains for low-capacity offices and losses for high-capacity ones (Buccioli et al., 2020). Consistent with these patterns, discretion tends to be beneficial in higher-capacity settings but not in lower-capacity ones (Bosio et al., 2022).

Capacity also affects innovation and long-run outcomes. In U.S. military R&D, deaths of responsible officers shortly before award reduce subsequent innovation, highlighting the importance of the design phase (Decarolis et al., 2021). In Russia, buyer identity explains substantial price variation, and reforms appear to benefit low-capacity settings while constraining high-capacity ones (Best et al., 2023). Cross-country evidence similarly shows that implementation quality explains more variation in outcomes than de jure rules, particularly where discretion is broad and capacity is low (Bosio et al., 2022). The consistent lesson is that the same rule can have opposite effects across agencies. Flexibility can help capable buyers adapt, innovate, and manage complex projects, while in low-capacity environments it can widen scope for waste and capture. Reform payoffs are therefore conditional. Rules that discipline weak bureaucracies may hinder strong ones, and discretion that empowers skilled officials may be misused without monitoring. Procurement systems should calibrate discretion to observable

agency capacity, invest in capacity when flexibility is intended, and rely on simpler standardized procedures where capacity is limited.

6.3.2. Pre-award design levers that implement discretion. Discretion and capacity, introduced above as high-level determinants of performance, translate in practice into a limited set of pre-award design choices under buyers' control. These choices directly affect participation, prices, and execution risk along two margins. First, they shape entry by altering search, eligibility, and bid-preparation costs. Second, they shape implementation by determining what is verifiable, the intensity of monitoring required, and the scope for strategic behaviour.

For recurrent needs, buyers choose between standalone tenders and repeated purchasing instruments, especially framework agreements and dynamic purchasing systems (DPS). Frameworks prequalify a fixed supplier set for a given duration and govern call-offs, while DPS remain open to entrants and rely on simplified mini-competitions for call-offs (Dimitri, Piga, and Spagnolo, 2006; Albano and Sparro, 2008). The appeal is to economize on repeated tendering costs and accelerate purchasing, ideally shifting competition to the call-off stage. Reviews report large price effects for centralization and framework-type reforms, sometimes above 50%, but estimates often bundle reforms and may reflect centralization rather than the instrument itself (Fazekas and Blum, 2021); where gains are driven by centralization, classic evidence attributes much of the improvement to reductions in passive waste (Bandiera et al., 2009). A key trade-off arises under uncertainty: because frameworks lock in terms while suppliers face cost risk and buyers retain an outside option, bids can include a winner's-curse-like premium relative to auctions run as needs arise. Gur, Lu, and Weintraub (2017) show this premium can be mitigated by monitoring outside-market prices with matching requirements, indexing prices to common cost components, or allowing restricted flexibility that lets the winner match spot prices without exceeding the contracted level. Overall, repeated instruments can reduce duplication, but performance hinges on sustaining contestability over time and having capacity to run call-offs transparently and monitor suppliers (Bandiera et al., 2009; Bosio et al., 2022; Fazekas and Blum, 2021; Gur, Lu, and Weintraub, 2017).

Specifications are a second lever linking discretion to both entry and execution. Detailed technical requirements can reduce uncertainty and opportunism, but narrow choices (brand references, proprietary technologies, unique features) can shrink the eligible pool and effectively preselect incumbents. Rigid specifications can also raise renegotiation risk when conditions change, consistent with adaptation costs under incomplete contracts (Bajari and Tadelis, 2001; Bajari, Houghton, and Tadelis, 2014). Performance-based approaches specify outputs rather than inputs, supporting innovation and quality but increasing evaluation and monitoring demands; when quality is hard to verify, they require capacity to assess heterogeneous proposals, monitor outcomes, and enforce compliance. The core trade-off is clarity versus flexibility: clarity simplifies bidding and enforcement, while flexibility can broaden participation and innovation at the cost of greater verification needs (Lember et al., 2014).

A third set of choices concerns contract size, lotting, and centralization. Lot design determines how much to aggregate versus split by product, geography, or time. Smaller lots can expand SME access and contestability but increase coordination costs and may sacrifice scale economies (Grimm et al., 2006; Albano and Sparro, 2010). Empirically, splitting can increase entry and lower prices (Castellani et al., 2018) yet also fragment PP and raise transaction costs (Caires et al., 2023), and it can be used strategically to remain below procedural thresholds that trigger more competitive or centralized procedures (Castellani et al., 2018). The associated centralization-decentralization trade-off is therefore about capability and oversight as much as scale.

Centralization can standardize procedures and exploit scale, and in Italy centralized purchasing reduced prices substantially, largely by reducing passive waste (Bandiera et al., 2009); cross-country reviews also report sizable gains, though bundled and context dependent (Fazekas and Blum, 2021). Decentralization can tailor tenders to local needs, but in low-capacity settings it can raise prices, while centralized pathways can add formalism and delay in emergencies (Caires et al., 2023). Fragmentation can also weaken oversight and rivalry when it yields repeated small awards to the same suppliers. Related strategic behaviour appears in subcontracting, including bid shopping, whose welfare effects depend on whether it attracts additional subcontractors when bid-preparation costs are high (Deneckere and Quint, 2025). Many systems therefore adopt hybrids (centralized PP for standardized high-value items and local discretion for context-specific needs), and performance depends on enforcement because weak rules invite circumvention (Bandiera et al., 2009; Castellani et al., 2018).

Information, publicity, and timing form a final lever operating primarily through search and participation costs. With asymmetric information, credible PP information can partly substitute for complex incentives when monitoring is limited (Laffont and Tirole, 1993). Transparency rests on clear rules, accessible documentation, non-discriminatory publication, and ex post monitoring and accountability (Bosio et al., 2022), and participation responds with first-order effects on outcomes (Krasnokutskaya and Seim, 2011). Causal evidence shows sizable participation effects from information reforms: in Italy, stronger advertising just above a threshold increased bidders by about 9.3% and raised winning rebates by about 7%, implying savings around €35k per €500k contract and larger entry by non-local firms (Coviello and Mariniello, 2014a). E-PP platforms can also increase participation and lower prices by standardizing information and reducing transaction costs (Decarolis and Giorgiantonio, 2015; Lewis-Faupel et al., 2016). However, transparency must be credible: in Uganda, information alone did not raise participation unless paired with trust-building signals (Colonnelli et al., 2024). Disclosure can also facilitate collusion when it becomes predictable (Coviello and Mariniello, 2014a), implying that transparency-driven entry gains should be balanced against coordination risk and paired with monitoring, auditing, and credible sanctions (Arrowsmith, 2010; Bosio et al., 2022).

Taken together, these pre-award choices are the concrete channels through which discretion affects outcomes: they determine who participates, how competition is sustained over time, and how costly it is to verify performance and deter manipulation. This is why similar formal discretion can be productive in high-capacity settings with credible oversight, but harmful in low-capacity settings where design can restrict entry or evade accountability.

7. Conclusions and Future Research Agenda

This survey organizes the economics of PP around a procurement-cycle framework that integrates two literatures often reviewed separately: PP as a channel through which public spending becomes real economic activity, and PP as an institutional process that generates a wedge between authorized spending and realized social value. Viewed through this lens, PP outcomes are jointly shaped by rules and enforcement, the scope and discipline of discretion, administrative capacity, and market structure. Together, these features determine participation, allocation, execution, and how PP demand propagates to firm-level, sectoral, and regional outcomes.

Four patterns emerge consistently across settings. First, the effects of PP spending are highly heterogeneous: composition, execution location, and production networks shape transmission from spending to output and employment, and contract-based shocks often propagate through upstream linkages beyond direct winners. Second, the PP wedge is pervasive and frequently

large. Delays, cost overruns, renegotiation, and quality shortfalls are common, arising both from active distortions (corruption, favoritism, political connections) and from passive weaknesses (limited capacity, weak project preparation, and monitoring). Third, tender and contract design strongly influence participation and prices, but award price alone is a noisy measure of VfM when quality and execution risks are salient. At the firm level, contract wins typically raise sales and ease financial constraints, while evidence on productivity and capability upgrading remains mixed and context-dependent. Fourth, strategic PP can advance innovation, environmental, and inclusion objectives, but effects are most robust when objectives are explicit, measurable, and supported by complementary policies and capabilities.

Looking ahead, policy relevance is highest where evidence remains fragmented or thin. Table 1 summarizes priority first-order research questions along the PP cycle and highlights cross-cutting needs for improved outcome measurement, welfare-relevant evaluation, and understanding of long-run market structure.

Table 1. Priority research questions for advancing the economics of public procurement

Stage	Research questions
Demand planning	For a given public buyer, how do planning choices (timing, bundling, budget use) affect bidder entry, prices, and execution outcomes? How do changes in funding rules or budgeting constraints shift what projects get procured and when? What is the impact of opening the PP to foreign firms?
Tender preparation and advertisement	Which tender and advertising choices increase qualified entry without raising collusion or low-quality participation? Does a public buyer's mix of tender designs predict its average prices, delivery performance, and dispute rates?
Bidding and contract allocation	How do price-only versus scoring auctions affect award prices, winner selection, and strategic bidding (including collusion indicators)? When do scoring auctions improve delivered quality without increasing total or lifecycle costs?
Contract execution	Which contract terms and enforcement environments reduce change orders, delays, and cost growth while preserving efficient flexibility? Which objective and subjective quality measures best capture delivered performance and predict downstream failures?
Economic consequences	What are the effects of PP participation on firms, and how do outcomes differ for winners and non-winning bidders? Does PP increase supplier concentration over time or sustain entry and competition?

Together, these questions point to a common need: moving beyond award-stage prices toward integrated evidence that links planning choices, tender design, allocation, and execution to delivered performance and downstream economic outcomes. Progress will likely come from three complementary directions. First, richer measurement of execution and quality, including standardized indicators that distinguish legitimate adaptation from opportunistic renegotiation and that capture delivered value rather than contracted intent. Second, data linkages that connect contracts to firms, workers, and places, allowing researchers to trace spillovers through supply chains and local labor markets and to study how PP shapes market structure over time. Third, credible identification strategies around institutional variation, staggered reforms, thresholds, and enforcement changes, ideally paired with designs that illuminate mechanisms and trade-offs. A natural institutional complement to this research agenda would be a “*What Works for Public Procurement*” platform: a living repository that curates rigorous evaluations of PP reforms, effects of PP and strategic objectives, standardizes outcome definitions and reporting, and translates findings into implementable guidance for buyers and policymakers. By lowering the costs of learning across contexts and making evidence cumulative, such a platform would help the field converge on which reforms improve VfM, under which conditions, and at what trade-offs.

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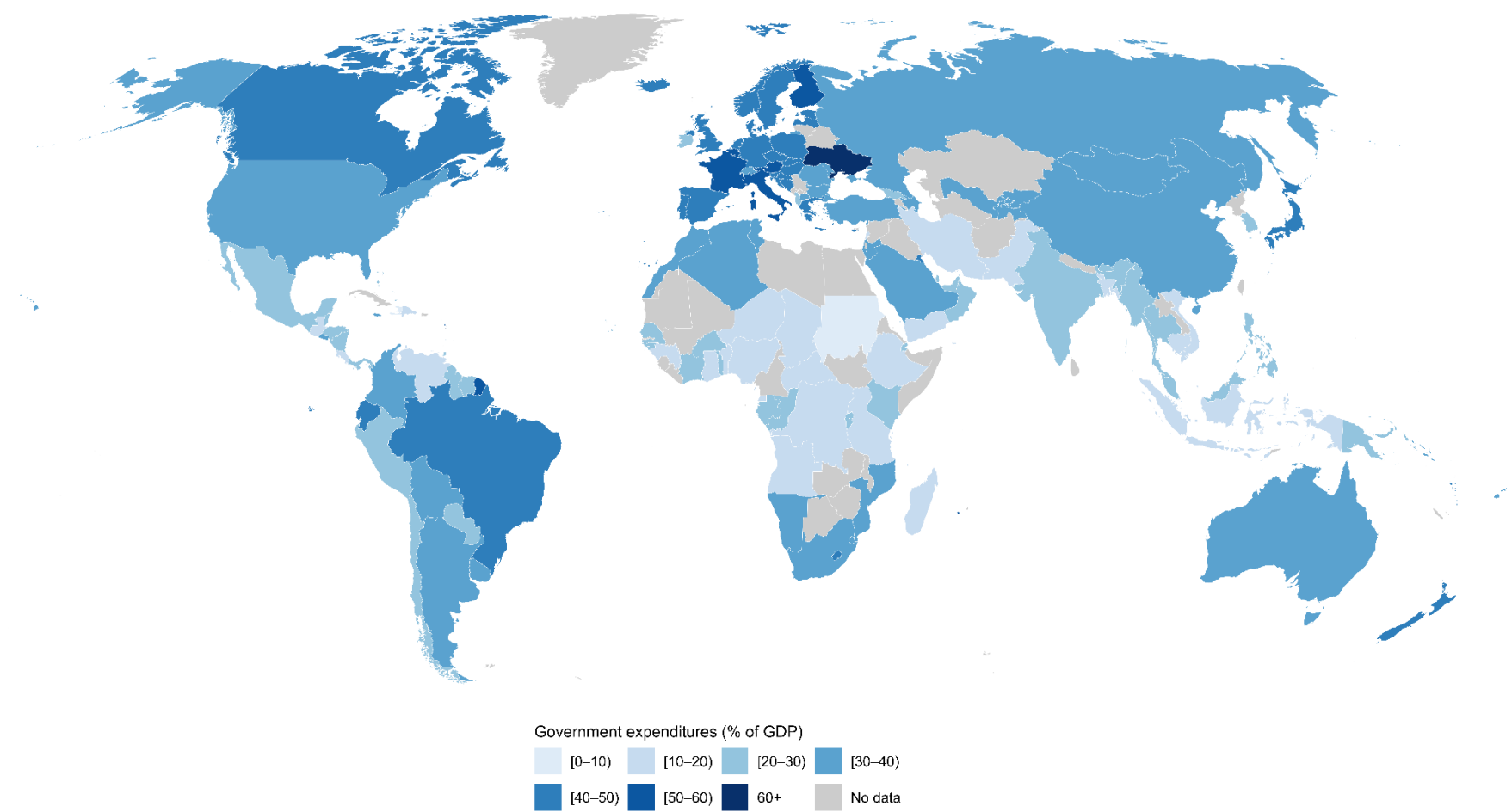
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Appendix

The Economics of Public Procurement

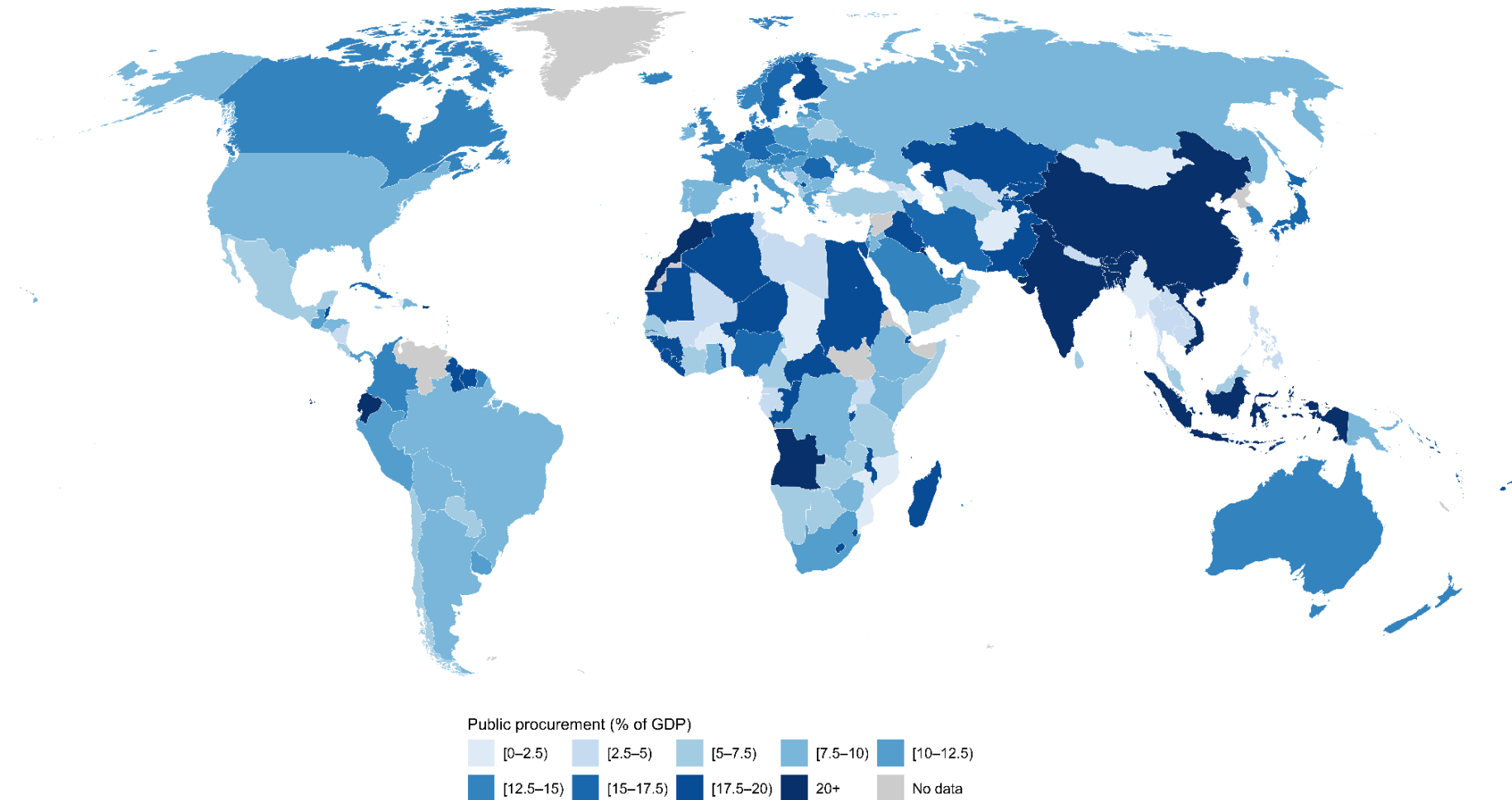
Francesco Decarolis and Stjepan Srhoj

Figure A1. Government expenditures (G) as % of GDP (Y)



Note: Map based on International Monetary Fund estimates of the government expenditures (G) as % of GDP in 2023 (Public Finances in Modern History Database, 2025).

Figure A2. Public Procurement as a % of GDP



Note: Map shows model-based estimates of public procurement spending (% of GDP) from Open Contracting Partnership and Spend Network, How Governments Spend (2020), combining published contracting data with official totals where available and regression imputations otherwise; estimates align with OECD figures. Accessed 1 Nov 2025.

Appendix A. Brief historical evolution of public procurement

Public procurement has deep historical roots. Since the earliest states began organizing the delivery of food, labour, and materials for collective needs, they combined in-house production with the acquisition of goods and services from external suppliers, including foreign states and merchants. Early precedents include the Code of Hammurabi (c. 1754 BCE), which imposed liability for negligent canal maintenance (Laws 53–54); the biblical account of Solomon’s contract with Hiram of Tyre, exchanging wheat and olive oil for timber and skilled labour to build the Temple (1 Kings 5:6–11); and palace records from the Bronze Age city of Ebla (c. 2400 BCE), which document organised deliveries of grain and olive oil as part of state-managed exchange systems.

In modern times, France pioneered one of the first unified national frameworks with the Law of 31 January 1833 and the Ordinances of 1836 and 1837 mandating competition, public advertisement, and contractual oversight (Truchet, 2012; Kalflèche, 2004). Building on such national experiments, the twentieth century saw the development of international frameworks aimed at harmonising and professionalising procurement across jurisdictions. The first binding multilateral rules were the Government Procurement Code adopted in the Tokyo Round of the General Agreement on Tariffs and Trade (GATT, 1979; in force 1981) and the Government Procurement Agreement under the World Trade Organization (WTO, 1994; in force 1996).

These principles were further elaborated in the United Nations Commission on International Trade Law (UNCITRAL) Model Law on Procurement of Goods and Construction (1993/1994), a non-binding legislative template widely used to promote fair and transparent procurement (Arrowsmith, 2004; Yukins and Nicholas, 2023), and in its later incarnation as the UNCITRAL Model Law on Public Procurement (1994, revised 2011), supported by a detailed Guide to Enactment. Today, global best practices are further shaped by instruments such as the Recommendation on Public Procurement of the Organisation for Economic Co-operation and Development (OECD, 2015), which sets out twelve integrated principles for efficient and accountable procurement systems, and the World Bank Procurement Regulations for Investment Project Financing Borrowers (2025), which govern all Bank-financed investment projects through enforceable rules emphasising open competition, integrity, and value for money.

The global normalisation of procurement principles such as competition, transparency, and value for money has also shaped how scholars approach the topic, marking a shift from a concern primarily rooted in other disciplines to an emerging field of economic inquiry.

Appendix B. Why the Economics of Public Procurement is a Distinct Subfield

B.1 Problem domain (Abbott criterion 1)

The economics of public procurement satisfies the three criteria that Abbott (1988, 2001, 2005) proposes for recognising a distinct subfield. First, it claims a specific problem domain: the design and consequences of public procurement systems, understood as multi-stage interactions among central government, public buyers, and suppliers that generate a wedge between planned and realised outcomes. Second, it has developed abstract analytical tools tailored to this domain. These span models of demand planning and the political economy of procurement, mechanism- and market-design approaches to tendering and bidding, contract-theoretic analyses of execution and renegotiation, and empirical methods that detect collusion and corruption, quantify implementation frictions, and estimate the causal effects of procurement rules and reforms on firms, markets, and aggregate outcomes. Third, it has built institutional structures that sustain cumulative research, including a rapidly growing body of theoretical and empirical work in leading journals, dedicated workshops and special issues, and increasingly rich national and cross-country datasets that anchor an active international research community.

The economics of public procurement claims a specific problem domain within economics: how the state, as a buyer, uses procurement institutions and market-based mechanisms to transform public spending into social and economic value. Within the field of public economics, it is closely related to fiscal policy, because procurement is an important channel through which public spending is translated into real activity, and the literature sometimes frames this role in terms of “the transmission of fiscal policy”, even as many recent studies explicitly describe themselves as contributing “*to the literature on the links between public procurement and economic performance*” (e.g. Bonfim et al., 2025, p. 1901). It is also closely related to industrial policy, because governments increasingly use procurement to target particular sectors, technologies, and regions. The economics of public procurement further draws on political economy, which studies how electoral incentives, lobbying, and bureaucratic agency shape public decisions, and on the economics of regulation, which analyses how the state constrains and steers market behaviour. On the micro side, it connects to industrial organization, including competition economics, mechanism design, and market design, since procurement markets are arenas in which firms compete for contracts under buyer-designed rules. These component areas are recognised as subfields of broader fields such as public economics, microeconomics, and industrial organization. The economics of public procurement is not merely their intersection, but an applied subfield organised around this specific problem domain, with its own organising questions about how procurement institutions and contract allocation shape economic outcomes.

B.2 Analytical toolkit (Abbott criterion 2)

Having established the specificity of the problem domain, we now turn to the second of Abbott’s criteria, the development of abstract analytical tools tailored to that domain. For economists, public procurement is an analytically specific setting in which contract allocation is shaped by contractual incompleteness (the impossibility of specifying and enforcing all contingencies *ex ante*), uncertainty on both buyer and bidder sides about costs, quality, and future states of the world, and strategic interactions that span the entire procurement cycle. These interactions begin before tenders are issued, through lobbying, suppliers’ efforts to secure future profits and growth, and politicians’ attempts to build political capital; in settings with weak institutions, this can also involve illicit personal enrichment by public officials. They continue through public buyers’

choices over the intensity and form of supplier competition (such as entry conditions, lot structure, and the use of framework agreements), the weighting of multiple objectives in award criteria (for example, price, quality, and social or environmental goals), and firms' bidding and collusive decisions, including whether to participate at all, how aggressively to bid, whether to form, join, or resist cartels, and how to price given expectations about monitoring, enforcement, and the scope for renegotiation. These interactions also shape downstream firm outcomes such as investment, hiring, learning, innovation, survival, and growth. This combination of contractual incompleteness, uncertainty, and multi-stage strategic interaction has required extending auction and contract theory (Laffont and Tirole, 1993; Bajari and Tadelis, 2001), developing new models of renegotiation (Guasch, Laffont and Straub, 2008), and creating empirical methods for detecting cartels (Conley and Decarolis, 2016; Kawai and Nakabayashi, 2022; Kawai et al., 2023) and corruption (Bandiera, Prat and Valletti, 2009; Olken and Pande, 2012; Colonnelli and Prem, 2022; Burguet, Ganuza and Montalvo, 2016; Decarolis and Giorgiantonio, 2022; Decarolis et al., 2025), as well as strategies for identifying the causal effects of procurement reforms (Athey, Levin and Seira, 2011; Lewis and Bajari, 2011; Decarolis, 2014). A promising body of robust evidence is emerging on the determinants of procurement performance (Decarolis et al., 2020; Best et al., 2023) and on the heterogeneous consequences of contracts and bundles of contracts for firms and markets (Ferraz et al., 2015; Di Giovanni et al., 2022; Ilzetzki, 2024; Gabriel, 2024). Together, these contributions constitute an analytical toolkit that is recognisably tailored to public procurement.

B.3 Institutional structure and data (Abbott criterion 2)

Turning to Abbott's third criterion, the economics of public procurement has developed an emerging institutional structure that sustains cumulative research. Leading general-interest economics journals now regularly publish studies that advance the economics of public procurement, and several journals have dedicated special issues.¹ There is a marked expansion in economics research on public procurement, with the steepest growth occurring in the last five years (see Figure A3). Launched in 2018, the Economics of Public Procurement Workshop has run for eight years, fostering a cohesive international research community. At the same time, the field is attracting increasing attention from scholars in adjacent areas such as innovation and industrial policy (for example, Mazzucato et al., 2025), which further broadens its scope and relevance.² These intellectual and institutional advances have been reinforced by major improvements in data infrastructure. On the one hand, rich within-country micro-level datasets now link procurement records to other administrative sources such as firm accounts, employment, ownership, and even criminal investigation data (Decarolis et al., 2020; Kawai et al., 2023; Best et al., 2023). On the other hand, new cross-country resources have emerged, including multi-country indicators on procurement law and practice (Bosio et al., 2022), harmonised tender records from sources such as Tenders Electronic Daily, and newly constructed cross-country micro-data (Fazekas et al., 2024). As governments expand digital procurement systems

¹ List of studies published in first tier general economics journals can be observed in the reference list. Example of special issue is in the *European Journal of Political Economy*: <https://www.sciencedirect.com/special-issue/322895/determinants-and-effects-of-sustainable-public-procurement>.

² Workshop programme, keynotes, list of presenters and host universities are available: <https://drive.google.com/file/d/1xIxu15HL99kxEw3FIQPUWvKnjss1IAXc/view?usp=sharing> [accessed 28th August 2025]. Other examples include the sixth interdisciplinary symposium on public procurement, link: <https://publicprocurementsymposium.it/2025> [accessed 26th September 2025]. In addition, a multidisciplinary journal, the *Journal of Public Procurement*, provides a consistent outlet for research with a policy and management orientation.

and administrative data pipelines, the availability and quality of data is likely to continue rising, enabling new empirical and institutional work. Taken together, these theoretical, empirical, institutional, and data developments satisfy Abbott's third condition and place the economics of public procurement on a trajectory similar to other now-established applied subfields. Building on these developments, this survey advances the subfield by offering a conceptual framework and synthesizing recent theoretical and empirical contributions.

Appendix C. Why This Survey Now? What Existing Reviews Cover and What is Missing

Existing reviews examine important aspects of public procurement but each covers only part of the field. Rejeb et al. (2024) provide a bibliometric map showing rapid growth across themes such as corruption, small-firm participation, innovation, digitalization, and sustainability. Croom and Brandon-Jones (2005) synthesize early evidence on e-public procurement, highlighting anticipated benefits including lower transaction costs, improved transparency, and administrative efficiencies. Fazekas and Blum (2021) review public procurement reforms and find that centralized purchasing and framework agreements deliver by far the largest savings for public buyers, often above 50%, while most other interventions, even when well implemented, reduce costs by only about 5–10%. They also conclude that the evidence base on the effects of procurement reforms, including widely adopted measures such as centralization and e-procurement, is still mostly low to medium in quality and largely non-causal. These reviews support our core claim that the economics of public procurement is still an emerging subfield. If rigorous evidence is limited even for foundational reforms focused on cost efficiency, it is unsurprising that evidence is even thinner in other domains where public procurement is used to pursue non-price objectives.

For example, innovation-focused surveys argue that public procurement can create demand, shape technological trajectories, and support strategic policy objectives (often framed as mission-oriented) (Edler and Georghiou, 2007; Mazzucato, 2011; Mazzucato et al., 2025; Bangert et al. 2025), while Chiappinelli et al. (2025) review a range of design levers in public procurement of innovation and conclude that no single configuration dominates across contexts, with innovation effects hinging on choices such as how procurement phases are bundled, how much discretion buyers have, and how competitive and centralised procedures are. On the environmental side, studies of green public procurement find mixed implementation and performance and still offer only partial guidance on how procurement design and local institutions shape environmental outcomes (Cheng et al., 2018; Ortega Carrasco et al., 2025). Related macroeconomic reviews study fiscal multipliers at national and regional levels (Ramey, 2011a, 2016, 2019; Gechert, 2015; Chodorow-Reich, 2019), shedding light on government spending broadly but not on public procurement institutions or micro-mechanisms. Taken together, these contributions resemble individual pieces of a broader puzzle: each advances a thematic agenda, but important pieces remain missing and no survey yet assembles these pieces into a unified economic framework. This survey fills that gap by organizing the field around the public procurement cycle, integrating theoretical and empirical evidence, and identifying research priorities for the next phase of work.

Appendix D. Initial literature search

To gauge both the size of the broader “procurement and economics” literature and the more focused body of work reviewed in this paper, we ran two Web of Science Core Collection searches. The first search is deliberately broad, combining “public procurement” with “economic*” to provide an overall bibliometric snapshot of publication and citation patterns (Figure A3). The second search implements a more detailed search protocol (Table A1): it uses a wider set of procurement-related keywords, restricts results to journal articles, and filters the output to a curated set of leading journals to construct the candidate corpus for screening and inclusion (Figure A4).

Table A1. Web of Science search code

Topic (Query): public procurement OR State procurement OR municipal procurement OR public sector procurement OR government purchase* OR public purchas*
Document Type (Query): Article
Publication titles (Query): American economic review OR quarterly journal of economics OR Econometrica OR Journal of Political Economy OR Review of Economic Studies OR Annals of Statistics OR American Economic Journal: Applied Economics OR American Economic Journal: Macroeconomics OR Biometrika OR Econometric Theory OR Economic Journal OR International Economic Review OR Journal of Business and Economic Statistics OR Journal of Econometrics OR Journal of Economic Literature OR Journal of Economic Perspectives OR Journal of Economic Theory OR Journal of International Economics OR Journal of Labor Economics OR Journal of Monetary Economics OR Journal of the American Statistical Association OR Journal of the European Economic Association OR Journal of the Royal Statistical Society. Series B: Statistical Methodology OR Quantitative Economics OR RAND Journal of Economics OR Review of Economics and Statistics OR Theoretical Economics OR American Economic Journal: Economic Policy OR American Economic Journal: Microeconomics OR American Economic Review: Insights OR American Journal of Agricultural Economics OR Annual Review of Economics OR Brookings Papers on Economic Activity OR Cambridge Journal of Economics OR Canadian Journal of Economics OR Computational Statistics and Data Analysis OR Ecological Economics OR Econometric Reviews OR Econometrics Journal OR Economic Development and Cultural Change OR Economic Inquiry OR Economic Policy OR Economic Theory OR Economica OR Economics Letters OR Energy Economics OR Journal of Development Economics OR Journal of Economic Behavior and Organization OR Journal of Economic Dynamics and Control OR Journal of Economic Growth OR Journal of Environmental Economics and Management OR Journal of Health Economics OR Journal of Human Capital OR Journal of Human Resources OR Journal of Industrial Economics OR Journal of Institutional Economics OR Journal of Law and Economics OR Journal of Law, Economics, and Organization OR Journal of Legal Studies OR Journal of Mathematical Economics OR Journal of Multivariate Analysis OR Journal of Population Economics OR Journal of Public Economics OR Journal of Risk and Uncertainty OR Journal of the Association of Environmental and Resource Economists OR Journal of the Royal Statistical Society. Series A: Statistics in Society OR Journal of the Royal Statistical Society. Series C: Applied Statistics OR Journal of Time Series Analysis OR Journal of Urban Economics OR Labour Economics OR Land Economics OR Oxford Bulletin of Economics and Statistics OR Oxford Economic Papers OR Public Choice OR Real Estate Economics OR Review of Economic Dynamics OR Review of Income and Wealth OR Scandinavian Journal of Economics OR Scandinavian Journal of Statistics OR Social Choice and Welfare OR World Bank Economic Review OR Academy of Management Journal OR Academy of Management Annals OR Academy of Management Review OR Administrative Science Quarterly OR Journal of Management OR Academy of Management Perspectives OR British Journal of Management OR Business Ethics Quarterly OR Journal of Management Studies OR Journal of Finance OR Journal of Financial Economics OR Review of Financial Studies OR Journal of Corporate Finance OR Journal of Financial and Quantitative Analysis OR Journal of Financial Intermediation OR Journal of Money, Credit and Banking OR Review of Finance OR Research Policy OR Journal of Product Innovation Management OR Journal of Operations Management OR International Journal of Operations and Production Management OR Journal of Supply Chain Management OR Production and Operations Management OR Management Science OR Operations Research OR European Journal of Operational Research OR IEEE Transactions on Evolutionary Computation OR Mathematical Programming OR Organization Science OR Human

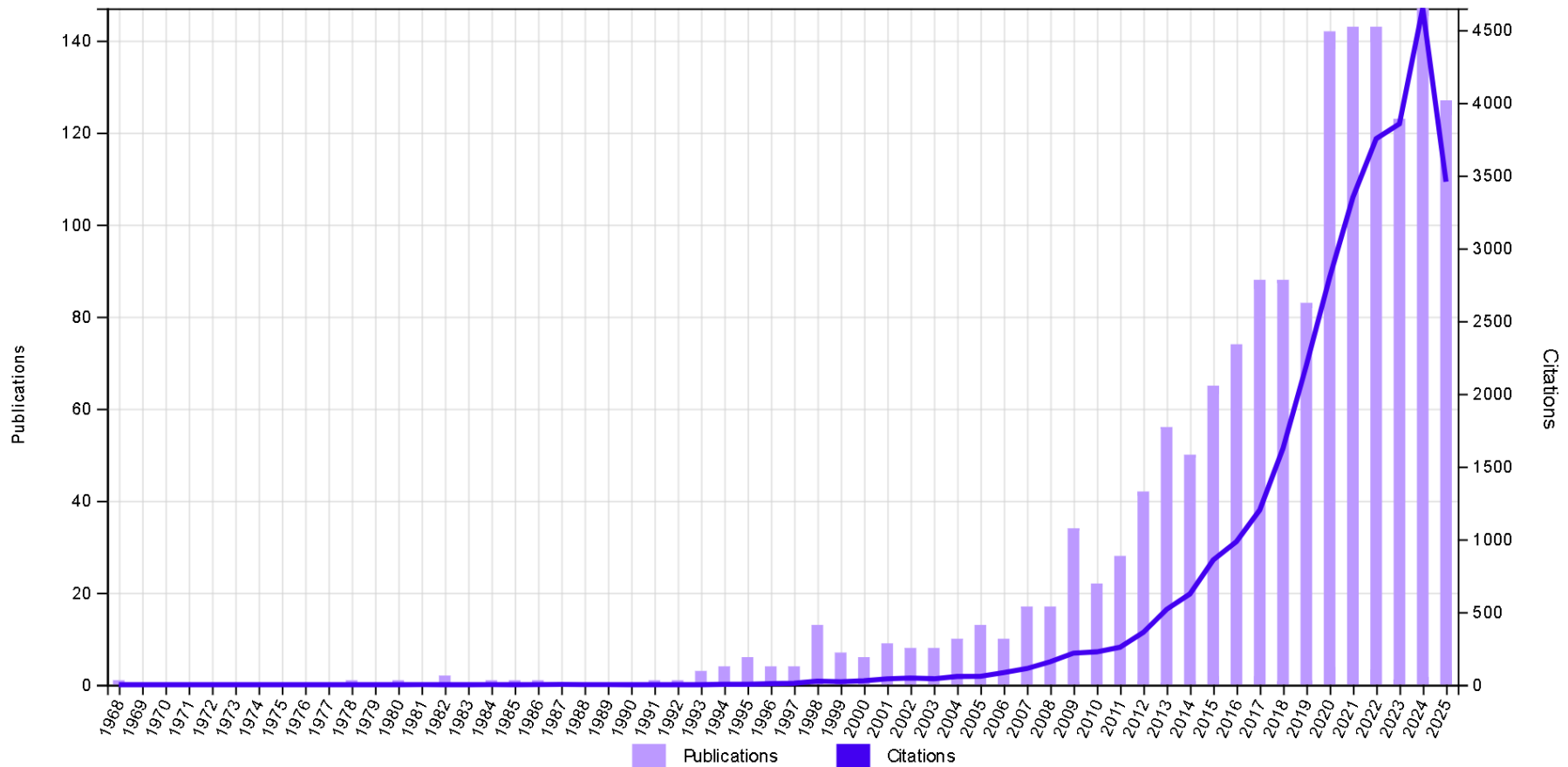
Relations OR Leadership Quarterly OR Organization Studies OR Organizational Research Methods OR Public Administration Review OR Governance OR Journal of Public Administration Research and Theory OR Public Administration OR Public Management Review OR Economic Geography OR Journal of Economic Geography OR Regional Studies OR Strategic Management Journal OR Global Strategy Journal OR Entrepreneurship Theory and Practice OR Journal of Business Venturing OR Strategic Entrepreneurship Journal OR Small Business Economics

Table A1 gives the Web of Science search code used to obtain initial list of studies for potential inclusion in the review. There are three parts of the search code. Firstly, after initial literature search and reading seminal studies, the wording of the topic is different, while many studies use the wording “public procurement”, there are important studies that do not use public procurement in the title or abstract, but instead use the terms “state procurement”, “regional procurement”, “municipal procurement”, “public sector procurement”, “government purchas*” and “public purchas*”. I therefore include all these six terms. Secondly, we are interested in published articles and not in other types of documents such as conference proceedings. Thirdly, as there are too many studies published (27,972 in WoS indexed journals), and studies are of heterogeneous quality, which is why we focus only on studies of higher quality. To do so, we use Academic Journal Guide and select journals in the field of economics that have 4*, 4- and 3-star ranking. For example, the 4* includes top 5 (Quarterly Journal of Economics, American Economic Review, Econometrica, Review of Economic Studies, Journal of Political Economy), 4 star journals includes journals such as Review of Economics and Statistics or Economic Journal, and 3 star journals includes journals such as European Economic Review, Journal of Public Economics, or Journal of Law, Economics and Organization. In addition, we include 4* and 4 star journal in related field, such as Finance, Management, Entrepreneurship, etc. where there are also studies published on public procurement.

To benchmark the profession’s attention to public procurement, we use the “Economics Literature Search” tool developed by Goldsmith-Pinkham (2024). We query the term “public procurement” in the Quarterly Journal of Economics, the American Economic Review, the Journal of Political Economy and the NBER Working Papers, and compute its annual frequency as a share of all items published in each venue (reporting both raw and length-adjusted measures). At present, Econometrica and the Review of Economic Studies are not available in the tool’s coverage and are therefore not included.

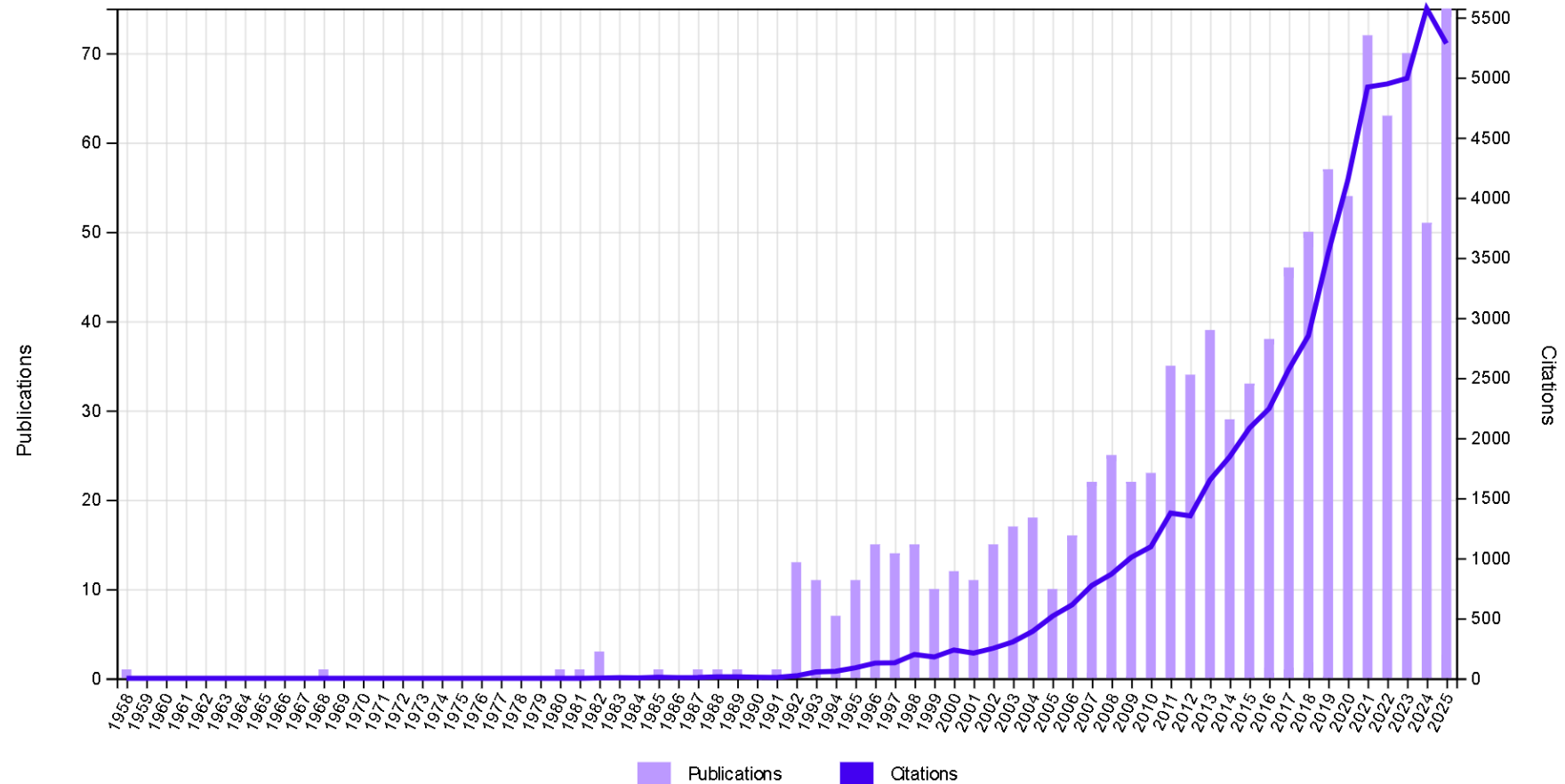
The resulting time series shows that “public procurement” was a very small share of output in the early 2000s, with a few temporary spikes (e.g., around 2003 and 2008–2012). From the mid-2010s onward, the share increases gradually, then jumps sharply after 2020, reaching roughly 1%+ by 2021–2023 and accelerating further in 2024–2025 (peaking around 2.3% raw and 1.9% length-adjusted in 2025). The length-adjusted line closely tracks the raw series, indicating the rise reflects more papers on the topic, not just longer papers that mention it. Taken together, this pattern suggests a clear take-off in top-economics attention to public procurement after 2020, consistent with the consolidation of an emerging subfield rather than isolated episodic interest (Figure A5).

Figure A3. Web of Science search results for “public procurement” and “economic” (SSCI and ESCI databases)



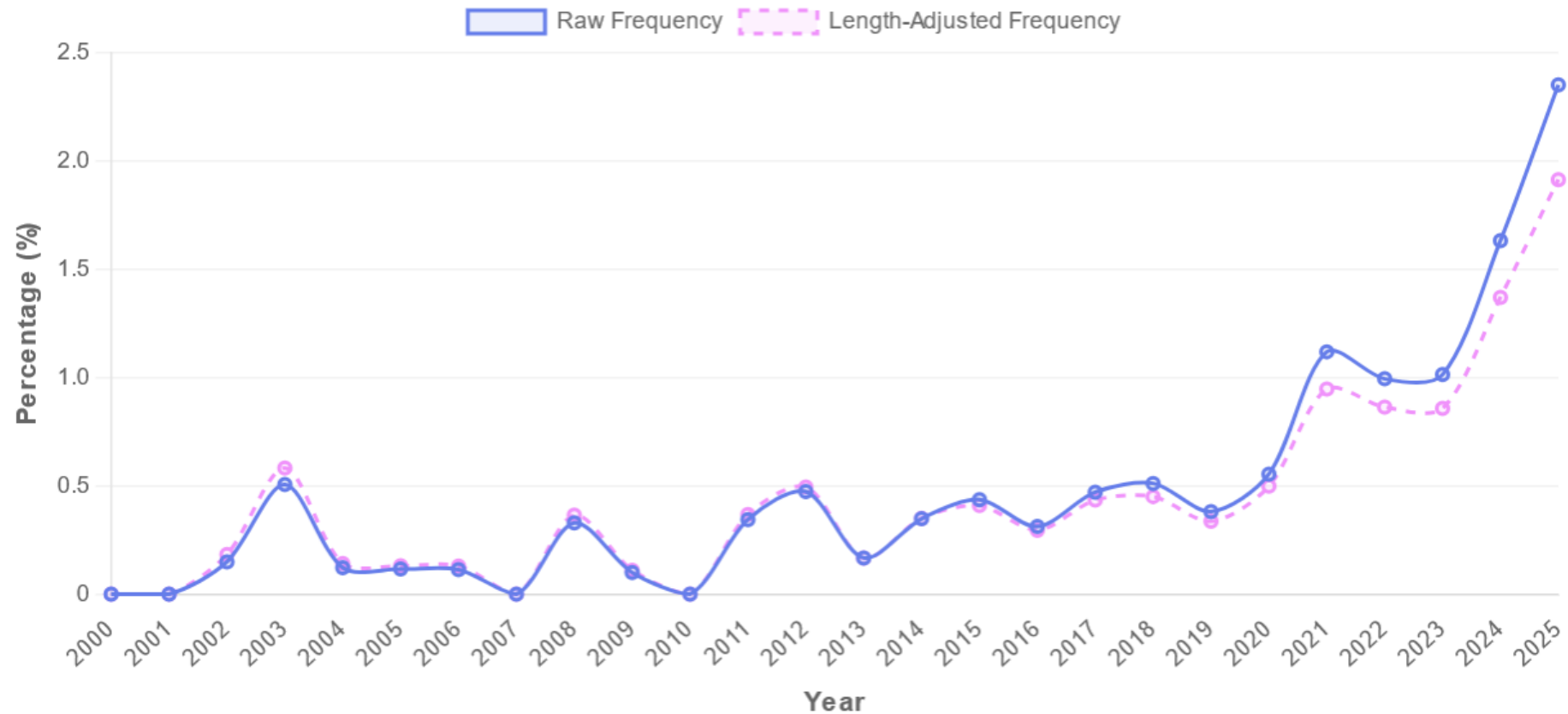
Note: Figure A3 presents bibliometric evidence on the academic literature connecting public procurement and economics. The search, conducted on 27 September 2025, identified 1,605 articles indexed in the Web of Science Social Sciences Citation Index (SSCI) and Emerging Sources Citation Index (ESCI) databases containing the terms “*public procurement*” and “*economic*” in any field. These articles have been cited 31,642 times by 24,254 other papers, resulting in an average of 19.7 citations per article and an H-index of 79.

Figure A4. Refined Web of Science search results (SSCI and ESCI databases)



Note: Bibliometric snapshot based on a Web of Science Core Collection (Clarivate) search for the topic/query used in this study (timespan 1955–2025; retrieved 1 Dec 2025): 1,045 records; 56,615 total citations (55,191 excluding self-citations); 46,533 citing articles (46,057 excluding self-citations); 54.18 citations per item on average; h-index = 120. In Web of Science, “without self-citations” excludes citations originating from items within the analyzed results set, and all counts are contingent on the specific query, index coverage, and timespan settings.

Figure A5. Frequency of ‘Public Procurement’ in QJE, AER, JPE and NBER (Share of Publications)



Note: To benchmark interest in public procurement in the broader economics profession, we also use the “Economics Literature Search” tool developed by Goldsmith-Pinkham (2024). We query the term “public procurement” in “All fields” of the Quarterly Journal of Economics, the American Economic Review, the Journal of Political Economy and the NBER Working Papers, and compute its annual frequency as a share of all items published in each venue (reporting both raw and length-adjusted measures). At present, *Econometrica* and the *Review of Economic Studies* are not available in the tool’s coverage, and are therefore not included. Source: Goldsmith-Pinkham, P. (2024). Tracking the Credibility Revolution across Fields. arXiv:2405.20604.

Appendix E. Conceptual Framework and Fiscal Transmission: Additional Details

How do public procurement institutions shape the way government expenditures are translated into economic activity and longer-run patterns of productivity and structural change? We begin from the standard open-economy national income identity,

$$Y = C + I + G + NX \quad (1)$$

Where aggregate output Y equals private consumption C , private investment I , government expenditure G , and net exports NX (exports minus imports). Public procurement is a component of G capturing the part of used to purchase goods, services and works from external suppliers. In macroeconomic theory, the role of G in determining output has attracted sustained intellectual attention. Early work emphasized government expenditures as a tool for fiscal stimulus and countercyclical demand management (Keynes 1936), a view later formalized and refined in modern New Keynesian models (Mankiw and Romer 1991; Rotemberg and Woodford, 1997; Woodford, 2003; Christiano et al., 2005; Galí, 2015). In this tradition, government purchases affect output directly by increasing G in aggregate demand and indirectly by altering private consumption and investment decisions, as well as by shaping price dynamics through inflation, real interest rates, and markups (see, e.g., Woodford 2003). Over the past century, the scale of G has risen substantially in many economies, and in recent years it has reached high levels: for example, G accounted for approximately 57% of GDP in France, 45% in Brazil, 41% in Japan, 37% in Australia and Russia, and 36% in the United States (see Figure A1).

The macroeconomic literature provides the natural starting point for understanding how procurement translates planned spending into realized demand. Although this work typically treats government purchases as an aggregate category, it implicitly embeds the procurement process within the measurement of G . In the fiscal multiplier literature, the object of interest is typically the effect of G on output Y , summarized by a spending multiplier: how many dollars Y rises after a \$1 increase in G . In standard macro (national-accounts) data, G is measured as government purchases (government consumption plus government investment) and therefore excludes transfers.³

Aggregate evidence places public-purchase multipliers in a central range around 0.7–1.5, while meta-regression results suggest a baseline (“reference”) purchases multiplier of roughly 0.7–1.0 (about 0.73) and emphasize substantial heterogeneity across estimates (Ramey, 2011b, 2016, 2019; Gechert, 2015). This heterogeneity is partly contextual: evidence on state dependence is mixed, with some specifications delivering larger multipliers in slack periods or when monetary policy is constrained, but with results that are not uniformly robust (Ramey and Zubairy, 2018). It is also partly structural, consistent with leakage in open economies, as multipliers tend to be lower in countries with higher import-to-GDP ratios (Gechert, 2015). A further, central dimension is composition across fiscal instruments. Synthesizing 1,069 multiplier estimates from 104 studies, Gechert (2015) reports that relative to the reference purchases multiplier, government investment multipliers are higher (about 0.6, mean around 1.36), government consumption multipliers are lower (mean around 0.80), and tax cuts and transfers are smaller by roughly 0.3–0.4 units. Subsequent evidence, however, suggests these rankings are not universal: Boehm

³ Much of the multiplier literature therefore asks whether the composition of government purchases G matters for $\Delta Y/\Delta G$, building on early structural VAR approaches (e.g., Blanchard and Perotti, 2002) and subsequent work on state dependence (Auerbach and Gorodnichenko, 2012, 2013). A central methodological refinement is Ramey’s emphasis on fiscal foresight: because policy is often anticipated before outlays move, timing-based VAR shocks can be contaminated by expectation effects. Ramey (2011a) addresses this by constructing defense “news” shocks from narrative sources to date revisions in expected future defense spending, improving identification of exogenous purchase shocks and motivating later work that isolates unanticipated policy variation.

(2020) finds a consumption multiplier around 0.8 but an investment multiplier close to zero for temporary investment surprises, while Clemens, Michelsen, and Rieth (2025) estimate investment multipliers around 2 for Germany using a narrative instrument with shocks dominated by additional public construction. Finally, composition interacts with the time horizon. Antolín-Díaz and Surico (2025) show that while the short-run effect is around one, the long-run multiplier can rise above one as spending responses tilt toward more persistent components, especially public R&D, whereas government investment is comparatively shorter-lived and government consumption is modest at most horizons.

In national accounts, the closest aggregate proxy for public procurement sits inside government purchases G , but the standard breakdown is not procurement-clean. Government “consumption” mixes contract-based purchases with the public wage bill, while government “investment” captures capital formation and therefore misses much recurring contracting in operations, maintenance, and other current purchases. As a result, aggregate G blurs procurement spending with payroll and maintenance, motivating empirical designs that measure procurement directly using contract-level data on awards and (when available) the location of production and delivery.⁴ Because procurement spending is implemented through contracts performed in particular places, procurement shocks are naturally geographically concentrated. This makes regional and cross-sectional designs a useful complement to aggregate national-accounts approaches, since they map exposure to contract spending to local outcomes more directly, though identification still must contend with spillovers and general-equilibrium effects. In the canonical U.S. defense setting, Nakamura and Steinsson (2014) exploit cross-state differences in exposure to national defense buildups and estimate a relative multiplier near 1.5 (a 1 pp increase in military spending as a share of state GDP raises state GDP by about 1.5 pp).

Cross-state dispersion largely reflects exposure: buildups deliver much larger shocks to states with a bigger pre-existing defense footprint (defense contractors and defense-oriented manufacturing), and prime-contract spending closely tracks in-state production (using “military shipments” data), alleviating concerns that contracts are merely booked locally but executed elsewhere. Results are similar using procurement contracts alone versus total defense outlays, implying contracts capture the main local impulse even when personnel pay co-moves. Using city-level DoD spending, Auerbach, Gorodnichenko, and Murphy (2024) estimate about \$1 of GDP per \$1 spent in the recipient city and roughly \$0.50 in same-state spillovers. Exploiting census-induced, windfall-financed reallocations that raise German municipal purchases by roughly 5 pp. of base-year purchases on a sustained basis, Hager and Huber (2025) document a “*dynamism drain*”: procurement demand boosts employment in slack municipalities in the short run, but is followed by weaker medium-run employment and output-per-worker growth in boom areas and in industries that sell directly to government.

Regional multiplier reviews emphasize that such effects are often large and persistent (Chodorow-Reich, 2019).⁵ A related question is whether such regional effects are mechanically

⁴ National-accounts categories bundle procurement and non-procurement items. Government final consumption expenditure comprises both intermediate consumption (purchases of goods and services used as inputs in government production) and compensation of employees, so movements in “consumption” cannot be interpreted as procurement-only. Gross fixed capital formation (government investment) is typically more procurement-intensive, but it is conceptually tied to new capital formation and major upgrades, and thus excludes much routine procurement embedded in current operations, including maintenance and many service contracts. For context, FY1985 U.S. DoD outlays were split roughly into procurement (about 31%), military personnel (27%), operations and maintenance (30%), and RDT&E (12%) (DoD Comptroller, 1984 ‘Green Book’, Table 1-9; shares computed).

⁵ In many regional studies that exploit defense build-ups, the shock is constructed from defense public procurement contracts allocated geographically, sometimes augmented with military personnel compensation; authors report similar results across the two measures (Davis, Loungani and Mahidhara, 1997; Dupor and Guerrero, 2017; Hooker

larger when spending is financed “*from outside*” the region (federal or foreign aid) rather than locally. In a New Keynesian model of regions in a monetary union, Pennings (2022) finds that at business-cycle frequencies, outside-financed purchase multipliers are only modestly larger than locally deficit-financed multipliers (around 0.07–0.16 higher), suggesting that the deployment of spending and local economic structure are likely to matter more than financing source for short-run regional purchase multipliers.

Our contribution in this section is not to revisit the aggregate fiscal multiplier literature, but to clarify how the institutional mechanics of public procurement shape the transmission of G to the real economy. To formalize the distinction between procurement and other public outlays, let total G consist of public procurement expenditure G^{PP} and all other public outlays G^0 such as public employee wages, transfers, and other public costs. Define $s \in [0, 1]$ as the public procurement share of G , so:

$$G^{PP} = sG \quad , \quad G^0 = (1 - s)G. \quad (2)$$

This decomposition allows us to study how the composition of G , not only its level, shapes aggregate outcomes. Public procurement is quantitatively important. As a share of GDP, public procurement is 14.5% in France, 16.2% in Japan, 13.2% in Australia, 9.3% in Russia and 9.4% in United States (see Figure A2).⁶ In the median country, the parameter s (the share of G spent through public procurement) is about 30%, and typically ranges from 20% to 40%; for example, it is 25.6% in the United States, 25.1% in Russia, and 32.2% in France.

To move from accounting identities to transmission, we express output Y as a function of public procurement and non-procurement spending. Let μ_{PP} denote the output multiplier on public procurement expenditure, and let μ_{NP} denote the multiplier on non-procurement public outlays, both defined as marginal effects on output holding other components constant:

$$\mu_{PP} = \frac{\partial Y}{\partial G^{PP}} \quad , \quad \mu_{NP} = \frac{\partial Y}{\partial G^0} \quad (3)$$

Equation (3) defines μ_{PP} as the public procurement multiplier associated with the chosen portfolio of projects. We interpret μ_{PP} as an expected potential public procurement multiplier: potential because it captures the output and productivity effects this portfolio would generate if implemented as planned, and expected because it is chosen *ex ante* under uncertainty about how successful the projects will be. For example, a government may commit €200 million to procure autonomous vehicles; *ex ante*, μ_{PP} reflects the anticipated gains from safer transport, learning, and spillovers, even though the realised effects may turn out higher or lower depending on technological performance and adoption. Projects are not selected solely to maximise short-run output: governments also procure for national security, environmental and social goals, and redistribution. These choices over what to buy, where, and for whom can raise or lower μ_{PP} relative to a purely output-maximising benchmark.

and Knetter, 1997; Nakamura and Steinsson, 2014). To situate these shocks within the public budget, public procurement typically accounts for about 25–30% of total government expenditure and roughly 10–17% of GDP across advanced economies, whereas other large components of G include public-sector wages (about 14–22% of G) and social protection transfers (around 40+% of G in the European Union and nearly 50% in the United States), alongside interest payments, subsidies, and administrative operating costs. This internal heterogeneity underscores why the economic effects of G depend on composition, not only on its aggregate level.

⁶ We provide comparison for a few countries where international statistics seemed credible. Discrepancies include China and India where G was estimated at 33% (China) and 29% (India) of Y , while $s \cdot G$ is estimated at 35% (China) and 20% (India) of Y . In both cases, public procurement seemed too high. We believe more credible estimates will be collected as the subfield consolidates.

E.1 The heterogeneous multiplier: z and ς

Equation (3) summarises the transmission of public procurement using a single multiplier μ_{PP} . This scalar representation is useful as a starting point, but it masks systematic differences in how procurement affects output across sectors and across macroeconomic conditions. To capture this heterogeneity, we allow the procurement multiplier to vary with the sector z in which demand is injected and the state of the economy ς . We define the sector- and slack-specific public procurement multiplier as

$$\mu_{PP}(z, \varsigma) = \frac{\partial Y}{\partial G_z^{PP}}, \quad (4)$$

which captures the output response to public procurement spending directed to sector z under macroeconomic conditions indexed by ς . The fiscal-multiplier literature is informative for public procurement because it typically studies government purchases, defined in national accounts as government consumption expenditures and gross investment, including both the wages and salaries of public employees and the acquisition of goods, services, and public works from firms, while excluding transfers and interest (Ramey, 2011a).⁷ Most empirical multiplier studies use this purchases concept rather than total government expenditure G (Ramey and Zubairy, 2018) and show that the effect of government purchases on output depends on macroeconomic slack ς . Fazzari et al. (2015) find substantially larger multipliers when capacity utilisation is low (peak values around 1.6), whereas Ramey and Zubairy (2018), using long-run U.S. data and local projections, estimate more stable multipliers in the range 0.6–1 with limited state dependence outside periods when monetary policy is constrained; Ramey’s (2011a, 2011b, 2016, 2019) surveys conclude that short-run multipliers for temporary, deficit-financed government purchases most plausibly lie between about 0.8 and 1.5, with larger values mainly when monetary policy is highly accommodative.⁸ From our perspective, the contract-based component of government purchases is closest to public procurement, so this evidence suggests that the procurement multiplier should vary with business-cycle conditions.

At the same time, spending shocks do not propagate uniformly across the economy, making the sectoral dimension z a first-order determinant of the multiplier: government demand is highly concentrated in a small set of industries and sector-specific shocks generate heterogeneous responses in output, hours, wages, and productivity (Ramey and Shapiro, 1998; Nekarda and Ramey, 2011); federal procurement is granular, with a smaller number of suppliers and sectors accounting for large share of transactions (Cox et al., 2024); multipliers vary with the composition of purchases because industries differ in price stickiness, adjustment costs, and supply elasticities (Boehm, 2020); and sectoral value-added multipliers are strongly increasing in industries’ network centrality, with input–output linkages and price-rigidity heterogeneity accounting for a large share of aggregate amplification (Bouakez et al., 2025). These patterns are consistent with the broader production-network literature, which shows that shocks to upstream or high-centrality sectors propagate strongly through input–output linkages (Acemoglu et al., 2012) and

⁷ National accounts do not separately identify “procurement” as a distinct spending category; instead, the contract-based acquisition of goods, services, and public works from firms is embedded within government consumption and investment alongside the wages and salaries of public employees. We reserve the term public procurement for this contract-based component, which is nevertheless tightly linked to the government purchases series used in the multiplier literature

⁸ For illustration, a short-run multiplier of 0.8 means that a temporary increase of 1 euro in government purchases raises aggregate output by 0.80 euro, whereas a multiplier of 1.5 implies that the same 1 euro increase raises output by 1.50 euros. Multipliers below (above) one therefore indicate that the output response is less (more) than one-for-one with the initial fiscal impulse

that network structure amplifies heterogeneity in the magnitude and persistence of fiscal responses (Barattieri et al., 2023). Taken together, this literature implies $\mu_{PP} = \mu_{PP}(z, \varsigma)$ rather than as a single constant.

E.2 The efficiency wedge: ω

At the same time, not all planned procurement spending is effectively translated into realized economic activity. Let $\phi \in [0, 1]$ denote implementation effectiveness, capturing the share of planned public procurement that materializes as delivered output. When $\phi = 1$, the procurement program fully translates authorized spending into the intended quantities and timing of delivery, conditional on the chosen projects. When $\phi < 1$, part of the intended economic impulse is lost because institutions or markets prevent authorized spending from becoming delivered value. We summarize these implementation losses by the procurement loss wedge

$$\omega = 1 - \phi \quad (5)$$

Taken together with the definition of μ_{PP} (equations 3 and 4), the equation (5) implies that μ_{PP} is an expected potential multiplier and that the realised multiplier is $\phi\mu_{PP} = (1 - \omega)\mu_{PP}$ where ϕ is implementation effectiveness and ω the procurement loss wedge. A higher ω corresponds to a larger gap between planned procurement spending and delivered output. The wedge ω captures implementation frictions – such as delays, leakage, cost overruns, and quality shortfalls – rather than the intrinsic productivity of the chosen projects.

These frictions are not merely theoretical. Public procurement relies on contractual relationships that require planning, tendering, and execution, which introduce meaningful time lags between fiscal decisions and realized output (Cox et al. 2024; Ramey 2011a). Consider a typical federal procurement project in the United States, such as a road maintenance or construction contract financed through the annual federal budget. After Congress approves budget authority, the responsible agency must complete internal planning and project design before soliciting bids. This process, moving from appropriation to an active tender, typically takes 6 to 12 months, depending on project complexity and administrative capacity. Once bids are evaluated and a winner is selected, the agency signs a contract and obligates funds, creating a legally binding commitment to pay. As shown by Cox et al. (2024) on the sample of federal procurement, only a portion of these obligated funds are paid quickly: about 20% of obligations are disbursed within the first quarter, roughly 50% within the first year, and the remainder over the following two to three years as work is completed and verified. The lag from budget approval to final outlays can therefore exceed one and a half to three years, underscoring the slow transmission of fiscal stimulus through the procurement channel. This limits procurement's effectiveness as a rapid countercyclical instrument but underscores its importance as a tool for achieving the government's longer-term strategic and developmental objectives.

To illustrate that implementation frictions can be substantial in practice, we use World Bank data from Bosio et al. (2022), based on a global survey of procurement lawyers, construction and engineering firms, and procuring entities about a standardized US\$2.5 million open-tender road resurfacing contract implemented by a central government agency. The benchmark contract is designed to represent a routine public road works tender that is comparable across countries and covers key stages of the procurement cycle, including entry and competition, award criteria, contract execution, payment, and corruption risks. These data suggest that the procurement loss wedge ω is sizeable in many settings, as delays, cost overruns, low-quality execution and limited effective competition appear to be pervasive rather than exceptional features of public

contracting.⁹ Understanding why ω is higher in some countries than in others is therefore a first-order question in the economics of public procurement: existing research highlights bureaucratic capacity as one determinant (Bosio et al., 2022), but cross-country differences in market-design choices (discretion versus price-only or mixed price-quality auctions), transparency (whether tender documentation is publicly available), and integrity conditions (abuse of technical specifications, corruption, and collusion) mean that otherwise similar procurement impulses can translate into very different realised outcomes, so that even a well-designed μ_{PP} can be substantially distorted in terms of timing, cost, quality, and resource allocation.¹⁰

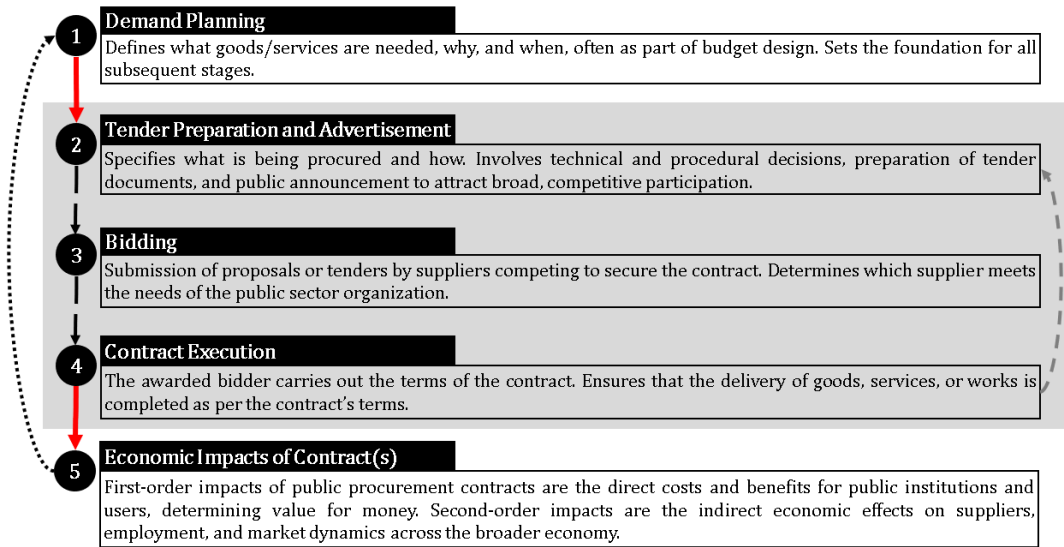
Public procurement law establishes the formal institutional environment that governs the key determinants of implementation effectiveness ϕ , while conduct in practice determines how much of this legal potential is realized (Bosio et al. 2022). In this sense, public procurement law is a central policy lever for reducing the procurement loss wedge ω , because it defines the rules, procedures, and constraints that determine the degree of buyer discretion, the intensity of supplier competition, and the strength of contractual enforcement, and it structures how existing administrative capacity can be deployed. From an economics perspective, however, the law is not a single abstract document but a bundle of concrete design levers, such as auction formats, scoring rules, participation and disclosure requirements, and enforcement tools, that define the market design of public procurement in the sense emphasised by Roth (2002, 2008). We therefore move to a finer level and organise the remainder of the survey around these specific mechanisms, in parallel with how the literatures on fiscal policy and industrial policy study particular instruments rather than entire legal codes.

These institutional and market determinants operate through a staged public procurement cycle. Figure A6 presents a five-stage framework for fiscal transmission through public procurement and highlights four central elements through which procurement influences economic activity: demand planning, the implementation core, economic consequences for the production side of the economy, and dynamic feedback across stages. In what follows we describe the stages in turn and relate them to these four elements.

⁹ Expert assessments indicate that cost overruns in public road works occur in more than half of cases in several major economies and in at least one-quarter of cases in a broad group of middle-income countries (Figure A7), that abuse of ex post renegotiation is common and can affect up to nine-tenths of contracts in some low-capacity environments (Figure A8), and that perceived low-quality delivery is widespread, often occurring in 50–90% of cases in many African countries and at non-trivial rates in several large middle- and high-income economies (Figure A9); limited competition is also a common concern across many African and Latin American countries and in major economies such as the United States, China, India, and Canada (Figure A10).

¹⁰ Countries differ markedly even in basic procurement-regime features: some rely heavily on buyer discretion, others on reverse first-price auctions based solely on price, and others on mixed price-quality criteria (Figure A11); they also diverge on whether tender documentation is made publicly available, directly affecting transparency and entry (Figure A12). Abuse of technical specifications in public road procurement is perceived as common in many countries (Figure A13), corruption in procurement is reported to be pervasive in many less developed economies and remains present in several large middle- and high-income countries (Figure A14), and collusion among bidders is frequently observed (Figure A15), underscoring that ω reflects not only administrative capacity but also the underlying rules, oversight structures, and competitive conditions under which procurement operates

Figure A6. The Economics of Public Procurement: A Five-Stage Conceptual Framework



Source: Authors, based on the literature.

Stage 1, demand planning, reflects *ex ante* beliefs of politicians and bureaucrats about how procurement supports service delivery, political goals, and domestic industry, and it is shaped by political incentives, lobbying, and administrative capacity; in this way it determines which projects enter the pipeline and the strategic orientation of spending. Stages 2 to 4 form the implementation core of public procurement, because they govern how authorised budgets are transformed into contracts and deliveries and they determine whether the state obtains the intended public services in terms of prices, timeliness, and quality; from a policy perspective, the primary objective of procurement is precisely this delivery of infrastructure, health care, education, security, and other services to users. Stage 5, economic consequences, forms the outcomes core of the framework and lies at the heart of the economics of public procurement. At the macro level, we are interested in how procurement shapes aggregate outcomes such as sectoral structure, output, employment, and distributional and regional patterns; for a given spending programme or bundle of contracts, the expected procurement multiplier μ_{PP} can be viewed *ex ante* as a composition-weighted average of contract-level effects across the contracts in the bundle, while *ex post* the realised fiscal impact of procurement is summarised by the composite multiplier $\phi\mu_{PP}$. At the micro level, we study how procurement mechanisms and the allocation of individual contracts $c_1, \dots, c_n$ across firms $f_1, \dots, f_h$ affect the production side of the economy, including firm performance, entry, investment, participation in production networks, and the allocation of activity across sectors and technologies; since every unit of procurement spending must flow through such contracts, these contract-level responses are the building blocks of the bundle-level multiplier μ_{PP} . Our focus is on the incremental effects of the procurement process on delivery and outcomes, relative to a benchmark in which goods and services are delivered on time and as specified. User-side benefits of the procured goods and services themselves, such as health outcomes or mobility gains, are mainly analysed in sectoral literatures such as health economics or transport economics and are outside the core domain of the economics of public procurement.¹¹

¹¹ If a hospital procures a critical medicine and procurement failures cause late or non-delivery, the resulting excess mortality is an outcome of the procurement process, whereas the health gains from an effective medicine that is delivered on time and as specified belong primarily to health economics and clinical medicine. Other examples include:

Procurement is not simply a linear sequence of actions but a dynamic system in which governments and firms co-evolve. Buyers update beliefs about feasibility and supplier capability in light of realised performance, rather than behaving with the “*incredible certitude*” described by Manski (2013), and firms adapt their bidding, investment, and capacity decisions to observed opportunities and enforcement. Embedding procurement in this staged and iterative cycle clarifies how public spending is translated into real economic activity and institutional learning, organises diverse evidence within a common structure, and highlights mechanisms and parameters that remain underexplored in the theoretical and empirical literature.

The staged framework clarifies where implementation can fail: even if demand planning (Stage 1) selects socially valuable projects, weak performance in tender preparation, supplier selection, and contract execution (Stages 2 to 4) can prevent authorised spending from being translated into timely, cost-effective, and high-quality delivery to users. In practice, such failures can result in outright non-provision of contracted works or services, delay delivery and shift social costs to end users who do not receive services when needed, induce contractors to cut quality in order to raise profits or minimise costs, and generate overpayments that waste public funds and shrink the fiscal envelope available for other projects. Weak performance in Stages 2 to 4 can also influence the production side of the economy, because these stages shape firms’ participation in procurement markets, their bidding strategies, and their investment and capacity decisions.

We conceptualize ϕ as a composite measure of how effectively a procurement system converts authorised spending into delivered output along four dimensions of performance: timeliness, cost efficiency, quality of delivery, and allocative efficiency. A higher value of ϕ (lower value of wedge ω) on each of these four dimensions. Formally,

$$\phi = \phi(A, D, M, \sigma), \quad \phi \in [0, 1], \quad \frac{\partial \phi}{\partial A} > 0, \frac{\partial \phi}{\partial M} > 0, \frac{\partial \phi}{\partial \sigma} > 0, \frac{\partial \phi}{\partial D} \text{ ambiguous} \quad (6)$$

where A denotes administrative capacity (primarily the ability to implement Stages 2–4, built on investments decided in Stage 1), D the degree of discretion (rules and practices governing decisions in Stages 2–4), M market conditions (the competitiveness and capability of the supplier base that bidders bring to Stage 3 and that is shaped by past outcomes in Stages 4 and 5), and σ the quality of technical specifications and performance contracting (design choices in Stage 2 that shape execution in Stage 4). Administrative capacity shapes delays, cost overruns, and enforcement of quality through staffing, training, information systems, and the ability to manage contracts over time. Market conditions, including the number and size distribution of potential suppliers, influence contract prices and the likelihood that efficient firms win; when the supplier base is thin or highly concentrated, even well-designed procedures have limited scope to reduce mark-ups or misallocation, so M primarily affects cost and allocative efficiency while also mattering for quality and deadlines. Specification and contract quality determine how precisely needs are defined, how credibly performance can be monitored and enforced, and how risks are

India’s Golden Quadrilateral which raised firm output in transport-intensive industries by roughly 15–20% (Chatterjee et al., 2025), and rail expansion in colonial India reduced trade costs by about 20% and increased real agricultural incomes up to 16% through improved market access (Donaldson, 2018). Governance and design are pivotal: infrastructure can entrench extractive institutions when political incentives favor rents (Acemoglu et al., 2001). Road capacity tends to induce near-proportional traffic increases without complementary policies such as congestion pricing (Duranton and Turner, 2011). Spatial spillovers can be large and unequal, affecting urban form, usage, and inclusion, as shown by suburbanization from highway expansion (Baum-Snow, 2007), trade-related infrastructure concentrating gains in developed regions (Redding and Venables, 2004), low uptake when user incentives are weak (Devoto et al., 2012), gendered labor-market effects from electrification (Dinkelman, 2011), and sprawl that can exacerbate exclusion (Glaeser and Kahn, 2004).

allocated over the life of the contract, and thus affect time, cost, and quality outcomes simultaneously.

Discretion is inherently context dependent. In high-capacity and high-integrity environments, more discretion can raise ϕ by allowing buyers to tailor tenders and contracts to complex needs and to prevent delays and quality shortfalls. In low-capacity or weak-accountability environments, the same discretion can lower ϕ by amplifying corruption, favouritism, and rent extraction, raising costs and misallocating contracts. These mechanisms show that procurement law and its implementation, by shaping A , D , M , and σ , determine not only compliance and integrity but also the fundamental ability of public procurement systems to translate fiscal authorisations into timely, cost-effective, high-quality, and efficiently allocated public output. Strengthening the institutional and market foundations of ϕ therefore offers a direct avenue for lowering the wedge ω and improving the macroeconomic effectiveness of public procurement.

E.3 Public procurement in fiscal transmission

Equation (7) is not intended as a structural national-accounts identity. Instead, it provides a local linear representation of output around a given baseline allocation, designed to organize the marginal effects of public spending composition. In this representation, C , I , and NX are treated as baseline components that are locally fixed with respect to the marginal procurement shock under consideration. All induced private responses of consumption, investment, and net exports are therefore summarized by the multiplier terms.

The multipliers μ_{PP} and μ_{NP} capture the total equilibrium response of output to marginal changes in procurement and non-procurement public spending, respectively, conditional on sectoral composition z and macroeconomic conditions ς . The substantive object of interest is thus the differential expression for ΔY derived below, rather than equation (7) interpreted as an exact accounting identity.

With this interpretation in mind, we express output as a function of the composition of government expenditure and the effectiveness with which procurement translates budget allocations into real economic activity. Letting s denote the share of government spending implemented through public procurement, we write:

$$Y = C + I + \mu_{PP}(z, \varsigma)[\phi(A, D, M, \sigma)]sG + \mu_{NP}(1 - s)G + NX \quad (7)$$

This expression highlights that public procurement influences aggregate output through three channels: the share of government expenditure implemented through contracts s , the sectoral and macroeconomic environment captured by $\mu_{PP}(z, \varsigma)$, and the implementation effectiveness $\phi(A, D, M, \sigma)$ with which authorised procurement spending is converted into delivered activity. Even if procurement spending is directed to high-multiplier sectors and deployed under favourable macroeconomic conditions, its ultimate impact depends on how effectively it is executed, as summarized by ϕ . A first-order change in output associated with changes in G and the share s is:

$$\begin{aligned} \Delta Y = & [\mu_{PP}(z, \varsigma)(\phi(A, D, M, \sigma))s + \mu_{NP}(1 - s)]\Delta G \\ & + [\mu_{PP}(z, \varsigma)(\phi(A, D, M, \sigma)) - \mu_{NP}]\Delta s \end{aligned} \quad (8)$$

We can then decompose fiscal responses into interpretable components through a few intuitive parameters:

$$\Delta Y = \underbrace{\mu_{NP}\Delta G}_{\text{baseline (non procurement)}} + \underbrace{s[\mu_{PP}(z, \varsigma) - \mu_{NP}]\Delta G}_{\text{composition effect}} \quad (9)$$

$$- \underbrace{s\mu_{PP}(z, \varsigma)\omega(A, D, M, \sigma)\Delta G}_{\text{efficiency wedge}} + \underbrace{[\mu_{PP}(z, \varsigma)(1 - \omega) - \mu_{NP}]G\Delta s}_{\text{recomposition wedge}}$$

The decomposition highlights four channels through which changes in G affect output:

(1) Baseline non-procurement channel: $\mu_{NP}\Delta G$

This term captures the output response to changes in G that do not flow through public procurement, such as payroll, transfers, or administrative costs. Meta-analytic evidence suggests that these components can have different effects from direct demand for goods and services: Gechert (2015) finds that public spending multipliers average around one and are about 0.3–0.4 units larger than tax and transfer multipliers, with public investment multipliers roughly 0.5 units above spending in general, while Boehm (2020) estimates short-run multipliers of about 0.8 for government consumption but close to zero for government investment, implying substantial crowding out of private investment. Together, these findings indicate that the non-procurement component $\mu_{NP}\Delta G$ need not coincide with the procurement component $\mu_{PP}\Delta G$, since different spending items and delivery channels can generate systematically different output responses.

(2) Composition effect: $s[\mu_{PP}(z, \varsigma) - \mu_{NP}]\Delta G$

This term captures the gain (or loss) from shifting the composition of G toward public procurement, holding total G constant. If public procurement-intensive spending has a higher multiplier than non-procurement spending, shifting resources toward public procurement increases output; otherwise, it lowers output. This channel reflects heterogeneity across sectors z and macroeconomic states ς , and the possibility that procurement reallocates demand toward sectors with stronger domestic linkages, learning potential, or supply chain depth (Bouakez et al., 2025; Bombardini et al., 2024).

(3) Efficiency wedge: $-s\mu_{PP}(z, \varsigma)\omega(A, D, M, \sigma)\Delta G$

This term reflects the fact that planned procurement spending may not fully translate into realized economic activity. Institutional frictions in procurement, including delays, weak competition, low supplier capability, and weak contract enforcement, cause part of the fiscal impulse to dissipate. A large empirical literature documents that these implementation losses are common. For example, Decarolis (2014), Coviello and Mariniello (2014), Lewis-Faupel et al. (2016), Schoenherr (2019) and Bandiera et al. (2009) each provide evidence of substantial inefficiencies in public procurement across different settings. In our notation, these factors are summarized by the wedge $\omega(A, D, M, \sigma)$, which increases when administrative capacity is weak, discretion is poorly monitored, supplier competition is limited, or specification and contract quality are low, and which scales down the potential impact of procurement spending on output.

(4) Recomposition wedge: $[\mu_{PP}(z, \varsigma)(1 - \omega) - \mu_{NP}]G\Delta s$

This term captures the static effect of reallocating a given level of government expenditure toward public procurement, holding implementation effectiveness fixed. When the realized multiplier associated with procurement spending differs from that of non-procurement spending, a marginal increase in the procurement share s changes output even if total G is unchanged. This effect reflects differences in sectoral exposure, domestic linkages, and macroeconomic conditions

embedded in $\mu_{PP}(z, \varsigma)$, rather than any change in procurement efficiency or institutional performance (List, 2024).

The four fiscal channels map directly into the public procurement stages (Figure A6). The composition channel operates through Stage 1, where choices over what to buy and in which sectors determine s and z , informed by outcomes in Stage 5. The efficiency channel is governed by Stages 2–4, where A , D , M , and σ shape ϕ and thus the wedge ω between authorised spending and delivered activity. The recomposition channel captures how changes in s interact with these parameters, for example when expanding procurement strains capacity and lowers ϕ . Feedback from Stage 5 to Stage 1 closes the cycle, so realised consequences feed into future choices over s , z , and procurement design. Table A2 summarises how these channels map into the five stages.

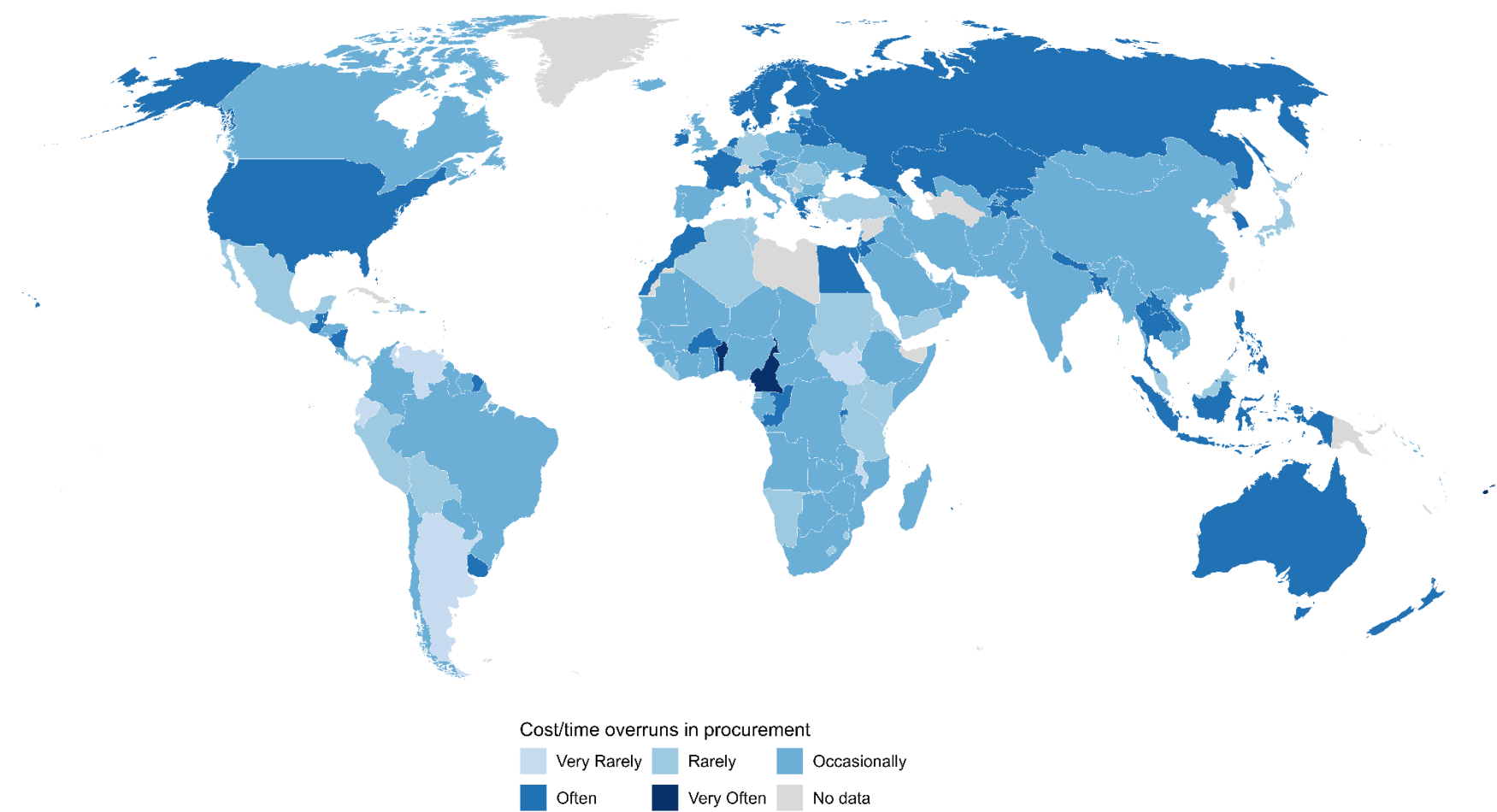
Relative to standard fiscal-policy analysis, which typically estimates a single multiplier for government purchases that effectively corresponds to the composite $\phi \mu_{PP}$, this framework separates the sectoral response $\mu_{PP}(z, \varsigma)$ from implementation effectiveness $\phi(A, D, M, \sigma)$ and anchors both in specific stages of the procurement process. In applied macro data it is difficult to identify μ_{PP} and ϕ separately, so existing fiscal studies usually recover only $\phi \mu_{PP}$ rather than the underlying procurement mechanisms. As a result, most of the current evidence on how procurement works is micro, focusing on contracts, bidding, prices, quality, and firm outcomes, while macro procurement multipliers remain relatively underdeveloped. The remainder of the survey therefore organises the micro-level literature around the five-stage framework and these fiscal channels, bringing in the available macro evidence on $\phi \mu_{PP}$ where it helps connect procurement institutions to aggregate fiscal transmission.

Table A2. Canonical mapping between fiscal channels and procurement stages

Fiscal Channel / Expression	Economic Interpretation	Procurement Stage(s)	Mechanism	Key Research Questions
Baseline non-procurement effect $\mu_{NP}\Delta G$	Output effect of government spending outside public procurement	Outside procurement cycle	Benchmark fiscal multiplier	How does public spending outside procurement transmit to output?
Composition effect $s[\mu_0(z, \varsigma) - \mu_{NP}]\Delta G$	Reallocation of spending toward procurement changes sectoral demand and multiplier composition	Stage 1 (Demand planning) and Stage 5 (Economic consequences)	Sector choice, domestic linkages, innovation/intensity, industrial upgrading	Which sectors/contracts yield higher multipliers? Does procurement stimulate upgrading or crowd out private activity?
Efficiency wedge $-s\mu_0(z, \varsigma)\omega(A, D, C, \sigma)\Delta G$	Loss in fiscal transmission due to institutional and market frictions	Stages 2–4 (Preparation, Bidding, Execution)	Transparency, competition, discretion, enforcement, supplier capability	How do rules and practices affect delivery? Which frictions matter most?
Recomposition wedge $[\mu_0(z, \varsigma)(1 - \omega) - \mu_{NP}]\Delta G$	Static effect of reallocating spending toward public procurement when procurement and non-	Stage 1 (Demand planning)	Changes in sectoral composition and demand propagation; no change in procurement efficiency.	When does reallocating spending toward procurement raise output? When do scale effects (learning or congestion) alter

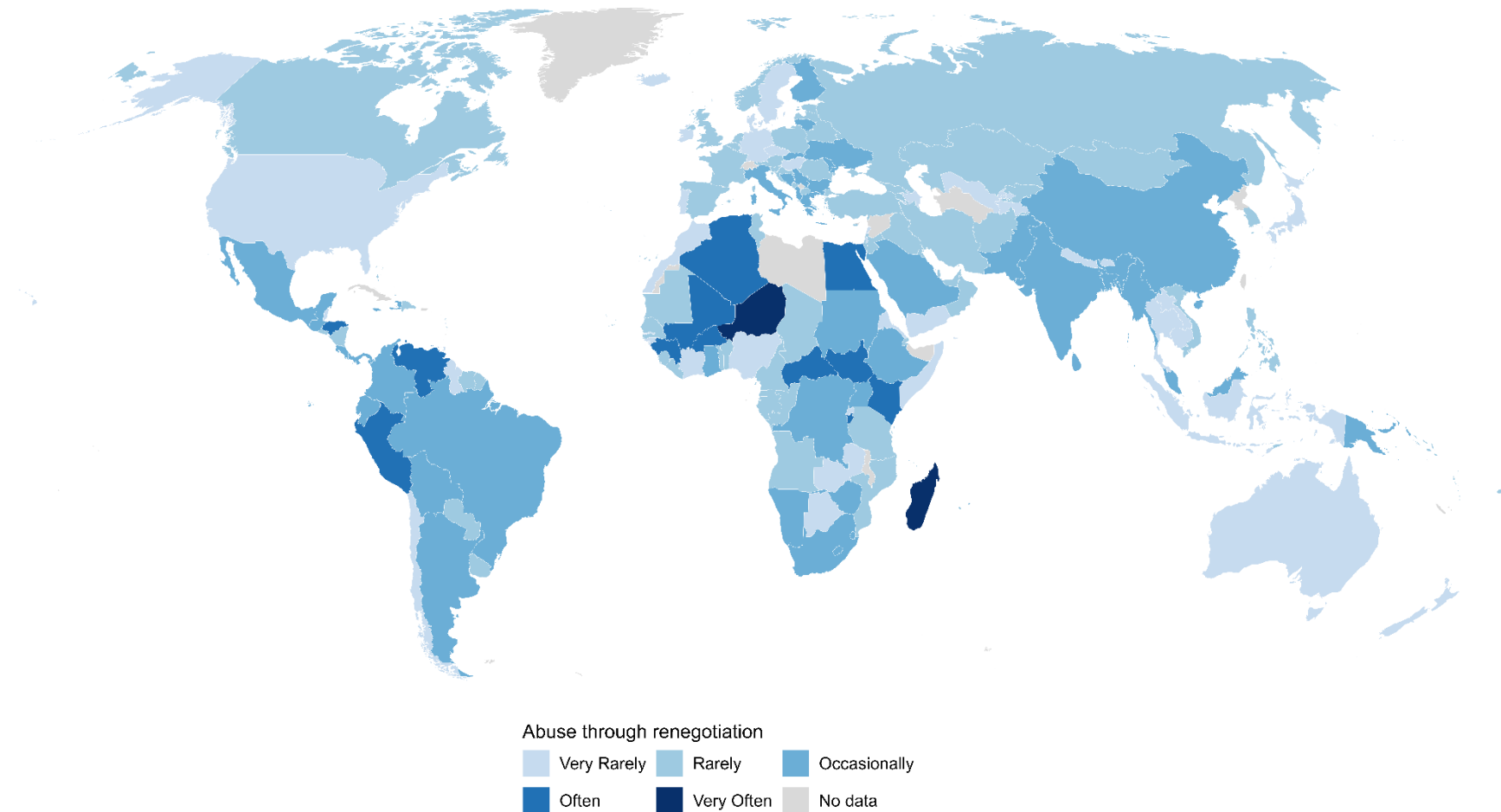
	procurement spending have different realized multipliers, holding implementation effectiveness fixed.			procurement efficiency beyond this static effect?
Feedback and learning (implicit dynamic)	Updating budget priorities and constraints based on realized outcomes	Stage 1 ↔ Stage 5 feedback loop	Bayesian updating, performance learning, policy revision	How do governments learn from procurement performance? How do firms adapt?

Figure A7. Perceived frequency of cost overruns in public road works contracts



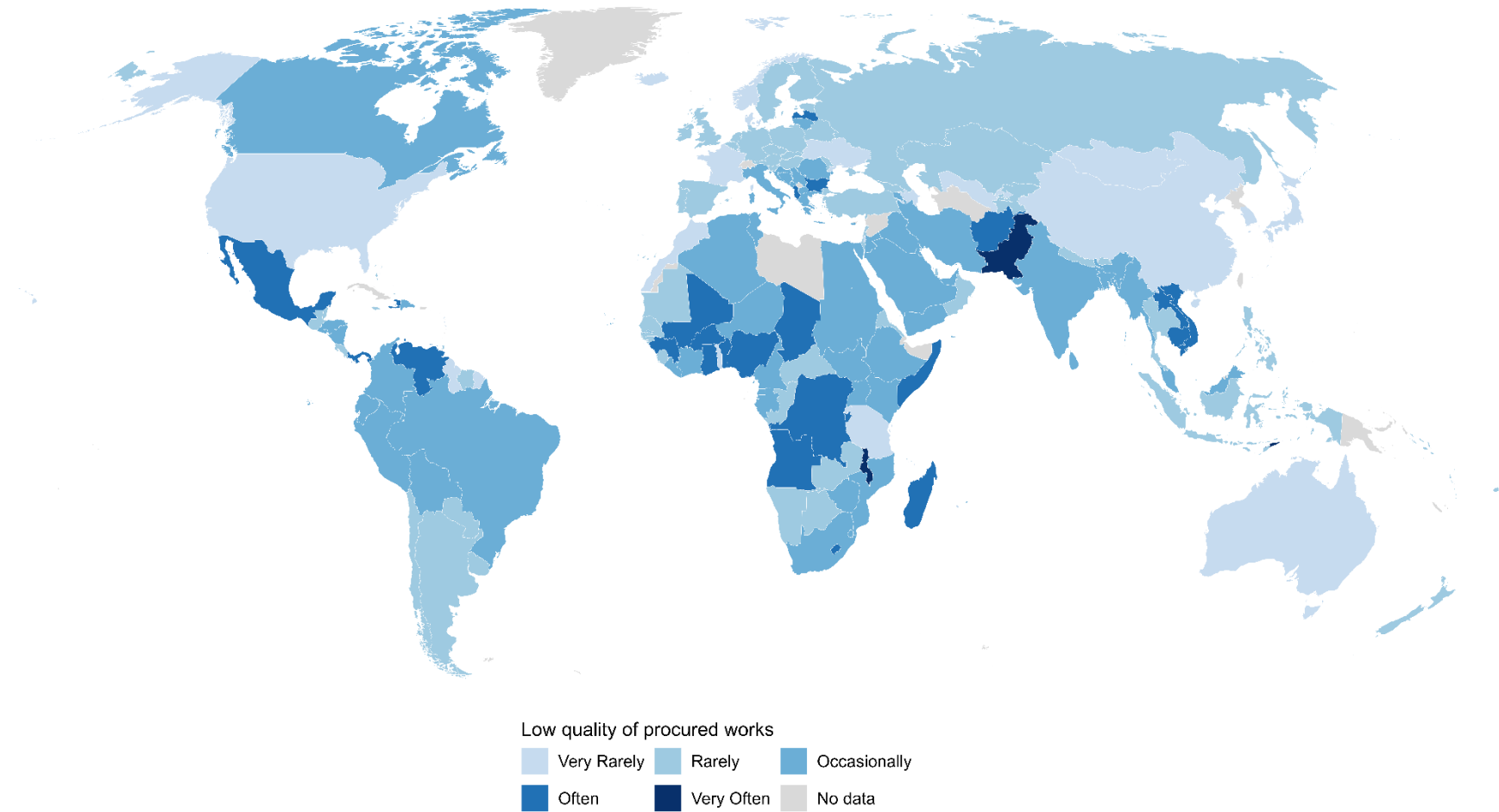
Note: Authors' representation using Bosio et al. (2022). Expert survey on how often a standardized US\$2.5m open-tender road resurfacing contract is delivered within budget (five-point scale from <10% to >90%). Lower values imply more frequent cost overruns. Applies to this benchmark road contract, not all procurement

Figure A8. Perceived frequency of abusing ex post renegotiation in public road procurement



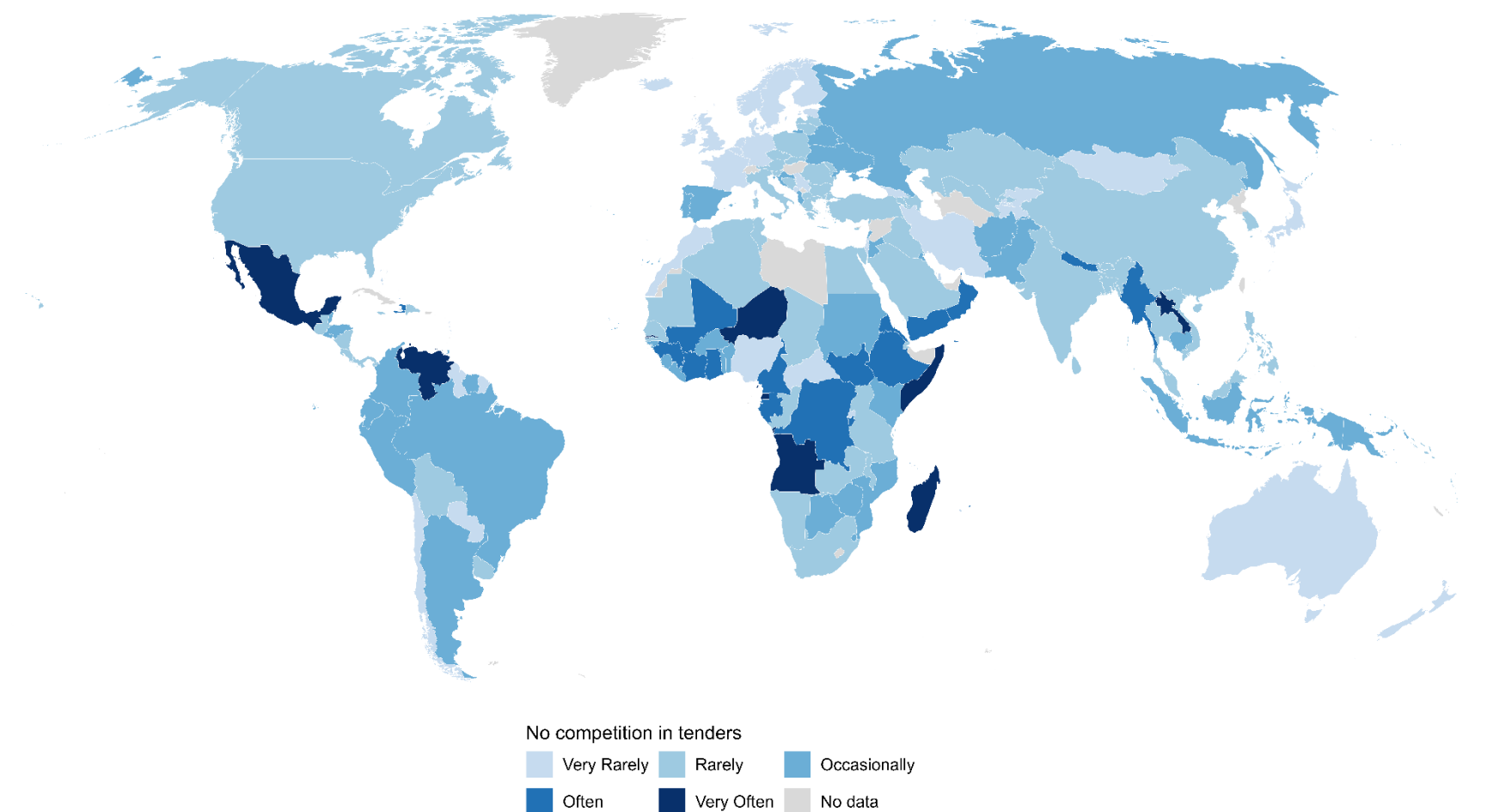
Note: Authors' representation based on data from Bosio et al. (2022). This figure reports country-level perceptions of how often the contract renegotiation process is abused to increase the price or scope of a standard public road works contract without launching a new competitive tender. The indicator is based on expert answers to the Bosio et al. (2022) World Bank survey question: *"How often are the following strategies used by private sector companies to circumvent public procurement rules? Abuse the renegotiation process to increase the price or the scope of the project without another competitive process."* Respondents are typically 8–12 professionals per country (public procurement lawyers, construction and engineering firms, and procuring entities), who answer with reference to a hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to be representative of a routine road works contract. Answers are given on a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90% of cases). The figure summarizes these practice-based perceptions at the country level for this benchmark class of road procurement contracts, not for all public procurement in the economy. The underlying data were collected between November 2018 and August 2019 as part of the World Bank Doing Business 2020 cycle.

Figure A9. Perceived frequency of low quality in public road works contracts



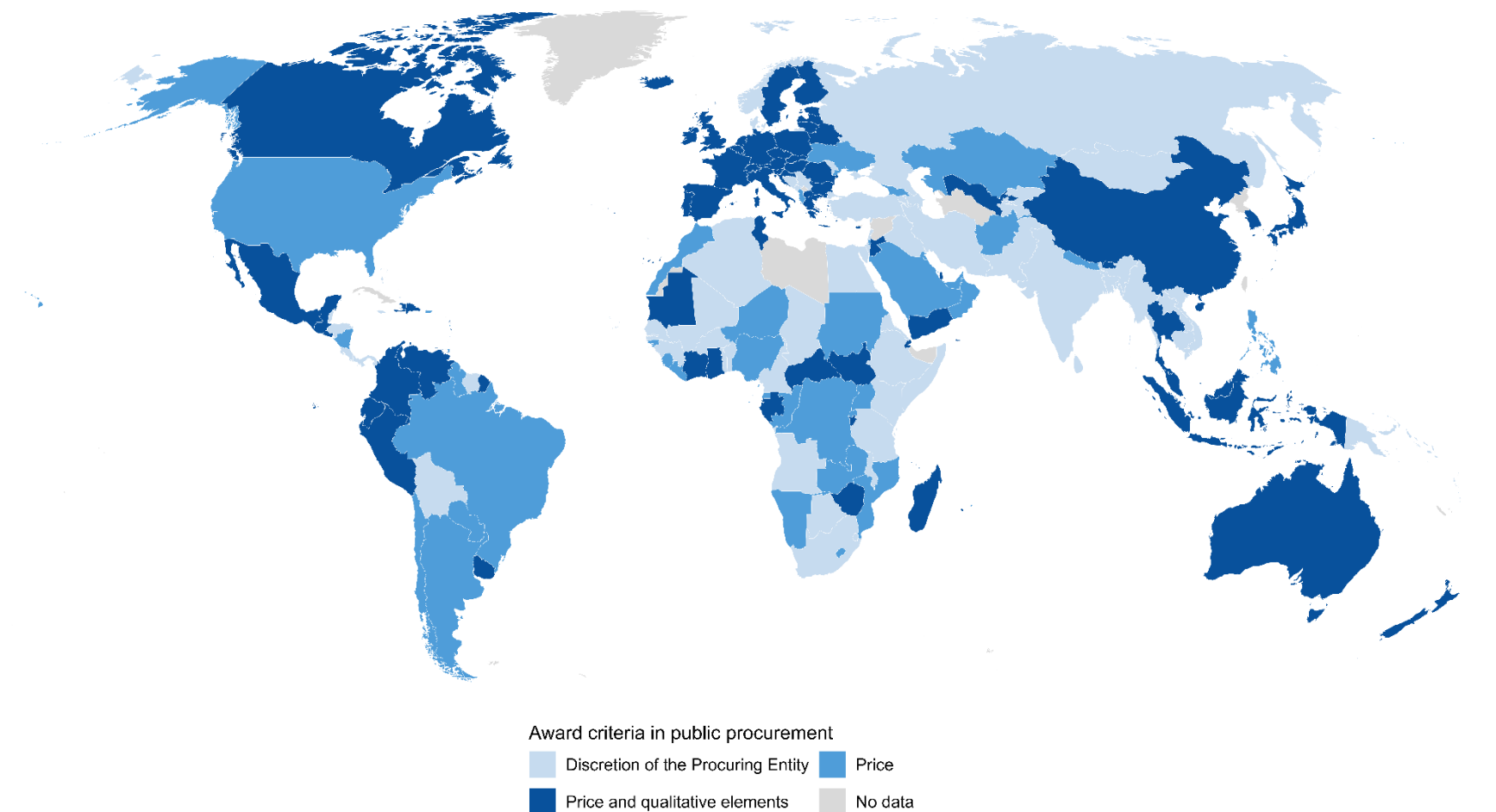
Note: Authors' representation based on data from Bosio et al. (2022). The figure reports country-level perceptions of how often public road works contracts are delivered with lower quality or with different technical specifications than those submitted during the tender process. The underlying data come from expert responses to the question: "*How often is the contract executed with less quality or with different technical specifications than those submitted during the tender process?*" Responses refer to a standardized hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in each country. Survey respondents (procurement lawyers, construction and engineering firms, and procuring entities) answer on a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90%). The figure summarizes perceived practice for this benchmark class of road contracts rather than all public procurement in the country.

Figure A10. Perceived frequency of absence of competition in public road procurement



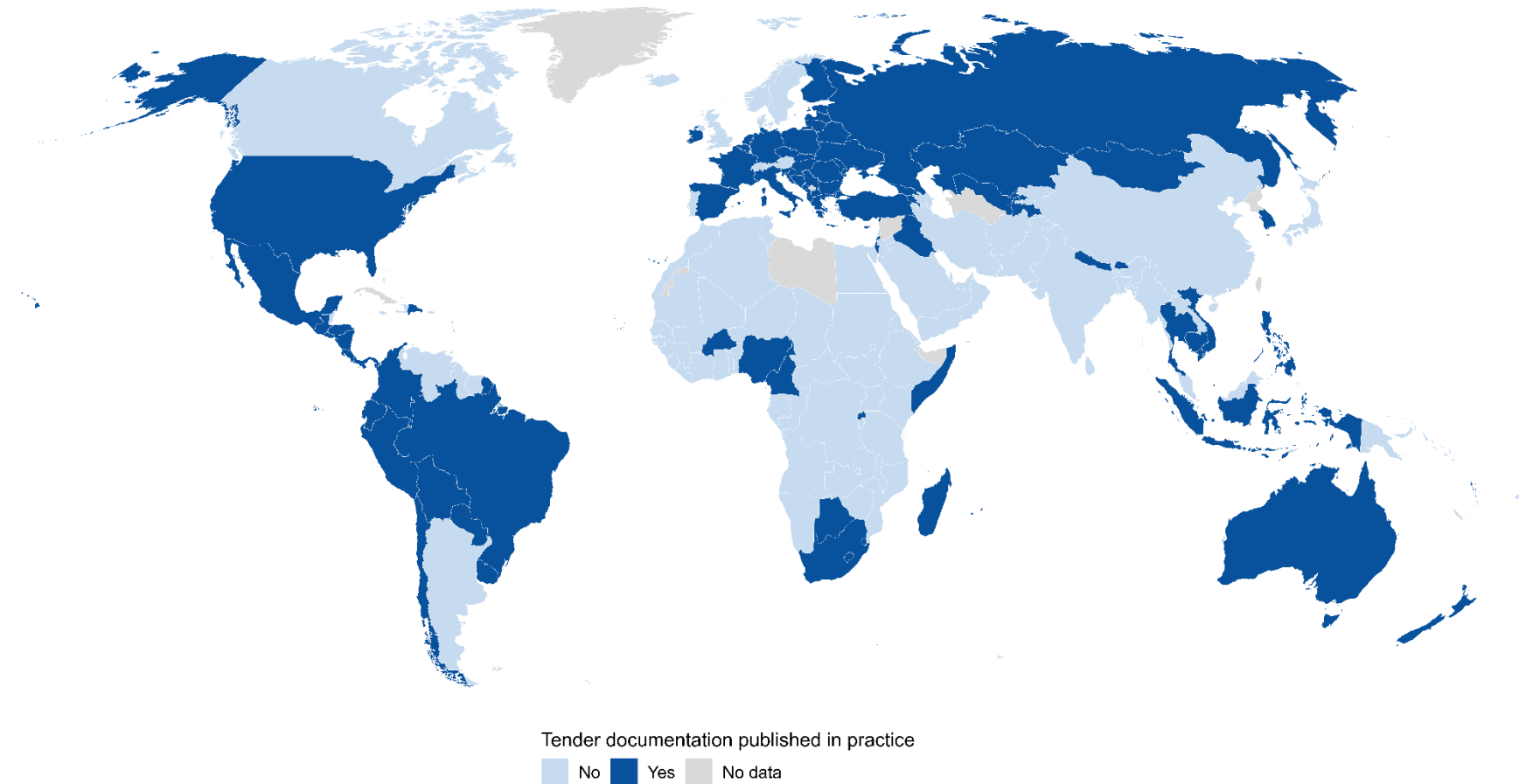
Note: Authors' representation based on data from Bosio et al. (2022). The figure reports country-level perceptions of how often procuring entities use non-competitive procurement methods instead of open tendering to restrict market entry in public road works. The underlying data come from expert responses to the survey item: "Use non-competitive procurement methods instead of open tendering to restrict market entry." Responses refer to a standardized hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in each country. Survey respondents (public procurement lawyers, construction and engineering firms, and procuring entities) answer on a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90%). The figure summarizes perceived practice for this benchmark class of road contracts rather than all public procurement in the country; higher reported frequency corresponds to a greater perceived absence of competition.

Figure A11. Award criteria in public road procurement



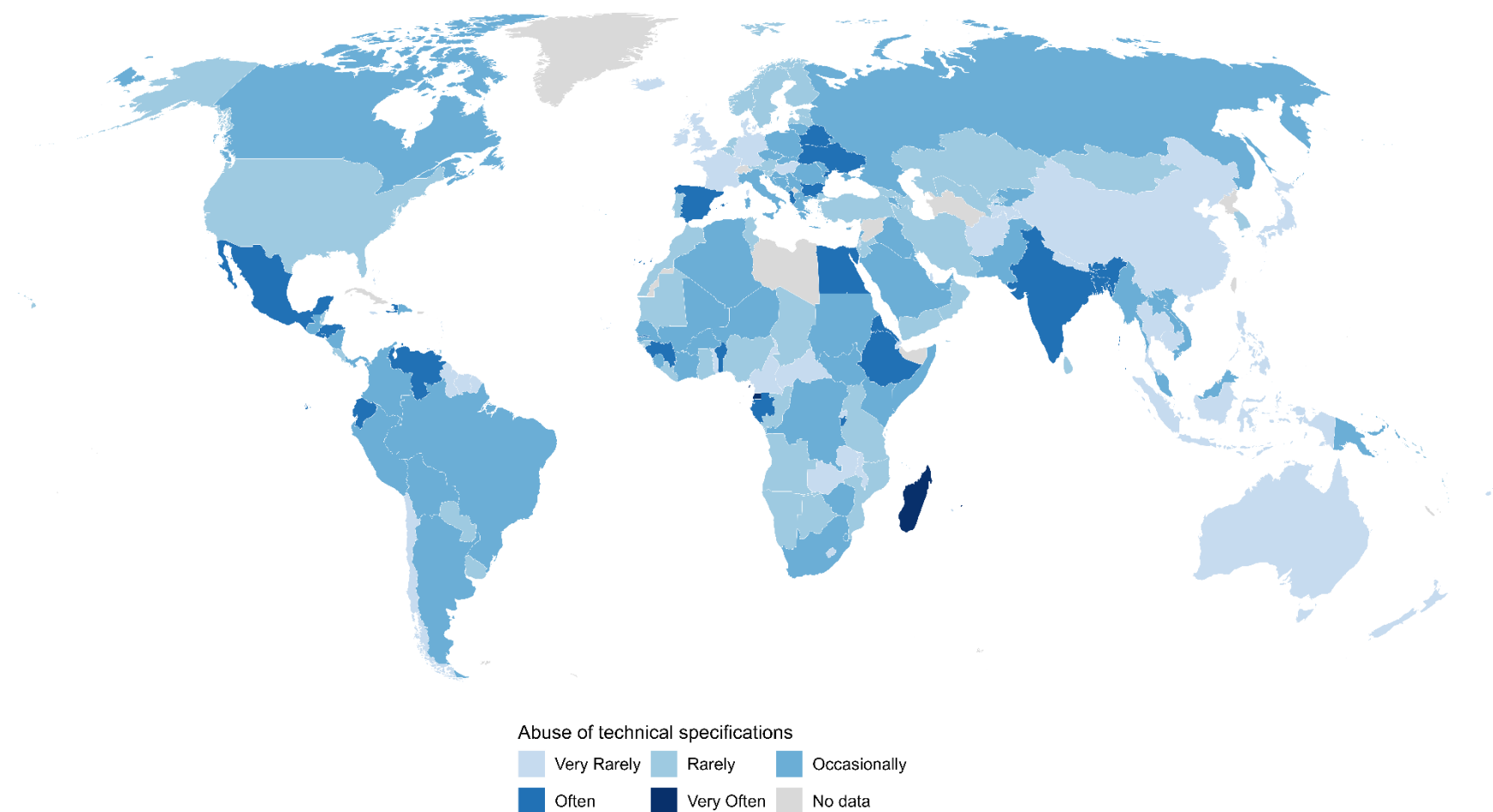
Note: Authors' representation based on data from Bosio et al. (2022). The figure classifies countries according to the award criterion that is typically used to select the winning bid for a standardized US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a routine public road works contract in each country. Survey respondents (public procurement lawyers, construction and engineering firms, and procuring entities) report which criterion is most commonly applied in practice, choosing among three options: (i) discretion of the procuring entity, (ii) price only, and (iii) price and qualitative elements (for example, technical quality or other non-price attributes combined with price). The map summarizes these practice-based responses at the country level for this benchmark class of road contracts and should not be interpreted as a measure of award criteria used in all public procurement in the country.

Figure A12. Publication of tender documentation in public road procurement



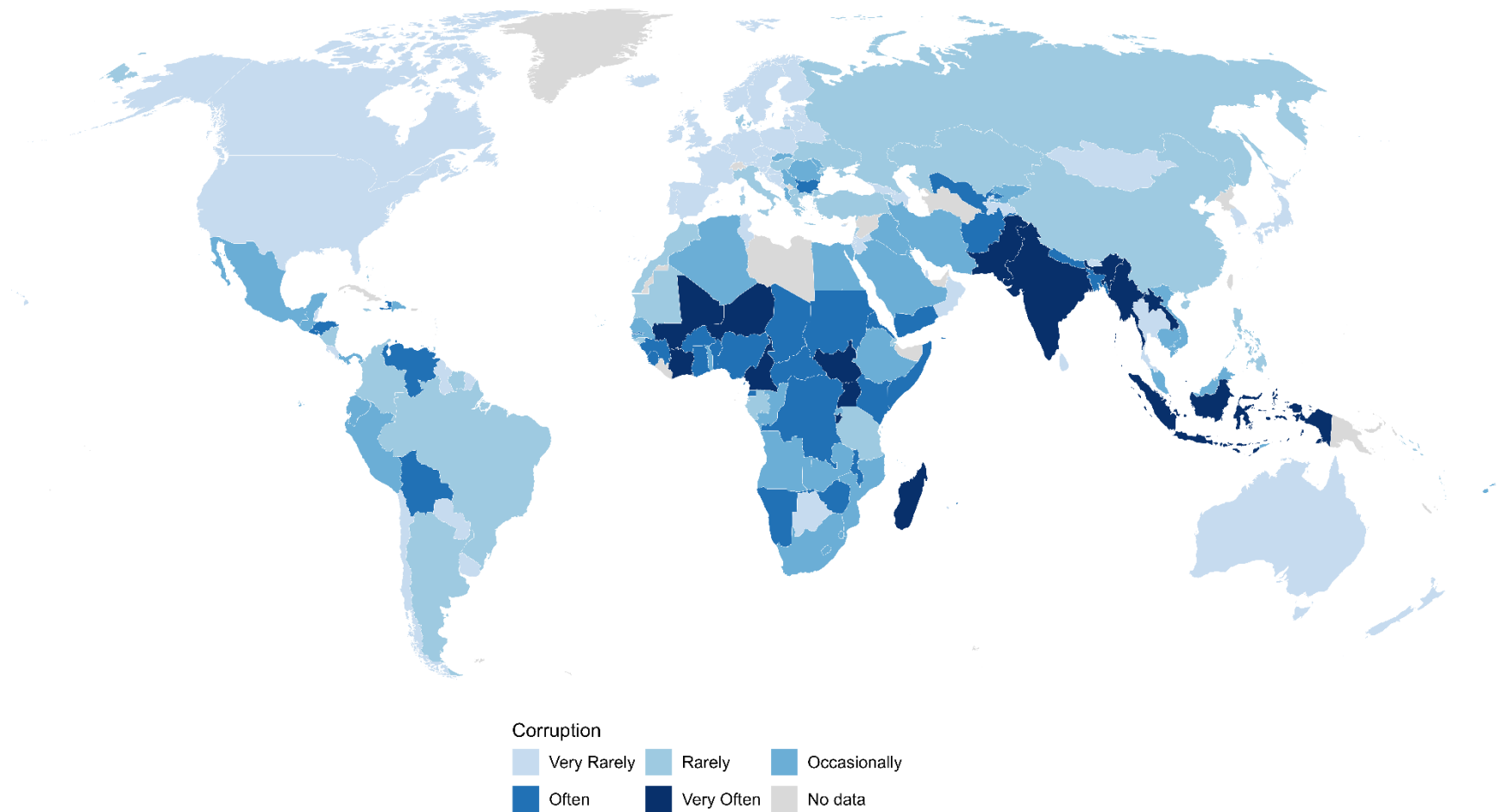
Note: Authors' representation based on data from Bosio et al. (2022). The figure reports, for each country, whether tender documentation is reported to be published in practice for a standardized US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in that country. Expert respondents (public procurement lawyers, construction and engineering firms, and procuring entities) answer a yes/no practice question on whether the full tender documentation is made publicly available (for example, online) for this type of contract. The map classifies countries according to these practice-based responses for the benchmark road contract and should not be interpreted as a measure of documentation practices for all public procurement in the country.

Figure A13. Perceived frequency of abusing technical specifications in public road procurement



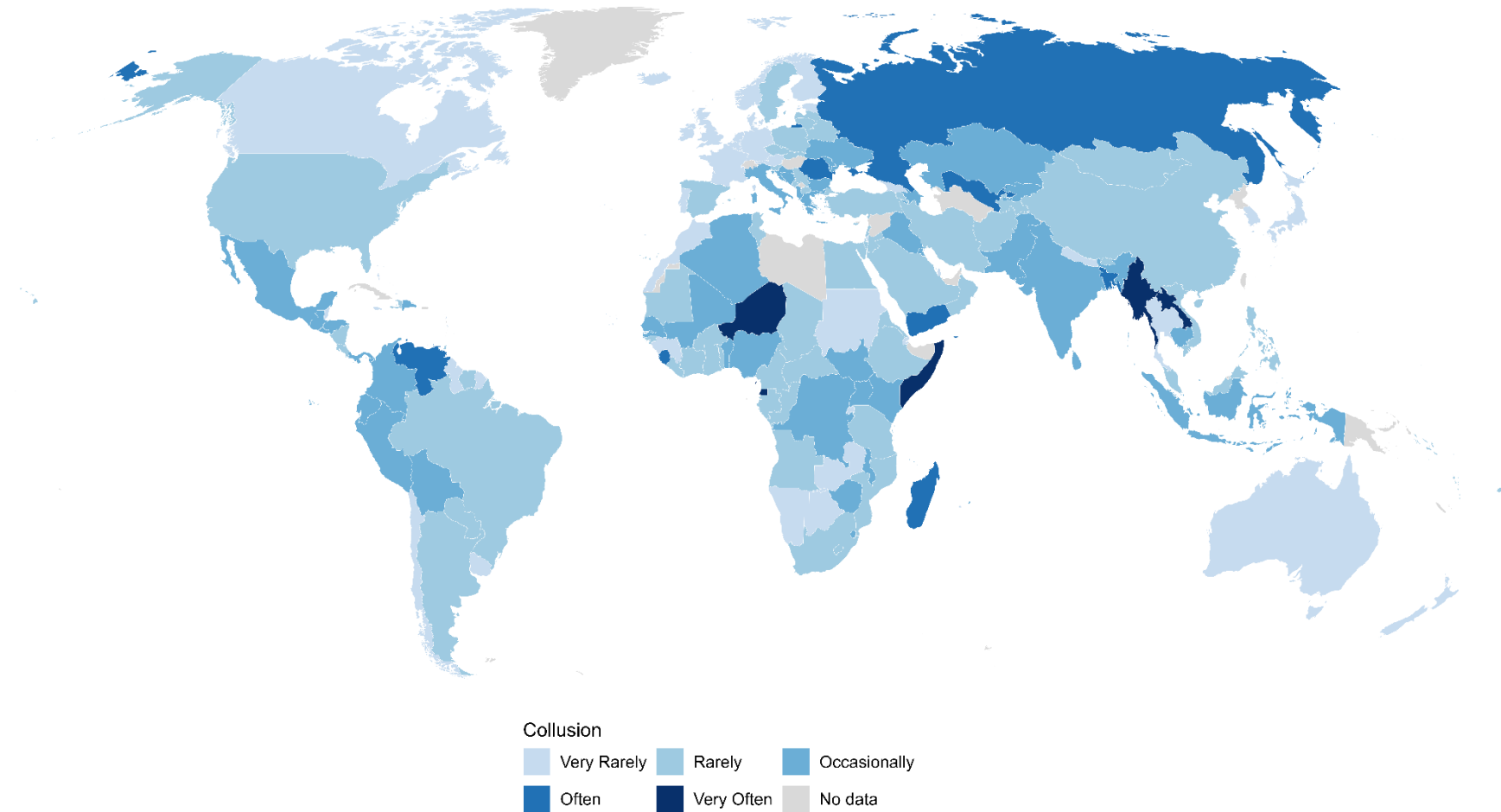
Note: Authors' representation based on data from Bosio et al. (2022). This figure reports country-level perceptions of how often procuring entities define technical specifications so as to benefit a specific bidder in public road procurement. The underlying data come from expert responses to the Bosio et al. survey question: "How often does the Procuring Entity define technical specifications to benefit a specific bidder?" Respondents answer with reference to a hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in each country. Survey respondents are procurement lawyers, construction and engineering firms, and officials from procuring entities, who report on actual practice rather than the de jure legal framework. Answers are given on a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90% of cases). The figure summarizes these practice-based perceptions for this benchmark class of road contracts, and should not be interpreted as a measure of all public procurement activity in the country.

Figure A14. Perceived frequency of corruption in public road procurement



Note: Authors' representation based on data from Bosio et al. (2022). The figure reports country-level perceptions of how often private sector companies resort to corruption in public road procurement. The underlying data come from expert responses to the outcome question: *"How often would private sector companies resort to informal payments in return for help from public officials in circumventing public procurement rules?"* Responses refer to a standardized hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in each country. Survey respondents (public procurement lawyers, construction and engineering firms, and procuring entities) answer on a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90%). The figure summarizes perceived practice for this benchmark class of road contracts and should not be interpreted as a direct measure of corruption in all public procurement in the country.

Figure A15. Perceived frequency of collusion in public road procurement



Note: Authors' representation based on data from Bosio et al. (2022). The figure reports country-level perceptions of how often collusion occurs in public road procurement. The underlying data come from expert responses to the survey block "How often are the following strategies used by private sector companies to circumvent public procurement rules?" and specifically to the item "Collusion between bidders". Responses refer to a standardized hypothetical US\$2.5 million open-tender road resurfacing contract implemented by a central government agency, designed to represent a typical public road works contract in each country. Survey respondents (public procurement lawyers, construction and engineering firms, and procuring entities) report on actual practice for this type of road contract using a five-point frequency scale: very rarely (<10% of cases), rarely (10–25%), occasionally (25–50%), often (50–90%), and very often (>90%). The figure summarizes these practice-based perceptions for this benchmark class of road works contracts and should not be interpreted as a direct measure of collusion in all public procurement in the country.

Figure A16. Allocation of Public Procurement by Broad Spending Categories

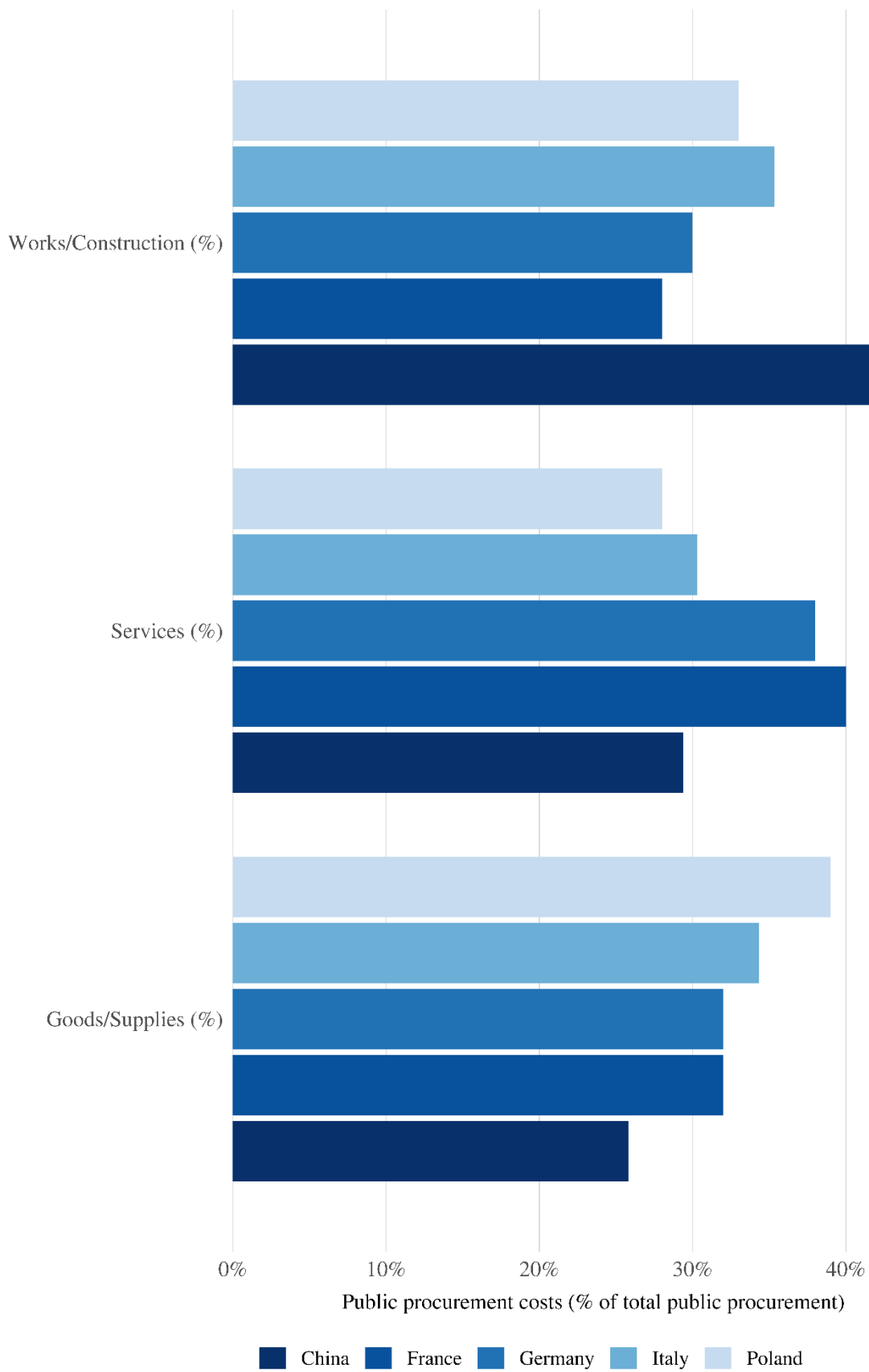
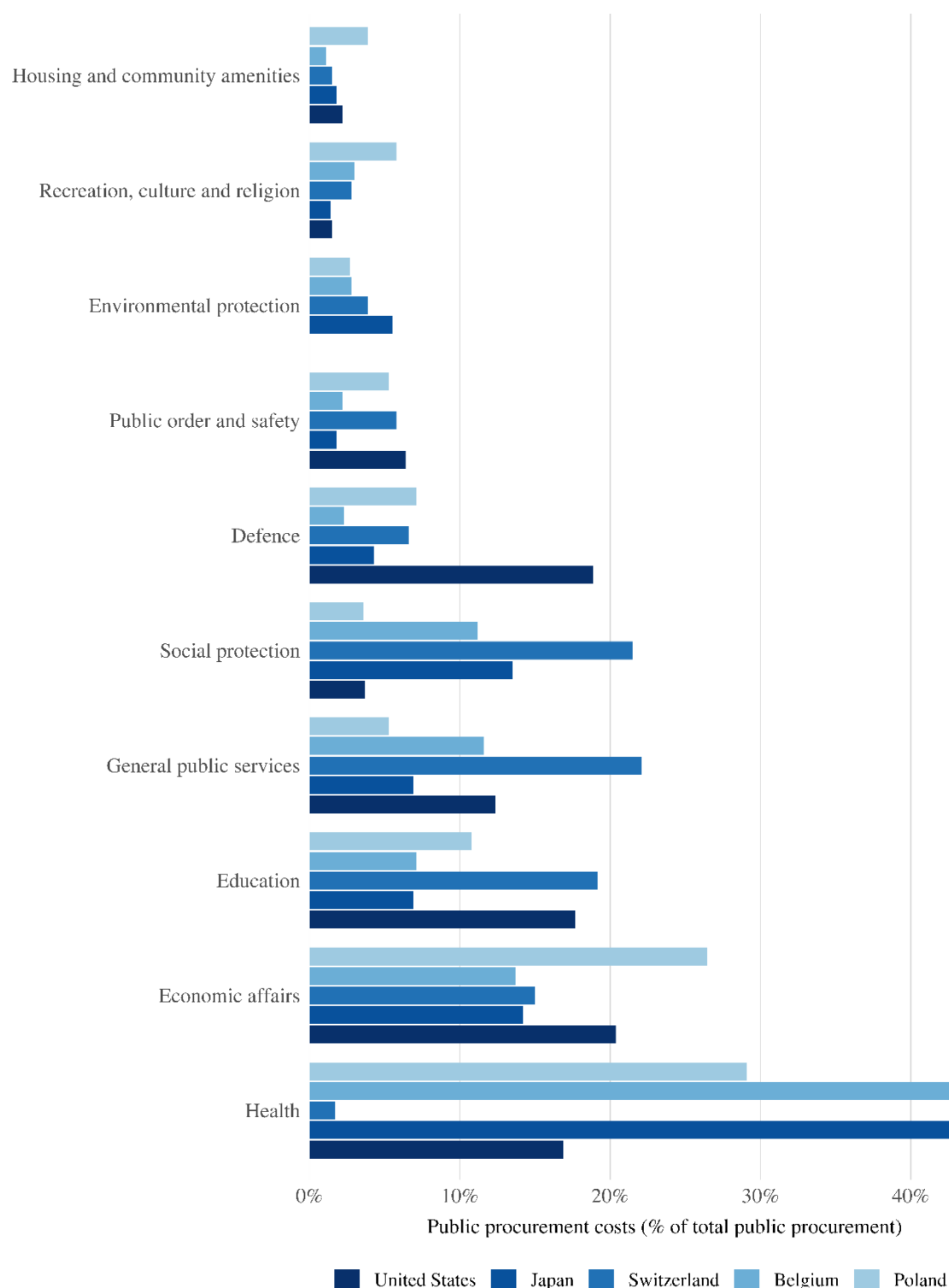


Figure A17. Allocation of Public Procurement by Purpose of Government Function



Note: The OECD reports procurement by function, using the Classification of the Functions of Government (COFOG), to enable cross-country comparison. Sectoral shares (for example, Defence at 20%) reflect the purpose of spending, not who spends it or what is purchased. The same item, such as vehicles, may fall under different sectors depending on use: military vehicles fall under Defence, while school buses are classified under Education. This method supports analysis of how procurement aligns with policy priorities. Definitions of each function are provided below. Source: OECD (2023), Government at a Glance.

Explanation A1. Note on Sectoral Composition of Public Procurement (Based on OECD Reports)

To enable meaningful cross-country comparisons, the OECD reports procurement by purpose using the Classification of the Functions of Government (COFOG). This framework groups spending into ten functional categories, reflecting the intended outcome of procurement rather than the institution responsible or the type of product acquired. According to the latest data from OECD Government at a Glance 2023, the composition of procurement spending varies across countries but typically concentrates in a few core sectors.

The ten COFOG sectors used to classify procurement are outlined below:

1. **General Public Services:** Covers the core operations of government, including executive and legislative activities, fiscal and financial affairs, foreign aid, public debt management, and intergovernmental transfers. It also includes services like public building administration and basic research.
2. **Defence:** Encompasses military and civil defence operations, foreign military aid, disaster preparedness, and related research and development. This includes the procurement of military equipment, infrastructure, and support services.
3. **Public Order and Safety:** Involves policing, fire protection, judicial services, and prison administration. It includes efforts to prevent crime, manage legal processes, and maintain public safety.
4. **Economic Affairs:** Supports economic development across sectors such as agriculture, energy, mining, manufacturing, construction, transport, and communication. It also includes labour market services, subsidies, and R&D related to economic activities.
5. **Environmental Protection:** Focuses on sustainability and conservation, including waste and wastewater management, pollution control, protection of biodiversity and landscapes, and environmental research.
6. **Housing and Community Amenities:** Involves the development and maintenance of housing, community infrastructure, water supply systems, and public lighting. It also includes services aimed at improving local living environments.
7. **Health:** Covers the provision of health services such as hospitals, outpatient care, medical goods, disease prevention, public health programs, and health-related research.
8. **Recreation, Culture, and Religion:** Includes support for recreational and sporting activities, cultural institutions (e.g. museums, libraries), public broadcasting, religious services, and community events.
9. **Education:** Covers spending on schools, universities, and other educational institutions, ranging from pre-primary to tertiary levels. It also includes student support services and educational research.
10. **Social Protection:** Encompasses welfare programs such as pensions, unemployment benefits, disability support, family allowances, housing assistance, and measures to reduce social exclusion.

This functional breakdown offers insights into how procurement spending aligns with public policy goals across countries.

Table A3. Size of public procurement of innovation in selected European Union Member States

Country	Government expenditures (G) as % of GDP	Public procurement (PP) as % of GDP	PP of innovation as % of total public procurement	PP of innovation as % of GDP	PP of innovation as % of G
Austria	52.0	13.6	6.5	0.9	1.7
Belgium	54.6	14.3	9.1	1.3	2.4
Bulgaria	37.7	7.5	2.0	0.2	0.4
Croatia	47.4	12.1	7.9	1.0	2.0
Cyprus	40.2	6.1	1.4	0.1	0.2
Czech Republic	43.7	12.7	4.1	0.5	1.2
Denmark	46.8	14.2	13.5	1.9	4.1
Estonia	43.1	13.7	13.9	1.9	4.4
Finland	56.6	17.7	16.5	2.9	5.2
France	57.0	14.5	11.4	1.6	2.9
Germany	48.4	15.5	9.4	1.5	3.0
Greece	49.9	10.3	6.6	0.7	1.4
Hungary	49.1	12.1	5.1	0.6	1.3
Ireland	22.7	7.5	17.2	1.3	5.7
Italy	53.8	10.2	7.0	0.7	1.3
Latvia	42.0	10.5	6.0	0.6	1.5
Luxembourg	48.1	11.8	4.5	0.5	1.1
Netherlands	43.2	19.5	8.8	1.7	4.0
Poland	46.7	11.0	4.3	0.5	1.0
Portugal	42.3	9.0	7.8	0.7	1.7
Romania	36.6	17.0	4.5	0.8	2.1
Slovak Republic	47.9	13.9	5.6	0.8	1.6
Slovenia	46.5	11.6	5.8	0.7	1.4
Spain	45.3	9.6	8.0	0.8	1.7
Sweden	47.5	16.3	13.1	2.1	4.5

Note: authors based on IMF, Eurostat, OCP and Spend Network data.

Appendix F. Auction-theory terminology and mechanism choice in procurement

F.1. Glossary of terms and benchmark assumptions

Independent private values (IPV): each bidder knows a valuation or cost that is private and statistically independent of rivals' values. Second-price sealed-bid (Vickrey) auction: bidders submit sealed bids; the highest-value bidder wins and pays the second-highest bid, which makes truthful bidding a dominant strategy. Revenue equivalence: under IPV, risk neutrality, symmetric bidders, and common support, standard auction formats yield the same expected revenue to the seller and the same allocation in expectation. Common value: the item has a single true value for all bidders, but each observes a noisy signal. Affiliated values: bidders' signals are positively correlated, so a high signal for one bidder makes high signals for others more likely. Winner's curse: in common or affiliated value settings, the winner tends to have the most optimistic signal and risks overpaying unless bids are shaded. Risk neutrality: bidders maximize expected profit without weighting risk. Common support: bidders' valuations are drawn from distributions with the same support so any value one bidder might have is possible for others. Sealed bid: bids are submitted once and privately, with no opportunity to observe or respond to rivals before award.

F.2. Why dominant-strategy and "richer" mechanisms are rare in procurement

Second-price and VCG-style mechanisms are rare in procurement not only because they are complex, but because they can be difficult to justify and police in audited, protest-prone environments. They can create politically contentious bid-payment gaps, raise privacy and disclosure concerns, and become fragile under budget constraints in ways that undermine dominant-strategy incentives. They can also exhibit non-monotonic revenues and be vulnerable to collusion and false-name or shill bidding (Ausubel and Milgrom, 2006). Multidimensional contracting further weakens any presumption that richer mechanisms dominate. With multi-dimensional private information and endogenous post-award actions, scoring rules shape both selection and execution incentives, and evidence from contract auctions shows that multi-attribute formats can exacerbate adverse selection and interact with moral hazard such that simpler single-attribute mechanisms can outperform richer ones in revenue without improving downstream performance (Kong et al., 2022).

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